Stationarity and Ergodicity of Local Dynamics of Interacting Markov Chains on Large Sparse Graphs

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Abstract

Finite-state interacting Markov chains are a well-studied class of stochastic processes on networks that have been used to model diverse phenomena, such as gas particles and magnetism from physics, voting, reputation systems, and the spread of epidemics. The nature of the interaction of the dynamics of these chains is governed by an underlying graph. As such, these processes are closely related to graphical models from statistics and machine learning and Gibbs measures from statistical mechanics. In studying these models, which for physical applications requires large collections of interacting chains, it is often of interest to autonomously characterize the typical dynamics of a given chain. Current approximations used in practice include Monte Carlo methods, the dynamic cavity method, and the mean field approximation, the latter of which is asymptotically exact on sequences of dense graphs. In a recent work by Lacker, Ramanan, and Wu, a coupled non-Markovian system of equations referred to as the *local recursion* is introduced, characterizing the dynamics of a typical particle and its neighborhood on infinite tree-like graphs. This thesis investigates the local recursion as an approximation to the dynamics of a particle and its neighborhood on a finite graph. It also investigates conditions needed for ergodicity of the local recursions. To do the latter, we first study a simplification of the local recursion as a chain of infinite order called the *frozen local recursion*. Infinite order chains are a class of processes whose ergodic properties have been studied; we apply the results of several papers on infinite-order chains to establish results on the frozen local recursion. We then introduce a reformulation of the local recursion as a nonlinear Markov process, a well-known class of processes with existing theory on stationarity and ergodicity. Finally, we discuss a nonlinear approximation of the nonlinear Markov formulation, which we call the k-approximation, in terms of the previous k timesteps of the local recursion's history. Under suitable conditions we demonstrate existence of stationary distributions for the k-approximation, and we also discuss some of its other properties. We also give numerical evidence of ergodicity for the k-approximations, as well as a numerical comparison of the k-approximations and the nonlinear Markov chain to other approximation methods. Throughout the work, we pose questions to guide future research.

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1 Introduction

Interacting particle systems are classes of stochastic processes used to model many natural phenomena in statistical physics and other fields like network queueing theory. These particle systems model entities via nodes on a graph, and their relationships by edges in the graph. To each node we assign a state that evolves randomly throughout time in a fashion governed by the underlying graph, or interaction network. We consider particle systems that evolve in a locally interacting, time-homogeneous, discrete-time Markovian fashion, which we can refer to as *interacting Markov chains*. Since we restrict to this class of processes in this thesis, we interchangeably refer to them as interacting particle systems throughout.

We are interested in studying the long-time behavior of these particle systems on graphs, in particular, the local behavior of such processes around a node or a neighborhood of a node. In this vein, two such approximations have been developed for this purpose: the mean field approximation and the local recursions, the focus of this thesis. It is known that the mean field approximation is asymptotically exact in the limit of dense graphs whose vertex counts go to infinity [3, 20], and given the ease of implementation, it has become a mainstay in the analysis of interacting particle systems. See Section 3 for further detail on this limit. For sparse graphs, an alternate method for interacting diffusions called *the local recursion* has been developed by Lacker, Ramanan, and Wu, and was further explored in the thesis [32].

While it is known that the local recursions are asynmptotically exact on finite time intervals, the question of approximating the long-time stationary behavior of these systems has not yet been fully analyzed. In this work, we analyze ergodicity of the local recursion, and in particular, if the ergodic properties of the full particle system are reflected in those of the local recursion. Inspired by [15] we introduce a novel reformulation of the local recursion as a nonlinear Markov chain, obtained by viewing the state space as histories of the process identified with the unit interval via a certain base expansion. Then, we introduce a novel approximation called the k-approximation, inspired by the k-Markov approximation [10] and the τ-approximation of [32]. We prove existence of stationary distributions of the k-approximations in the total variation topology using fixed-point theorems. Separately, we also establish stationarity and ergodicity for a simplified model of the local recursion based on the theory of infinite order chains. See the Appendix Section A.3 for a review of probability distances and convergence, and other theorems used throughout the thesis.

Finally, we give numerical evidence analyzing the convergence of the k-approximations to the local recursions, and results comparing the local recursion to the mean field approximation for a variety of processes and graphs. We find that the local recursion and the k-approximations are capable of capturing more nuanced behavior than the

mean field approximation, and are overall more robust approximations than the mean field, in many cases. We discuss the tradeoffs further in Section 6.6.

This report is organized as follows. In Section 2 we introduce interacting particle systems and several examples. In Section 3 we introduce the mean field and local recursions. Section 4 provides an outline of our analysis of the local recursion and surveys infinite-order chains and nonlinear Markov chains: related work we applied to our problem. In Section 5, we begin by introducing the reformulation of the local recursion as a nonlinear Markov chain, and the novel k-approximation. We then outline our results established on the ergodicity of the local recursion and of the k-approximations. Finally, in Section 6 we present numerical results exploring the thesis of this report, and we close with a discussion. Background results and proofs of our results are relegated to the appendix.

2 Interacting Particle Systems

Collections of random variables have been studied throughout the entire history of the field of probability theory. Beginning in the simplest case, collections of independent, identically distributed variables were among the earliest collections of random variables to be studied, culminating in some of the most well-known results in the field: the law of large numbers and central limit theorem. Stochastic processes - collections of random variables indexed by time - have also been studied, and are important models for natural time-evolving phenomena, such as the motion of a microscopic particle in fluid, or the price of a stock. The most widely studied model for stochastic processes has been the random walk and its continuous time counterpart, Brownian motion. However, in many cases it is useful to model collections of random variables with more complicated relationships that may model spatial properties or causal dependencies. The dependency structures can be captured by a graph, whose nodes represent the random variables and edges provide information about the conditional dependency structure between these variables. Such models are called *probabilistic graphical models*, and have been studied extensively in computer science and statistics.

Interacting particle systems are the dynamic analog of probabilistic graphical models: collections of random variables on an underlying graph that vary in time. Such models have been used to study a number of phenomena in statistical physics and epidemiology [21, 27]. We list a few well-known examples of interacting particle systems and static graphical models in Table 1. More specialized examples have been used to model phenomena such as reputation in social networks, TCP connections, queueing networks, and load balancing [20, 22]. In each of these examples, there is an underlying graph capturing the dependence of the infinitesimal evolution of each particle on the other particle.

Application	Model
Gas particles	Hardcore model (static) and MCMC algorithm (dynamic)
Epidemic spread	Graph SIR model, contact process (dynamic)
Magnetism	Glauber dynamics for Ising Model (dynamic)
Voter Consensus	Voter Model (dynamic)

Table 1: Examples of interacting particle systems and the systems they model

While interacting particle systems can be defined in continuous time, we focus on the discrete time case. To rigorously define an interacting particle system, we introduce a general undirected graph $G = (V, E)$ where V is the set of nodes of the graph and E its edges. Then a *discrete time interacting particle system* is a $\mathcal{X}^V$-valued stochastic process $\{X(t)\}_{t\in\mathbb{N}}$ where $\mathcal{X}$ is the state space of the process, and $X(t) = (X_v(t))_{v\in V}$ records the state of each particle at time t. The following notation will be used throughout the report:

- Denote by $X_S(t) = (X_v(t))_{v\in S}$ for any subset $S \subseteq V$,
- In the case we need to discuss histories of the process, let $X_S[t] := (X_v(s))_{v\in S, s\in[0,t]\cap\mathbb{N}}$.
- Let $X_S[t, t_0]$ denote the history of the process on S between times t_0 and t, which we can write:

$$X_S[t, t_0] := (X_v(s))_{v\in S, s\in[t_0,t]\cap\mathbb{N}}.$$

- We denote the set of neighbors of v by $\mathcal{N}_v$, or interchangeably ∂v as the set $\{w \in v : (w, v) \in E\}$. If $w \in \partial v$, we write $w \sim v$.

We restrict consideration to interacting particle systems that satisfy the following conditions:

1. The state space $\mathcal{X}$ is finite.
2. The collection $\{X_v(0)\}_{v\in V}$ are i.i.d. from some measure ν on the measurable space $(\mathcal{X}, 2^{\mathcal{X}})$. That is, we have i.i.d. initial conditions.
3. The evolution of each particle $X_v(t)$ at a given time only depends on the states of its neighbors, plus i.i.d. noise. More precisely, let $\{\xi_v(t)\}_{v\in V, t\in\mathbb{N}}$ be i.i.d. random variables in some Polish space $(\mathbb{X}, \mathbb{F})$, representing the noise. Then we assume that $X_v(t)$ evolves according to the following update:

$$X_v(t+1) = F(X_v(t), X_{\partial v}(t), \xi_v(t+1)), \tag{1}$$

for some measurable function $F : (\mathcal{X}, \mathcal{X}^{\partial v}, \mathbb{X}) \to (\mathcal{X}, 2^{\mathcal{X}})$. We refer to this assumption as the *local interactions property*, as in [32]. Notice that the form

of the update implies that the system evolves in a *time-homogeneous Markovian* fashion, that is $\{X(t)\}_{t\in\mathbb{N}}$ is a discrete-time homogeneous Markov process. Equivalently, for any $0 < s < t < \infty$, $X(t)$ is independent of $\{X(s')\}_{s'<s}$ given $X(s)$, or

$$X(t) \perp\!\!\!\perp \{X(s')\}_{s'<s} \mid X(s). \tag{2}$$

Throughout most of this report, we focus on the case where G is a rooted regular tree of degree κ. We denote the infinite regular tree of degree κ by T_κ. In the case $\kappa = 2$, T_κ is nothing but the line, which can be identified with the integers $\mathbb{Z}$. While it may seem that restricting our analysis to the case of trees might limit the power of our results, we argue that it is natural to first study the interacting particle systems and their approximations on the trees instead of general graphs. The simplest reason is that analysis on regular trees is more tractable than on general graphs. In fact, the symmetry properties of T_κ are crucial in the derivation of the local recursion that we present in Section 3.2. Furthermore, trees are natural models of *sparse* graphs as well as some types of random graphs; though the local recursion has been defined on random trees arising as local limits of sparse random graphs, understanding the local recursion on regular trees is an important first step. We also note more generally that the universal cover of any graph is a tree, and in some cases one can relate an interacting particle system on a graph to the same dynamics on its universal cover. See the introduction of [27] for further notes on these comments.

2.1 Examples

We introduce several examples of interacting particle systems below and a few algorithms for simulation. See Section 2 of [32] for some further examples. While many of these dynamics have continuous time counterparts, we again focus on the discrete time case. We note that the literature on interacting particle systems is very deep. See the book by Liggett [21] and the author's related works for an introduction of the theory of several of these processes in the continuous time case.

Ising Model

The Ising model originated in statistical physics as a model for magnetism. Given a graph $G = (V, E)$, traditionally a lattice in the physics literature, each node in the graph is assigned a *spin* in $\mathcal{X} = \{+1, -1\}$. Every configuration of particle spins, $\sigma \in \{+1, -1\}^V$ is given an associated energy $H(\sigma)$, where the function $H(\cdot) : \{+1, 1\}^V \to \mathbb{R}$ is called the Hamiltonian. The Hamiltonian is used to define a probability distribution π on the possible spin configurations on the graph, that makes it more likely to observe low energy states:

$$\pi(\sigma) = \frac{1}{Z} e^{-\beta H(\sigma)}, \tag{3}$$

where Z is a normalization constant that makes π a probability measure, and β is a parameter known as the inverse temperature. The general Hamiltonian for the Ising model is given by

$$H(\sigma) = -J \sum_{w \sim v} \sigma_v \sigma_w - h \sum_{v \in V} \sigma_v. \tag{4}$$

Here, h represents an external magnetic field such that if $h > 0$, then positive spin particles are more likely to be observed. For simplicity, we consider only the ferromagnetic case where $J = 1, h = 0$, so that our Hamiltonian is of the form

$$H(\sigma) = -\sum_{w \sim v} \sigma_v \sigma_w.$$

In this case, adjacent spins tend to be similar to each other, but as $\beta \to 0$, the spins start to behave more and more independently.

The normalization constant Z, and therefore the marginal probabilities, is typically incomputable for large graphs G, and so we must rely on Markov chain Monte Carlo (MCMC) methods to approximate the distribution π. There are many such MCMC methods such as Glauber dynamics or the more general Metropolis algorithm, and much theoretical work has gone into investigating mixing and other properties of these algorithms. For finite graphs G, we choose to introduce Glauber dynamics, which draws a sample from π given an input state σ, by updating with some probability the spin of a single node at each time step. At each time step we uniformly pick a vertex of the graph, and flip the spin at that node with probability

$$\frac{\exp(\beta S(\sigma, v))}{\exp(\beta S(\sigma, v)) + \exp(-\beta S(\sigma, v))} \tag{5}$$

where $S(\sigma, v) = \sum_{w : w \sim v} \sigma(w)\sigma(v)$. The process continues until convergence is reached, if the algorithm converges in finite time at all. To prove that these dynamics have a stationary distribution equal to π, it suffices to check the Kolmogorov cycle criterion or the reversibility condition - see Section 2.2.4 of [32] for an explicit proof. One only gets sufficiently close to the stationary distribution in order to approximately sample from the distribution π. The random time required to achieve such an approximate sample is known as *the mixing time*, and is typically difficult to estimate.

Algorithm 1 Glauber Dynamics

procedure GLAUBERDYNAMICS(G, σ, T, β)
  for $t \in \{1, \ldots, T\}$ **do**
    $\sigma' \leftarrow \sigma$.
    Pick v uniformly from V.
    $\sigma'_v \leftarrow -\sigma_v$ with probability

$$\frac{\exp(\beta S(\sigma, v))}{\exp(\beta S(\sigma, v)) + \exp(-\beta S(\sigma, v))}$$

$\triangleright$ Recall $S(\sigma, v) = \sum_{w:w\sim v} \sigma(w)\sigma(v)$

    $\sigma \leftarrow \sigma'$
  Return σ

In the sequel, we mainly focus on *parallel Glauber dynamics*, which is an adaptation of Glauber dynamics. In this case, there is no notion of a uniform measure over the countable set of nodes of the graph. Hence, instead of randomly choosing a node as with Algorithm 1, we update every node in the graph with some update probability p. Pseudocode for this algorithm is given in Algorithm 2. These dynamics have been studied in [2], although we do not know the precise ergodicity properties of parallel Glauber dynamics when $p < 1$.

Algorithm 2 Parallel Glauber Dynamics

procedure PARALLELGLAUBER(G, σ, T, β, p)
  for $t \in \{1, \ldots, T\}$ **do**
    $\sigma' \leftarrow \sigma$.
    for $v \in V$ **do**
      $\sigma'_v \leftarrow -\sigma_v$ with probability

$$\frac{\exp(\beta S(\sigma, v))}{\exp(\beta S(\sigma, v)) + \exp(-\beta S(\sigma, v))}$$

$\triangleright$ Recall $S(\sigma, v) = \sum_{w:w\sim v} \sigma(w)\sigma(v)$

    $\sigma \leftarrow \sigma'$
  Return σ

Potts Model

A simple extension of the Ising model to multiple states is the Potts model, where the spin of a particle can take values in $\mathcal{X} = \{1, 2, \ldots, q\}$. As with the Ising model, there is a measure of the same form as Eq. 3 induced by the Hamiltonian

$$H(\sigma) = \sum_{(i,j)\in E} \delta_{\sigma_i, \sigma_j}, \tag{6}$$

where the $\delta_{x,y}$ is defined by

$$\delta_{x,y} = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x \neq y. \end{cases} \tag{7}$$

While there is an analogous Hamiltonian for the antiferromagnetic case with external magnetic field, for simplicity we restrict our attention to the above Hamiltonian. As with the parallel Glauber dynamics, we consider parallel dynamics for the Potts model, which is exactly the same as 2 except the update probability is given by

$$\frac{\exp(\beta T(\sigma, v))}{\exp(\beta T(\sigma, v)) + \exp(-\beta T(\sigma, v))} \tag{8}$$

with $T(\sigma, v) = \sum_{w:w\sim v} \delta_{\sigma_w,\sigma_v}$

Contact Process

The contact process is a well-known model for epidemics. In contrast to other models such as the SIR model, the contact process is an SIS-type model where particles that become infected may recover, but can always be reinfected. If $G = (V, E)$ is a graph, then the contact process on the graph $\{X(t)\}$ is a $\{0,1\}^V$ valued stochastic process that evolves according to the following equations:

$$\mathbb{P}(X_v(t+1) = 1 \mid X_v(t) = 0) = \frac{p}{|\partial_v|} \sum_{w\in\partial v} X_w(t) \tag{9}$$

$$\mathbb{P}(X_v(t+1) = 0 \mid X_v(t) = 1) = q, \tag{10}$$

where p, q are parameters in $[0,1]$ called the infection and recovery rates, respectively. Intuitively, the model states that an infected particle recovers with probability q at each time step, and a susceptible particle is infected with a probability based on p and the fraction of infected neighbors it has. For an overview of the contact process on infinite regular trees in continuous time, see [27].

Example 2.1. We provide an explicit example of recasting the contact process on the 1-dimensional lattice $\mathbb{Z}$ in the assumed update form given in Eq. (1). Given $\xi_v(t)$ iid noise drawn uniformly from $[0,1]$, let $\{X_v(t)\}$ denote the contact process. The interacting particle system then evolves according to the equation

$$X_v(t+1) = \begin{cases} \mathbf{1}\left(\xi_v(t+1) < \frac{p}{2}(X_{v-1}(t) + X_{v+1}(t))\right), & \text{if } X_v(t) = 0 \\ \mathbf{1}\left(\xi_v(t+1) < 1 - q\right), & \text{if } X_v(t) = 1. \end{cases} \tag{11}$$

More compactly, we can express this as

$$X_v(t+1) = \mathbf{1}\left(\xi_v(t+1) < \frac{p}{2}(X_{v-1}(t) + X_{v+1}(t))\right)\mathbf{1}\left(X_v(t) = 0\right) \tag{12}$$

$$+ \mathbf{1}\left(\xi_v(t+1) < 1 - q\right)\mathbf{1}\left(X_v(t) = 1\right) \tag{13}$$

Voter Process

The voter process models majority opinion dynamics among a population. Given a graph $G = (V, E)$ the voter process $\{X_v(t)\}_{v \in V, t \in T}$ represents one of two opinions held by a voter at each node - hence $\mathcal{X} = \{0, 1\}$. The rules for the evolution of the voter process are simple: given a particle, at the next time step it adopts the opinion held by a majority of its neighbors. If there is a tie, it is broken by choosing at random one of the two opinions. Formally, then, the update rules for the voter process are given by:

$$\mathbb{P}(X_v(t+1) = 1) = \mathbf{1}\{\sum_{w \in \partial_v} X_w(t) > \frac{|\partial v|}{2}\} + \frac{1}{2}\mathbf{1}\{\sum_{w \in \partial_v} X_w(t) = \frac{|\partial v|}{2}\}. \tag{14}$$

We note that on the infinite regular graph T_κ for κ odd, this process is deterministic given the initial conditions, while for κ even the process is stochastic. The paper [17] studies the voter process and some questions about its long term behavior.

Oscillating Cellular Automaton Model

We introduce a stochastic cellular automaton inspired by [9] that exhibits convergence to oscillating behavior. Consider any regular tree of degree κ, T_κ. The binary oscillator cellular automaton (c.a.), as we call it, is a stochastic process $\{X_v(t)\}_{v \in V, t \in \mathbb{N}}$ with state space $\{0, 1\}^V$ that evolve via the following set of equations:

$$\mathbb{P}(X_v(t+1) = 0 \mid X_v(t) = 1) = 1 - \frac{p \sum_{w: w \in \partial v}(1 - X_w(t))}{|\partial v|} \tag{15}$$

$$\mathbb{P}(X_v(t+1) = 1 \mid X_v(t) = 0) = 1 - \frac{p \sum_{w: w \in \partial v} X_w(t)}{|\partial v|} \tag{16}$$

The model can be motivated by regarding it as an example of contrarian voter dynamics: indeed, a particle switches to the opposite opinion with high probability, lessened only if many of its neighbors has the opposite opinion. Locally, note that if all of the neighbors of a particle v are in the same state as the particle at time t, $X_v(t)$, then at the next time step $X_v(t+1) = 1 - X_v(t)$ with probability one. Then as soon as $X(t) = \mathbf{0}$ or $X(t) = \mathbf{1}$, the consecutive states will oscillate between the states $\mathbf{0}, \mathbf{1}$. The idea is that the system converges to one of the two states $\mathbf{0}, \mathbf{1}$ and then alternates between those two states by construction of the dynamics. See [9, Section 4.1] for further details and a many-state extension.

Oscillating behavior has been observed in myriad applications in the natural sciences: collective behavior in bird flocking, pedestrian movement, as well as in epileptical seizures [16]. The prototype model for coupled oscillation is the *Kuramoto model*, a collection of dynamic particles that exhibits synchronization behavior. The typical

Kuramoto model is given by N particles on a circle, with angles θ_i, that rotate around the circle according to the system of ODE

$$\frac{\mathrm{d}\theta_i}{\mathrm{d}t} = \omega_i + \frac{K}{N}\sum_{i=1}^{N}\sin(\theta_j - \theta_i), \quad \forall i = 1, \ldots, n. \tag{17}$$

Note that each particle has its own natural speed ω_i, but it has been observed that the collection of N particles eventually settle into phase lockstep: they eventually rotate around the circle at the same speed. While there are simplifications of the Kuramoto model to discrete time and discrete state, see e.g. [30, 16], many of the models have large state spaces, which makes our local recursion of Section 3.2 computationally intensive to simulate. Instead, we study the binary oscillator c.a. model defined above as a toy model for this synchronization behavior.

Stochastic Oscillation

We also introduce a simple two-state model, intended to exhibit stochastic frequency rather than the binary two state pattern characteristic of the oscillating c.a. model above. Each node oscillates with some random frequency, and this frequency depends on the neighboring nodes. The dynamics are given as in the following equation, and the state space is given by $\mathcal{X} = \{0, 1\}$.

$$\mathbb{P}(X_v(t+1) = 1 - X_v(t)) = q + p\frac{|w \in \partial v : X_u(t) = X_v(t)|}{|\partial v|}. \tag{18}$$

Here $q = \frac{1-p}{2}$, and $p \in (0, 1)$ is a parameter in the model.

3 Existing Approximations

In practice, if one wants to numerically understand the behavior of an interacting particle system $\{X(t)\}_{t\in\mathbb{N}}$ on $G = (V, E)$, it is straightforward to just simulate these and use Monte Carlo to obtain estimates for values such as $\mathbb{P}(X_v(T) = x)$ for desired $v \in V, T \in \mathbb{N}, x \in \mathcal{X}$. However, for large graphs it may be computationally infeasible to store G and simulate dynamics for long periods of time. In particular, if we were only interested in looking at the distribution of a certain node and its neighborhood, it would be inefficient to simulate dynamics on the entire graph in order to gain information about the local process. Hence, we consider approximations to the interacting particle system that model the dynamics of a typical particle in the network. The first we consider is the classical *mean field approximation* which models the dynamics of a single typical particle, and can be shown to be accurate on large dense graphs. The second approximation we consider is the *local recursion*, which models the dynamics of a particle and its neighborhood, and was developed primarily

for sparse graphs. Other methods include the dynamic cavity method, see e.g. [17], and moment approximation methods, although we do not study them in this current work. See [32, Section 3.2,3.3], for a deeper look at these approximation methods, how they compare to each other, and further references.

3.1 Mean Field Approximation

The mean field approximation has a deep history as an extensively studied model in physics, and has been developed in continuous time and discrete time [20, 3, 25]. The mean field approximation is designed for "weakly interacting" particle systems, where in a system of n nodes, the influence of any particle on another particle is of order $o(n)$. More precisely, consider an interacting particle system $\{X(t)\}$ on a graph $G = (V, E)$, whose transitions at a node $v \in V$ in fact depend on its neighbors via their *empirical measure*, a quantity we denote

$$M_{v,t} = \frac{1}{|\partial v|} \sum_{w \in \partial v} \delta_{X_w(t)}, \tag{19}$$

where δ_x is the dirac point mass measure, defined by

$$\delta_x(A) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A. \end{cases} \tag{20}$$

Building upon Eq. 1, we can then write the update for the interacting particle system as

$$X_v(t+1) = \tilde{F}(X_v(t), M_{v,t}, \xi_v(t)), \tag{21}$$

for a suitable function $\tilde{F} : \mathcal{X} \times \mathcal{P}(\mathcal{X}) \times \mathbb{X} \to \mathcal{X}$, where $\mathcal{P}(\mathcal{X})$ denotes the space of probability measures on $\mathcal{X}$. All of the interacting particle systems we considered in Section 2 are of this form. Then the mean field approximation amounts to assuming neighboring particles are independent and approximating $M_{v,t}$ by $\mathrm{Law}(X_v(t))$. Intuitively, we can think about this as a law of large numbers-type result, which grows more accurate as the number of neighbors $|\partial v|$ grows.

Rigorously, the mean field approximation relies on the following convergence result. Fix a sequence of dense increasing graphs G_n, in the sense that quantities d_n, the minimum degree of any node in the graph G_n, satisfies $d_n \to \infty$. On each G_n define an interacting particle system $\{X^{G_n}(t)\}$ each with the same dynamics, i.e., they each satisfy Eq. (21) for the same function $\tilde{F}$. For each $v \in V$ denote the empirical measure around v at time t by $M_{v,t}^{G_n}$. Define the *mean field process* $X^{MF}(t)$, a stochastic process on $\mathcal{X}$ that satisfies the update rule

$$X^{MF}(t+1) = \tilde{F}(X^{MF}(t), \mathrm{Law}\, X^{MF}(t), \xi_v(t+1)). \tag{22}$$

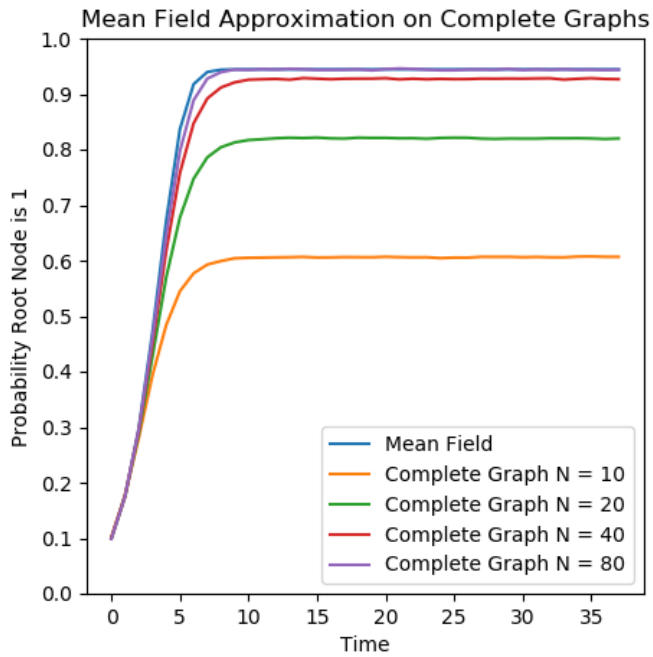


Figure 1: $p = 0.9, q = 0.05, 10^5$ MC samples. $\mathbb{P}(X_v(0) = 1) = 0.1$. The mean field approximation in comparison to the dynamics when $N = 80$ is so accurate that the lines overlap.

Such a process, whose dynamics depends not only on its past state, but also on the law of the past state, is referred to as a *nonlinear Markov chain*. See Section 4.1.1 for a further discussion. Then as $n \to \infty$, we have the convergence

$$M_{v,t}^{G_n} \to \mathrm{Law}(X^{MF}(t)), \tag{23}$$

and note that the limit does not depend on the initial node v. See [20] for further details. Then, the mean field approximation to an interacting particle system $\{X^G(t)\}_{t\in\mathbb{N}}$ on a fixed finite graph G amounts to approximating

$$M_{v,t}^{G} \approx \mathrm{Law}(X^{MF}(t))$$

for all $v \in V, t \in \mathbb{N}$, and assuming neighboring particles to be independent.

Implementing the mean field approximation relies on Eq. (22). Since our dynamics $\{X(t)\}$ is restricted to a finite state space, we can model the law of $X^{MF}(t)$ via a vector of length $|\mathcal{X}|$, and we can easily calculate $X^{MF}(t)$ for all t via a recursion. Simply, we could write by law of total probability, for every $x \in \mathcal{X}$:

$$\mathbb{P}(X^{MF}(t+1) = x) = \sum_{y\in\mathcal{X}} \mathbb{P}(X^{MF}(t+1) = x \mid X^{MF}(t) = y)\mathbb{P}(X^{MF}(t) = y). \tag{24}$$

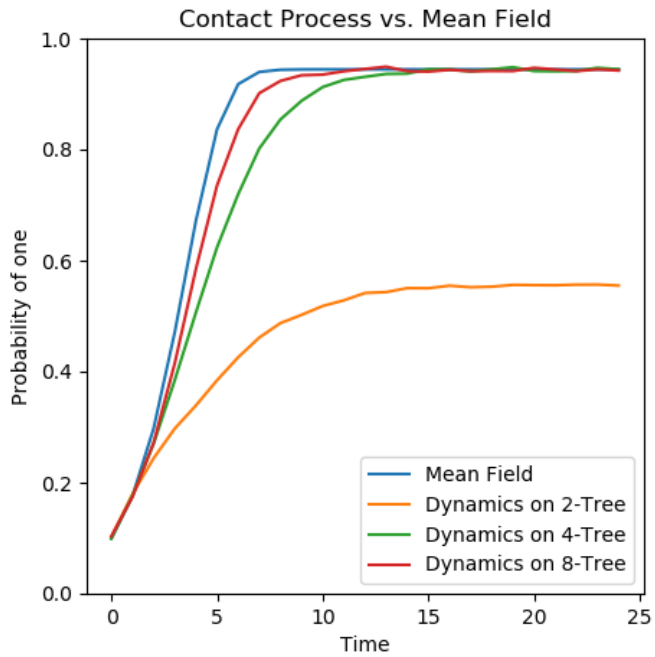


Figure 2: $p = 0.9, q = 0.05, 10^5$ MC samples. $\mathbb{P}(X_v(0) = 1) = 0.1$. These trees are each at a depth of 4.

Of course the conditional probability $\mathbb{P}(X^{MF}(t+1) = x \mid X^{MF}(t) = y)$ is dynamics-specific and depends on the form of the noise $\xi_v(t)$ in Eq. (22). We also emphasize that the conditional probability depends on the Law X^{MF}. Figure 1 shows the mean field approximation to the contact process compared with the contact process on complete graphs of increasing degree, recording the probability of the a given node being infected (at state 1). The dynamics were approximated via Monte Carlo with $N = 10^6$ samples. To be precise, we simulated the contact process according to Eq. (9), and random iid initial conditions, up to time $T = 25$. We then obtained estimates of the probabilities of a given node being 1 via the sample average over all simulated runs. To be precise, let $\{X_v^{(i)}(t)\}$, for $i = 1, \ldots, 10^6$, denote the simulated values for the state of node v at time t. We estimate $\mathbb{P}(X_v(t) = 1)$ by $\frac{1}{N}\sum_{i=1}^{N} X_v^{(i)}(t)$. Other quantities can be estimated similarly. Figure 2 shows a similar experiment, except with the mean field approximation compared to dynamics on κ-regular trees, of low depth. We again used 10^6 MC samples. In both cases we observe that as the degree of the graph increases, the mean field simulation becomes a better and better approximation for the dynamics on these high-degree graphs. This agrees with the convergence results in Eq. 23.

Example 3.1. We show how to derive the mean field approximation recursion for

the contact process. Using Eq. (24), we have that

$$\mathbb{P}(X^{MF}(t+1)=1) = \mathbb{P}(X^{MF}(t+1)=1 \mid X^{MF}(t)=1)\mathbb{P}(X^{MF}(t)=1) + \mathbb{P}(X^{MF}(t+1)=1 \mid X^{MF}(t)=0)\mathbb{P}(X^{MF}(t)=0).$$

But because $X^{MF}(t)$ satisfies Eq. (22) for the contact process dynamics, we know that

$$\mathbb{P}(X^{MF}(t+1)=1 \mid X^{MF}(t)=1) = (1-q),$$

and

$$\mathbb{P}(X^{MF}(t+1)=1 \mid X^{MF}(t)=0) = p\mathbb{P}(X^{MF}(t)=1).$$

This last equation holds since for $\{X_v(t)\}_{v\in V, t\in\mathbb{N}}$ the contact process, we have that

$$\mathbb{P}(X_v(t+1)=1 \mid X_v(t)=0) = \frac{p}{|\partial v|}\sum_{w\in\partial v} X_w(t),$$

and this is expressible in terms of the empirical measure $M_{v,t}$ via

$$\frac{p}{|\partial v|}\sum_{w\in\partial v} X_w(t) = p\int_{\mathcal{X}} x dM_{t,v}(x). \tag{25}$$

Hence the mean field approximation satisfies

$$\mathbb{P}(X^{MF}(t+1)=1 \mid X^{MF}(t)=0) = p\int_{\mathcal{X}} x d\mu = p\mathbb{E}[X^{MF}(t)],$$

where $\mu = \text{Law}\, X^{MF}(t)$. But the expectation $\mathbb{E}[X^{MF}(t)]$ is simply $\mathbb{P}(X^{MF}(t)=1)$.

We can then obtain that

$$\mathbb{P}(X^{MF}(t+1)=1) = (1-q)\mathbb{P}(X^{MF}(t)=1) + p\mathbb{P}(X^{MF}(t)=1)\mathbb{P}(X^{MF}(t)=0). \tag{26}$$

Writing $\mathbb{P}(X^{MF}(t)=1) = p_t$, we arrive at the simple recursion

$$p_{t+1} = (1-q)p_t + pp_t(1-p_t), \tag{27}$$

which is easily computed.

For further examples of mean field approximations to different interacting particle systems, refer to the Appendix, Section A.4. We record the mean field approximations for each of the dynamics in Section 2.

3.2 Local Recursion

While the mean field process provides an approximation for interacting particle systems on dense graphs, another approximation called the *local recursion* as discussed in [19, 32], covers the *sparse* regime. The local recursion models the behavior of a

typical particle and its neighborhood via a closed, self-consistent set of equations expressible only in terms of $X_{\mathcal{T}}(t)$, where $\mathcal{T}$ is a subset of nodes defined below. In order to achieve this, the derivation of the local recursion takes advantage of the symmetry properties of the regular tree and a certain conditional independence structure of histories of particles. Although we will not discuss the derivation of the local recursion, we record this conditional independence property as the following condition:

$$X_S[t] \perp\!\!\!\perp X_{V \setminus (S \cup \partial^2 S)}[t] \mid X_{\partial^2 S}[t], \tag{28}$$

where $S \subset V$ and $\partial^2 S$ is the *double boundary* of S, the set of nodes of distance at most 2 away from S; we can alternatively think of this set as exactly $\partial^2 S = \{v \in V : d_G(v, S) \in \{1, 2\}\}$, where $d_G(x, y)$ for $x, y \in G$ is the number of edges in the shortest path between x, y in G. Intuitively, we can think of Eq. (28) as a dynamic analog to the spatial Markov property from graphical models, not discussed here. Although the local recursion has also been introduced in a more general setting including diffusions on many classes of random graphs, we focus on the discrete time case on infinite regular trees T_κ of degree κ. To rigorously define the local recursions, we consider the following setup and notation:

- Without loss of generality fix a node v of T_κ as the root node, and let $\mathcal{T}$ denote the depth one subtree rooted at v.
- Relabel the nodes in $\mathcal{T}$ such that v is 0, and its neighbors are labeled $1, \ldots, \kappa$.
- Let $P_v^j(X[t]) := \mathbb{P}(X_v(t+1) = j \mid X[t])$ denote the transition probability of the interacting particle system, as a function of the history of the process. Note that by the locally interacting and Markov property encoded in Eq. (1), we can rewrite the conditional probability as $\mathbb{P}(X_v(t+1) = j \mid X_{v \cup \partial v}(t))$. In this way we can denote the quantity $P_v^j(X[t])$ by $P_v^j(X_{v \cup \partial v}(t))$, and regard P_v^j as a function $\mathcal{X}^{\kappa+1} \to [0, 1]$, for every $j \in \mathcal{X}, v \in \mathcal{T}$.
- $P_v(X_{v \cup \partial v})$ be the $|\mathcal{X}|$-length vector of transition probabilities whose entries are the transition probabilities for node v.

Then for any interacting particle system $\{X_v(t)\}_{v \in T_\kappa, t \in \mathbb{N}}$, we can define the local recursion as follows:

Definition 3.1 (Local Recursion)**.** *A weak solution to the local recursion with initial condition $\lambda_0 \in \mathcal{P}(\mathcal{X})$ is a probability space $(\overline{\Omega}, \overline{\mathcal{F}}, \overline{\mathbb{P}})$ on which is defined an $\mathcal{X}^{\mathcal{T}}$-valued stochastic process $\{\overline{X}(t)\}_{t \in \mathbb{N}_0}$ that satisfies the following:*

1. *$\overline{X}_v(0), v \in \mathcal{T}$ is iid with common law λ_0.*
2. *$\overline{\mathbb{P}}(\overline{X}_0(t+1) = j \mid \overline{X}[t]) = P_0^j(\overline{X}_{\partial v}(t))$.*

3. *For each* $v \in \mathcal{T} \setminus \{0\}$ *and* $t \in \mathbb{N}_0$,

$$\overline{\mathbb{P}}(\overline{X}_v(t+1) = j \mid \overline{X}[t]) = \mathbb{E}^{Y[t] \overset{(d)}{=} \bar{X}[t]} \left[P_0^j(Y_{\mathcal{T}}(t)) \mid Y_0[t] = \bar{X}_v[t], Y_1[t] = \bar{X}_0[t] \right]. \tag{29}$$

Given this, we can define $\overline{P}_v^j$ as the transition probabilities of the local recursion in analogy with the notation for the transition probabilities of the dynamics.

For further information and a derivation of the local recursions, see [32]. One immediate observation from Definition 3.1 is that the local recursion is more complex than its underlying dynamics $\{X_v(t)\}$, in two ways. Indeed, while the interacting particle system is Markovian, $\{\bar{X}(t)\}$ is decidedly non-Markovian because the transition probability in Eq. (29) depends on the history of the process itself up to time t, $\bar{X}[t]$, through a conditional law. In fact, we can view the local recursion as a time-inhomogeneous *chain of infinite order*, where the time-homogeneity is due to the dependence on the law of the process up to t. See Section 4.1 for further details. On the other hand, if we transform the state space of the process so that it evolves on the infinite dimensional sequence of states $\mathcal{X}^\infty$, we can view the process as a nonlinear Markov chain. In particular, since $\mathcal{X}$ is finite we can identify $\mathcal{X}^\infty$ with $[0,1]$ and study the nonlinear Markov chain with state space $[0,1]^d$ for some d. In Section 5.2 we introduce this novel reformulation of the local recursion as a nonlinear Markov chain, and analyze the ergodic properties of the local recursion via this nonlinear Markov chain reformulation. The difficulty in analyzing this reformulation is that the evolution of the process still depends on the conditional law of the process - this falls outside the typical framework for the theory of nonlinear Markov processes. See Section 4.1.1 for an overview of the theory.

The idea behind the local recursion is a convergence result of the dynamics on $\mathcal{T}$ to $\bar{X}_{\mathcal{T}}$ on sparse graphs, as the size of the graphs goes to infinity. In one particular case, let T_κ^d denote the regular tree of degree κ and depth d. Then consider interacting particle systems $\{X^{G_d}(t)\}$ on the graphs $G_d = T_\kappa^d$. As $d \to \infty$ we have that

$$X_{\mathcal{T}}^d[t] \Rightarrow \bar{X}_{\mathcal{T}}[t] \ \forall t < \infty,$$

so that the local recursions are asymptotically exact. The *local approximation*, on a fixed regular tree $G = T_\kappa^d$ with dynamics $\{X^G(t)\}$ then amounts to the following approximation:

$$X_{\mathcal{T}}^G[t] \approx \bar{X}_{\mathcal{T}}[t], \forall t. \tag{30}$$

Example 3.2. Consider the contact process on the 2-regular infinite graph, $\mathbb{Z}$. Following a notation similar to Example 2.1 We can define the local recursion as a

process $\{\bar{X}_v(t)\}_{v\in\mathcal{T},t\in\mathbb{N}}$ on $\mathcal{T} = \{-1, 0, 1\}$ as a subset of $\mathbb{Z}$. The local recursion satisfies the following dynamics:

$$\bar{X}_0(k+1) = \begin{cases} \mathbf{1}(\xi_0(k+1) < \frac{p}{2}(\bar{X}_{-1}(k) + \bar{X}_1(k))), & \text{if } \bar{X}_0(k) = 0 \\ \mathbf{1}(\xi_0(k+1) < 1-q), & \text{if } \bar{X}_0(k) = 1 \end{cases} \tag{31}$$

$$\bar{X}_1(k+1) = \begin{cases} \mathbf{1}(\xi_1(k+1) < \frac{p}{2}(\bar{X}_0(k) + \bar{Z}_2(k))), & \text{if } \bar{X}_1(k) = 0 \\ \mathbf{1}(\xi_1(k+1) < 1-q), & \text{if } \bar{X}_1(k) = 1 \end{cases} \tag{32}$$

$$\bar{X}_{-1}(k+1) = \begin{cases} \mathbf{1}(\xi_{-1}(k+1) < \frac{p}{2}(\bar{X}_0(k) + \bar{Z}_{-2}(k))), & \text{if } \bar{X}_{-1}(k) = 0 \\ \mathbf{1}(\xi_{-1}(k+1) < 1-q), & \text{if } \bar{X}_{-1}(k) = 1 \end{cases} \tag{33}$$

where

$$\bar{Z}_2(k) = \mathbb{E}^{Y\sim\mu}[X_1(k)|Y_0[k] = \bar{X}_1[k], Y_1[k] = \bar{X}_0[k]] \tag{34}$$

$$\bar{Z}_{-2}(k) = \mathbb{E}^{Y\sim\mu}[X_{-1}(k)|Y_0[k] = \bar{X}_{-1}[k], Y_1[k] = \bar{X}_0[k]]. \tag{35}$$

Here $\mu = \operatorname{Law} X[k]$.

While the local recursion has been shown to be asymptotically exact for finite times, one open question concerns the ergodic properties of the local recursion. In particular, suppose the interacting particle system approximated by the local recursion converges to a stationary distribution μ, and the local recursion also has a stationary distribution ν. Since the local recursion looks at the dynamics in a neighborhood of the root, we should intuitively expect that ν should approximate $\mu|_{\mathcal{T}}$, the marginal of μ restricted to the nodes in $\mathcal{T}$. See Sec. 5.2 for further details, where we pose further questions and investigate a few of them.

4 Outline and Literature Survey

We present a summary of our analysis of the problem and intertwine this with a literature survey of related work. Throughout this section, fix a regular tree of degree κ, $G = (V, E)$, and an interacting particle system $\{X_v(t)\}_{v\in V,t\in\mathbb{N}}$ on G with state space $\mathcal{X}$. Let $\{\bar{X}(t)\}$ be its local recursion.

4.1 Infinite Order Chains

To begin analyzing the local recursion the first idea is to reformulate the local recursion as a *chain of infinite order*, as mentioned in Section 3.2. The difficulty in analyzing this reformulation directly is that the local recursion as a chain of infinite order is time-inhomogeneous, since its evolution at each time step depends on the law of its history up to time t. To simplify this problem, we can consider 'freezing' the

dependence on the external measure μ, where μ is a fixed measure on the state space $\mathcal{X}^\infty$, the set of infinite sequences of elements in $\mathcal{X}$. For a more concrete example, consider the local recursion to the contact process on $\mathbb{Z}$ defined in Example 3.2. Instead of letting μ in Equation (35) vary as the law of $X[k]$, we fix this measure as a given measure on $\mathcal{X}^\infty$. We then call these dynamics the *frozen local recursion* μ. In this simplification, the first problem to study is whether or not the frozen local recursion has a stationary distribution, and under what conditions we can ensure existence of such a stationary distribution. We present the formulation of the local recursion as a chain of infinite order in Section 5.1.1, along with the question of ergodicity of the frozen local recursion for the contact process on $\mathbb{Z}$.

A *chain of infinite order* is a very general stochastic process in which the transition of the process depends on its infinite past. In this section, we denote a chain of infinite order by $\{Z(n)\}_{n\in\mathbb{Z}}$, where $Z(n)$ takes values in some finite set $\mathcal{X}$. As before, denote by $Z[n] := \{Z(s) : s \leq n\}$, which in this case is equivalent to the notation $Z[n-1, -\infty]$. Let $x_k^n = (x_k, \ldots, x_n)$ for $k > n$ be a sequence in $\mathcal{X}^{|n-k+1|}$. If two sequences of histories $x_{-\infty}^n, y_{-\infty}^n$ agree up to the mth previous time step, we write $(x_{-\infty}^n = y_{-\infty}^n)_m$.

Definition 4.1. *A homogeneous chain of infinite order is a stochastic process $(Z(n))_{n\in\mathbb{Z}}$ whose transitions probabilities depend on the infinite past, which we denote by*

$$Q_x(x_{-\infty}^{n-1}) := p(x \mid x_{-\infty}^{n-1}) := \mathbb{P}(Z(n) = x \mid Z[n-1] = x_{-\infty}^{n-1}). \tag{36}$$

The chain is time inhomogeneous if

$$\mathbb{P}(Z(n) = x \mid Z[n-1] = x_{-\infty}^{n-1}) \tag{37}$$

is a function of n, which we denote $p_n(x \mid x_{-\infty}^{n-1})$.

Definition 4.2. *Let $(Z(n))_{n\in\mathbb{Z}}$ be chain of infinite order. We call it stationary if for any sequence $x_{-\infty}^0$, and $n, n' \in -\mathbb{Z}$,*

$$\mathbb{P}(Z[n] = x) = \mathbb{P}(Z[n'] = x). \tag{38}$$

A measure μ on $\mathcal{X}^\infty$ is a stationary distribution if $\{Z(n)\}_{n\in\mathbb{Z}} \sim \mu$ is a stationary process.

The literature on chains of infinite order - also known under various names such as chains with complete connections, random Markov chains, g-measures, among others - is quite developed, having been first studied by Onisescu and Mihoc in [26] and Doeblin and Fortet [8]. In particular, the paper by Harris [15] gives several results on the stationary distributions of chains of infinite order that can be directly applied to the frozen local recursion. Harris treats the case of homogeneous chains of infinite order on a finite state space $\mathcal{X}$, and gives conditions for the existence of a stationary distribution of a chain of infinite order, given its transition functions; not only do

we have existence and uniqueness, but convergence to this stationary distribution is guaranteed at least in the Cesaro limit sense. The idea of the proof is to study the *associated Markov chain* of the chain of infinite order, whose state space consists of infinite sequences of elements - called *histories* - in the original chain. Harris encodes these sequences with a d-ary encoding scheme, making the state space of the associated Markov chain $[0,1]$:

Definition 4.3 (Associated Markov chain)**.** *Consider a chain of infinite order $\{Z(n)\}_{n\in\mathbb{Z}}$ with transition probabilities $Q_i(x)$ for $x = (x_1, x_2, \dots)$ a sequence of states. Define the functions $f_i(x) : [0,1] \to [0,1]$ by*

$$f_i(x) = Q_i(u), \tag{39}$$

where $x = 0.u_1u_2\dots$ is the $|\mathcal{X}|$-ary expansion and $Q_i(u)$ defined as in Eq. (36). The associated Markov chain to $\{Z(n)\}$ is the process $\{X(n)\}_{\mathbb{Z}}$ with transition probabilities

$$\mathbb{P}(X(n) = \frac{i+x}{|\mathcal{X}|} \mid X(n-1) = x) = f_i(x). \tag{40}$$

This construction is the basis for our introduction of the nonlinear Markov chain and the k-approximation of Sections 5.2, 5.3.

On the way to finding the existence of stationary distributions, it turns out we need the following conditions on our set of transition functions. We define for $x, y \in \mathcal{X}_{-\infty}^{n-1}$

$$\beta_m = \sup_{i,(x=y)_m} |Q_i(x) - Q_i(y)|, \tag{41}$$

which are **continuity rates**. The idea is intuitive: the continuity rates measure how much the transition probabilities can vary if the recent history of the process is fixed.

- We ask that $\lim_k \beta_k \to 0$, and

$$\sum_{m=0}^{\infty} \prod_{k=0}^{m} (1 - \frac{1}{2}|\mathcal{X}|\beta_k) = \infty. \tag{42}$$

  To the have the latter condition, it is enough that $\beta_k = O(|\mathcal{X}|/k)$.

- For some $a \in \mathcal{X}$, we have for $i \in \mathcal{X}, x \in \mathcal{X}_{-\infty}^{0}$,

$$\mathbb{P}(a|x) \geq \Delta > 0. \tag{43}$$

From [15], we have the following result.

Theorem 4.1 (Theorem 6 of [15])**.** *Consider transition functions $Q_i(x)$ that satisfy Equations (42), (43). Then:*

1. *There exists a homogeneous infinite order chain Z_n with a stationary distribution, whose transition functions are exactly the $Q_i(x)$.*

2. *This is the only such process.*

3. *We have continuity of the Z process in the following way:*

$$\lim_{k\to\infty} \mathbb{P}(Z(0) = a | Z(-1) = x_{-1}, \ldots, Z_(-k) = x_{-k}) = Q_i(x_{-\infty}^{-1}).$$

The proof follows by coupling (see Definition 4.4) two independent copies of the associated Markov process. One is able to show convergence of the law to a Cesaro limit, regardless of initial state. As a by-product of the above argument, if the continuity rates are *summable* then the infinite order chain has a stationary distribution. In fact, we also get convergence, given further assumptions:

Theorem 4.2 (Theorem 3 of [15])**.** *If the Equations 42, 43 hold and furthermore $\sum \beta_k < \infty$, then consider the associated Markov process $X(n)$. The quantity*

$$G_n(x, y) = \mathbb{P}(X(n) \leq x \mid X(0) = y) \tag{44}$$

converges in x to a attracting distribution $G(x)$ independent of y. In particular this establishes the existence of a stationary distribution for the associated Markov chain, and hence for the chain of infinite order.

We apply this technology to our problem in Section 5.1.1.

4.1.1 Markov Approximations

Consider the problem of simulating the frozen local recursion. In general, simulating chains of infinite order would lead to a exponential blowup in the amount of time and memory needed for simulation. It is then necessary to come up with algorithms which can provide good approximations to the frozen local recursions, and to develop methods that can inspect the error of approximation. One very natural approximation is that of a Markov approximation, in which the process is simulated starting at time 0, and at each consecutive step, only the last k steps in its history are taken into account. This so called *k-order Markov approximation* has been studied in [5] and [10]. In [5], Bressaud, Fernandez, and Galves study the properties of the order k Markov approximation and give conditions on the chain of infinite order that ensure the Markov approximation recovers the chain of infinite order in the limit as $k \to \infty$. The later paper [10] extends these results in a direct way to the non-homogeneous case, using essentially all of the same arguments. We present more general non-homogeneous case here. In order to ask how good this approximation scheme is, we use the so called Ornstein $\bar{d}$-metric, which measures the greatest probability of mismatch of the original chain and the approximation over any coupling between the two processes. First recall the definition of a coupling of two random variables:

Definition 4.4. *Given X, Y random variables on possibly different probability spaces $(\Omega_x, \mathcal{F}_X), (\Omega_Y, \mathcal{F}_Y)$, a coupling between X, Y is a random variable on $(\tilde{X}, \tilde{Y})$ on the product probability space $(\Omega_X \times \Omega_Y, \mathcal{F}_x \times \mathcal{F}_Y)$ such that $\tilde{X} \overset{(d)}{=} X, \tilde{Y} \overset{(d)}{=} Y$.*

Definition 4.5 ($\bar{d}$-metric)**.** *If X, Y are two (not necessarily stationary) stochastic processes, then define*

$$\bar{d}(X, Y) = \inf\{\sup_{n \in \mathbb{Z}} \mathbb{P}(\tilde{X}(n) \neq \tilde{Y}(n)) : (\tilde{X}, \tilde{Y}) \text{ is a coupling of } X, Y\}.$$

We now formally define the approximation scheme we analyze here:

Definition 4.6 (Canonical Markov Approximation of order k)**.** *Let $k \in \mathbb{N}$. Then if $\{X(n)\}_{n \in \mathbb{Z}}$ is a chain of infinite order, define the Markov approximation of order k as $\{X^{[k]}(n)\}_{n \in \mathbb{Z}}$, with the transition probabilities*

$$P_n^{[k]}(a|x_{n-k}^{n-1}) := \mathbb{P}(X_n^{[k]} = a | X^{[k]}[n-1, n-k] = x_{n-k}^{n-1}).$$

The approximation is quite intuitive: we can easily make the chain less dependent on its entire history by truncating its past after some point, and only using the information of some previous k time steps to evolve the chain forward. However, we can only expect that this approximation is good if original chain is not too sensitive to its far past; that is, if two histories x, y agree at the previous m states, $x \overset{(d)}{=} y$, then the transition probabilities of the infinite order chain should be close. To introduce this result, we first adapt the notion of continuity rates for the non-time homogeneous case:

Definition 4.7 (Non-homogeneous continuity rates)**.** *If $\{X(n)\}$ is a non-homogeneous chain of infinite order, define its continuity rates as*

$$\beta_s = \sup_{n \in \mathbb{Z}} \sup_{i \in \mathcal{X}} \sup_{(x=y)_m} \left| P_n(i|x_{-\infty}^{n-1}) - P_n(i|y_{-\infty}^{n-1}) \right|. \tag{45}$$

Given this, we then have the following result:

Theorem 4.3 (Theorem 1 of [10])**.** *Let X be a chain of infinite order with continuity rates $\beta_k \to 0$ as $k \to \infty$, and satisfies the weak non-nullness criterion that $\inf_{n \in \mathbb{Z}} \sum_{y_n \in \mathcal{X}} \inf_x \mathbb{P}_n(y_n | x_{-\infty}^{n-1}) > 0$. Then if $X^{[k]}$ denotes its order k Markov approximation, and if X has summable continuity rates, we have:*

$$\bar{d}(X, X^{[k]}) \leq C\beta_k \to 0. \tag{46}$$

The proof actually offers a more quantitative bound for the Ornstein distance between X and its approximation, in terms of the continuity rates and the quantity $\lambda_0 := \inf_{n \in \mathbb{Z}} \sum_{y_n \in \mathcal{A}} \inf_x \mathbb{P}_n(y_n | x_{-\infty}^{n-1})$. In fact, we have:

$$\bar{d}(X, X^{[k]}) \leq \frac{1 - (1 - \beta_k^*)^k}{k[\prod_{j=0}^{\infty}(1 - \beta_j^*)]^2},$$

where

$$\beta_0^* = 1 - \lambda_0$$
$$\beta_n^* = \min(\beta_0^*, \min |\mathcal{X}| \beta_n).$$

The proof bounds the distances between the chain and its approximation via a coupling. We define a maximal coupling of the two processes such that there is a positive probability of coincidence at each time step, and the probability of moving apart after having been coincident is bounded above by the continuity rates of the chain.

4.2 Nonlinear Markov chains

In Section 5.2, we investigate a reinterpretation of the local recursion as a nonlinear Markov chain, a process whose evolution at each time step not only depends on its current state, but its law as well. This is achieved by viewing the state space as sequences of elements in $\mathcal{X}$. Since $\mathcal{X}$ is finite, we can identify $\mathcal{X}^\infty$ with $[0,1]$ via identifying $\mathcal{X}$ with $\{0,1,\ldots,|\mathcal{X}|-1\}$ and mapping $(x_0, x_1, \ldots,) \in \mathcal{X}^\infty$ to $0.x_0x_1\ldots$, its base-$|\mathcal{X}|$ encoding. We can then study the ergodic properties of the local recursion via this nonlinear Markov chain reformulation and an associated nonlinear Markov chain we call the k-approximation. The idea mimics the construction of the *associated Markov chain* of Definition 4.3 We now review nonlinear Markov chains in this section, along with some methods for establishing ergodicity.

We can define a nonlinear Markov chain $X(t)$ more generally as a stochastic process on a measurable state space $(\mathcal{X}, \mathcal{F})$ such that for each $t \in T$ a denumerable index set,

$$X(t+1) = F(X(t), \mathrm{Law}(X(t)), \xi_{t+1}), \tag{47}$$

for some measurable function $F : \mathcal{X} \times \mathcal{P}(\mathcal{X}) \times \mathbb{X} \to \mathcal{X}$. Note that this is similar to Eq. (22), and indeed, the mean field process is a nonlinear Markov chain. The survey [18] provides an introduction to nonlinear Markov processes, although it mainly focuses on continuous time and space. Alternatively, we can characterize a nonlinear Markov process by the fact that its transition kernel at each time depends on the law of the process:

$$\mathbb{P}(X(t+1) \in A \mid X(t) = x) = P_{\mu_t}(x, A) \tag{48}$$

where $\mu_n = \mathrm{Law}(X_t) \in \mathcal{P}(\mathcal{X})$. This defines a function, which we refer to as a probability *kernel*, $P_\bullet$ which we can regard as a function from $\mathcal{P}(\mathcal{X}) \times \mathcal{X} \times \mathcal{F} \to [0,1]$. Of course, if the kernel $P_\mu(x, A)$ does not depend on the measure μ, then the process is Markov. Mirroring the theory for Markov processes, we impose the following conditions on the kernel:

- $P_\mu(x, \cdot)$ is a probability measure on $(\mathcal{X}, \mathcal{F})$ for every $x \in \mathcal{X}, \mu \in \mathcal{P}(\mu)$.
- $P_\bullet(\cdot, A)$ is a measurable function from $\mathcal{P}(\mathcal{X}) \times \mathcal{X} \to [0,1]$ for every $A \in \mathcal{F}$.

Nonlinear Markov chains have been used in the study of limits of weakly interacting Markov processes, which are applied in studying the mean field approximation from physics [20, 25, 3, 6], as well as social dynamics [12], and even Monte Carlo algorithms [1]. Examples of such processes are Fokker-Planck Equations [11], and McKean-Vlasov equations, which are diffusions in which the coefficients depend on the law of the process itself [7]. McKean-Vlasov processes in particular have a rich history arising from kinetic models in physics, and have been used in financial applications.

4.2.1 Stationary Processes and Ergodicity

As with the study of many processes, one is often interested in the long time behavior of nonlinear Markov processes. In this case, we say that a measure μ is *invariant measure*, or a *stationary distribution*, if it satisfies the relation

$$\mu = \mu P_\mu, \tag{49}$$

or equivalently,

$$\mu(A) = \int_{\mathcal{X}} P_\mu(x, A) d\mu(x), \tag{50}$$

for all $A \in 2^{\mathcal{X}}$, where $P_\mu(\cdot, \cdot)$ denotes the probability kernel of the nonlinear Markov process. Then, defining the operator $\Phi : \mathcal{P}(\mathcal{X}) \to \mathcal{P}(\mathcal{X})$ by $\Phi(\mu) = \mu P_\mu$, note that finding a stationary distribution of the Markov process is equivalent to finding fixed points of the operator. The ergodicity of a nonlinear Markov process then reduces to establishing continuity/ Lipschitz properties of Φ and then applying fixed point results such as the Banach fixed point theorem. More broadly applicable results such as the Schauder fixed point theorem can also be applied, although this only establishes existence of a stationary distribution.

As opposed to the ergodic properties of Markov processes, which are well understood, the ergodic properties of nonlinear Markov chains are strange. [24] provides examples of irreducible nonlinear Markov processes related to the random walk that have infinitely many invariant measures. As such, it should be expected that stronger assumptions about the properties of the process are required to obtain ergodicity. In the classical theory of Markov chains, often times the key properties of irreducibility, aperiodicity, and recurrence, are, along with ergodic theorems, enough to establish existence and convergence to a stationary distribution. The theory of Markov processes are similarly well-understood. For nonlinear Markov processes, we require stronger assumptions on the form of the process, or on convergence rates. As mentioned above, convergence and existence often follow from a fixed-point argument.

In [7], Butkovsky provides sufficient conditions for existence and uniqueness of a stationary distribution, which in fact guarantee *uniform ergodicity*: convergence to the invariant measure at an exponential rate, in the total variation distance. However, the conditions required for these results are restrictive. We require that

- There exists α such that for all $x, y \in \mathcal{X}$ we have that

$$\sup_{\mu,\nu\in\mathcal{P}(\mathcal{X})} d_{TV}(P_\mu(x,\cdot), P_\nu(y,\cdot)) \leq 2(1-\alpha).$$

- There exists $\lambda \in [0, \alpha]$ such that for all $x \in \mathcal{X}$ and $\mu, \nu \in \mathcal{P}(E)$, we have

$$d_{TV}(P_\mu(x,\cdot), P_\nu(x,\cdot)) \leq \lambda d_{TV}(\mu, \nu).$$

The first item is the analog of a "global Dobrushin condition" for nonlinear Markov chains - essentially, the transition kernels at two different states, with varying measure parameters, should not be mutually singular. The second condition asks for a fixed x, that the kernel at x should have Lipschitz continuous dependence (in the total variation distance) on the input measure parameter, although the constant needed in the inequality is bounded by the constant α in the first condition. We found that the restriction on λ as well as the existence of mutually singular transitions made it difficult to apply these methods to our problem.

One can also obtain existence and uniqueness of stationary distributions by assuming a certain form for the dependence of the probability kernel on the law of the process. For example, the kernel of a nonlinear Markov chain may depend on the measure via its first or second moment or the empirical distribution of the process. In [6], Budhiraja et al. treat particle dynamics modeled by the nonlinear Markov process with $\mathcal{X} = \mathbb{R}^d$,

$$X^i(t+1) = AX^i(t) + \delta f(X^i(t), \mu_t^N, \xi_{t+1}^i), \tag{51}$$

where X_t^i is the state of the ith particle at time t and μ_t^N is the *empirical measure* $\frac{1}{N}\sum_{i=1}^N \delta_{X_n^i}$. Here $A \in \mathbb{R}^{d\times d}, \delta \in \mathbb{R}$ are constants. Using Lipschitz assumptions on f and the form of the update, the authors prove uniform ergodicity of (51) in the 1-Wasserstein distance. As with [7], the authors establish that the operator $\Phi(\mu) = \mu P_\mu$ is a contraction in the 1-Wasserstein distance, and appeal to the Banach fixed point theorem. We note that this is markedly different from the form of the nonlinear Markov chain we establish in Section 5.2, in which the evolution depends on the conditional law of the process.

Other methods to establish ergodicity

In addition to the more hands-on approaching of establishing continuity properties of Φ and using fixed point theorems, ergodicity results can be produced from more general, and more technical, machinery. Analyzing ergodicity in general Markov chains in infinite dimension turns out to be difficult. This is due to degeneracy: the fact that the transition kernels of Markov processes in infinite dimension are often mutually

singular. We can even observe this in the associated Markov process defined in [15] above. However, several papers [14], [29], [31] have developed machinery to overcome this issue and derive variants of ergodic properties for Markov processes in infinite dimension.

In [14], Hairer, Mattingly, and Scheutzow weaken the standard coupling argument to a notion called *asymptotic convergence*, which is used to show uniqueness of ergodic measures, as well as existence and convergence, modulo some further assumptions. As stated in [29], instead of requiring an exact coupling in which two processes X_n, X'_n must eventually coincide, we can get away with an asymptotic coupling of the two processes in which we only need they they converge to each other. Developing on earlier work by Hairer [13], Hairer et al. [14] also give another ergodicity result that extends a classical result of Harris, which gives exponential convergence of a Markov chain to a unique invariant measure, under sufficient conditions. In analyzing McKean-Vlasov Processes in [7], Butkovsky uses these results from [14].

In a different vein, Tong and van Handel [29] levy inherently measure theoretic methods to study the ergodicity (and also, conditional ergodicity) in infinite dimensional Markov chains. The theory developed by Tong and van Handel is predicated on so called local versions of *zero-two* laws that appear frequently in the theory of Harris chains. Combined with the notion of asymptotic coupling developed in [14], or the notion of *locally mixing* developed by Föllmer, the local zero two laws lead to results on the ergodicity of Markov processes as well as non-Markov processes satisfying extra conditions.

5 Problem Formulation and Theoretical Results

Throughout this section, we fix some interacting particle system $\{X(t)\}_{t\in\mathbb{N}}$ on an infinite regular tree T_κ, with state space $\mathcal{X}$. That is, $\{X(t)\}$ is a $\mathcal{X}^{T_\kappa}$ valued stochastic process. Furthermore, denote by $\{\bar{X}(t)\}$ the local recursion of the interacting particle system, which is a $\mathcal{X}^{\mathcal{T}}$-valued process where $\mathcal{T}$ is the depth 1 subtree of T_κ. To recap the outline provided in Section 4, our analysis is as follows:

1. We interpret the local recursion as a chain of infinite order.
2. We simplify the analysis by fixing the input measure in the conditional law, and analyze the frozen local recursion. In particular, we investigate its stationary distributions and Markov approximations.
3. We reformulate the local recursion as a nonlinear Markov process, and analyze its ergodic properties.
4. We also introduce the k-approximation, another nonlinear Markov chain related to the local recursion., and analyze its ergodic properties.

The constructions and results in this section are novel, although we are inspired by the papers [15, 10, 32]. Indeed, some of our results build directly on theorems established there.

5.1 Infinite Order Chain Formulation

As mentioned in Section 4, we can view the local recursion as a chain of infinite order with state space $\mathcal{X}^{\mathcal{T}}$. Indeed, recall the local recursion as defined in Def. 3.1, a $\mathcal{X}^{\mathcal{T}}$-valued process with transitions at time t defined by

- $\bar{P}_0(\bar{X}_{\mathcal{T}}[t]) = P_0(\bar{X}_{\mathcal{T}}(t))$
- For each $v \in \mathcal{T} \setminus \{0\}$,

$$\bar{P}_v^j(\bar{X}_v[t]) = \mathbb{E}^{Y[t]\sim\mu}\left[P_0^j(Y_{\mathcal{T}}(t)) \mid Y_0[t] = \bar{X}_v[t], Y_1[t] = \bar{X}_0[t]\right],$$

where $\mu = \mathrm{Law}(X[t])$. We can specify the local recursion as a chain of infinite order by introducing the quantities

$$P_v^\mu(i \mid \bar{X}[t]) := \mathbb{P}^\mu(X_v(t+1) = i \mid \bar{X}[t]), \tag{52}$$

for $i \in \mathcal{X}$, which is given by

$$\begin{cases} P_0^i(\bar{X}(t)) & \text{if } v = 0 \\ \mathbb{E}^{Y[t]\sim\mu}\left[P_0^i(Y_{\mathcal{T}}(t)) \mid Y_0[t] = \bar{X}_v[t], Y_1[t] = \bar{X}_0[t]\right] & \text{if } v \neq 0. \end{cases} \tag{53}$$

We can then define the transition probabilities of the chain of infinite order for $\mathbf{x} \in \mathcal{X}^T$ as

$$P^\mu(\mathbf{x} \mid \bar{X}[t]) := \prod_{v=0}^{d} P_v^\mu(\mathbf{x}_v \mid \bar{X}[t]), \tag{54}$$

by independence of the evolution of each coordinate. Note that these transition probabilities depend on $\mu = \mathrm{Law}(X[t])$ and so the chain of infinite order is nonhomogeneous.

5.1.1 Frozen Local Recursion Analysis

As a first step to the analysis of ergodicity of the local recursion, we try to apply results on ergodicity of infinite order chains from [15]. The issue is that the infinite order chain formulation of the local recursion proper is time-inhomogeneous via its dependence on its own law at each time step. We then consider the *frozen local recursion* obtained by fixing the measure μ as given. Throughout this section, we focus on the case of the contact process on $\mathbb{Z}$, but note that the results can be generalized to other

processes. Recalling the contact process formulation given in Example 3.2, we freeze μ as a measure on $(\mathcal{X}^{-1,0,1})^\infty$ to obtain the following dynamics:

$$\bar{X}_0^\mu(k+1) = \begin{cases} \mathbf{1}(\xi_0(k+1) < \frac{p}{2}(\bar{X}_{-1}(k) + \bar{X}_1(k))), & \text{if } \bar{X}_0(k) = 0 \\ \mathbf{1}(\xi_0(k+1) < 1-q), & \text{if } \bar{X}_0(k) = 1 \end{cases}$$

$$\bar{X}_1^\mu(k+1) = \begin{cases} \mathbf{1}(\xi_1(k+1) < \frac{p}{2}(\bar{X}_0(k) + \bar{Y}_2(k))), & \text{if } \bar{X}_1(k) = 0 \\ \mathbf{1}(\xi_1(k+1) < 1-q), & \text{if } \bar{X}_1(k) = 1 \end{cases}$$

$$\bar{X_{-1}}^\mu(k+1) = \begin{cases} \mathbf{1}(\xi_{-1}(k+1) < \frac{p}{2}(\bar{X}_0(k) + \bar{Y}_{-2}(k))), & \text{if } \bar{X}_{-1}(k) = 0 \\ \mathbf{1}(\xi_{-1}(k+1) < 1-q), & \text{if } \bar{X}_{-1}(k) = 1 \end{cases}$$

where

$$\bar{Y}_2(k) = \mathbb{E}^{X\sim\mu}[X_1(k)|X_0[k] = \bar{X}_1^\mu[k], X_{-1}[k] = \bar{X}_0^\mu[k]]$$
$$\bar{Y}_{-2}(k) = \mathbb{E}^{X\sim\mu}[X_{-1}(k)|X_0[k] = \bar{X}_{-1}^\mu[k], X_1[k] = \bar{X}_0^\mu[k]].$$

To find conditions on this chain for which we have stationarity, we apply results from [15] as well as the bounds on the Markov approximation provided by [10],[5]. To do this it suffices to find conditions for which the chain has summable continuity rates, which we denote $\beta_k^{\bar{X}^\mu}$, and to check the irreducibility type (non-nullness) conditions, Equations (42), (43). This is encapsulated by the following proposition.

Proposition 5.1. *Let $\{\bar{X}^\mu(n)\}_n$ denote the local recursion for the contact process with frozen law μ. If $p \neq 1, q \neq 0, 1$ and μ is such that $\{\bar{X}^\mu\}_n$ has summable continuity rates, then $\bar{X}$ above has a stationary distribution (more specifically, there exists a stationary process with the same transition functions as those of $\bar{X}^\mu$) and the Orstein distance between $\bar{X}^\mu$ and its canonical order k-Markov approximation goes to zero as $k \to \infty$.*

A full proof would check equations (42), (43) and apply the theorems of Section 4.1. It then suffices to find suitable conditions on μ for which the frozen local recursion also has summable continuity rates. We conjecture that it is enough to have μ with summable continuity rates.

Conjecture 5.1. *If μ is a measure with summable continuity rates, then $\{\bar{X}(n)^\mu\}_n$ is a chain of infinite order with summable continuity rates.*

From here, we can ask a few interesting questions, assuming the hypothesis of Prop. 5.1.

1. How does the stationary distribution given in Prop. 5.1 change when the measure μ is perturbed slightly?

2. Are the k-Markov approximations ergodic? If so, do these stationary distributions converge to those of the local recursion?

5.2 Nonlinear Markov Chain Reformulation

Inspired by the papers [15], [10], we can recast the local recursion as a stochastic process on $[0,1]^{\kappa+1}$ via the identification of $\mathcal{X}^\infty$ with $[0,1]$. This recasts the local recursion as a nonlinear Markov chain introduced in Section 4.2. Nonlinear Markov chains are a class of processes whose ergodic properties have been studied, and so techniques from that literature can be brought to bear on this problem. The reformulation works by mapping histories of the process $\bar{X}_v[k] = (x_k, x_{k-1}, \ldots, x_0)$ to values in $[0,1]$ via the base-$|\mathcal{X}|$ expansion $x_v = 0.\phi(x_k)\phi(x_{k-1})\ldots\phi(x_0)$, where ϕ is some bijection mapping $\mathcal{X} \to \{0, \ldots, |\mathcal{X}|-1\}$.

To first define the nonlinear Markov chain formulation of the local recursion, we introduce some notation:

- Let $\mathcal{D}^\kappa_{|\mathcal{X}|,k}$ denote the set of $|\mathcal{X}|$-adic cubes of side length 2^{-k} in $[0,1]^\kappa$, that is, the set of cubes of the form
$$\prod_{i=1}^{\kappa}[a_i|\mathcal{X}|^{-k}, (a_i+1)|\mathcal{X}|^{-k}), \tag{55}$$
for $a = (a_1, \ldots, a_{\kappa+1}) \in \{0, \ldots, |\mathcal{X}|^k - 1\}^{\kappa+1}$. For $|\mathcal{X}| = 2$, we term these dyadic cubes. Then $\bigcup_k \mathcal{D}^\kappa_{|\mathcal{X}|,k}$ denotes the set of $|\mathcal{X}|$-adic cubes of any side length.
- Let $\sigma(\mathcal{D}^\kappa_{|\mathcal{X}|,k})$ be the sigma algebra generated by these $|\mathcal{X}|$-adic cubes, and $\sigma(\bigcup_k \mathcal{D}^\kappa_{|\mathcal{X}|,k})$ analogously.
- Denote by $\mathcal{M}^{\kappa+1}_{|\mathcal{X}|,k}$ the set of measures on the measurable space
$$([0,1]^{\kappa+1}, \sigma(\mathcal{D}^{\kappa+1}_{|\mathcal{X}|,k})), \tag{56}$$
topologized by total variation distance. That is, these measures are constant on cubes of the form (55).
- Denote by $\phi(\mathcal{X})$ the set $\{0, \ldots, |\mathcal{X}|-1\}$.

We then define the associated nonlinear Markov process as the stochastic process $\tilde{X}^t$ in the measurable space $([0,1]^{\kappa+1}, \sigma(\bigcup_k \mathcal{D}^{\kappa+1}_{|\mathcal{X}|,k}))$. Intuitively, each coordinate $\tilde{X}^t_v$ keeps track of the history of local recursion at node v via its base $|\mathcal{X}|$ encoding.

Definition 5.1 (Nonlinear Markov Chain interpretation of local recursion)**.** *The associated nonlinear Markov process is the stochastic process $\tilde{X}^t$ in the measurable space $([0,1]^{\kappa+1}, \sigma(\bigcup_k \mathcal{D}^{\kappa+1}_{|\mathcal{X}|,k}))$, whose transition kernel $\mathcal{P}^\mu(x,\cdot)$ is defined via*

$$\mathcal{P}^\mu(\mathbf{x},\cdot) = \mathrm{Law}\left(\frac{1}{|\mathcal{X}|}(\mathbf{x} + v(\mathbf{x},\mu))\right) \tag{57}$$

for $\mathbf{x} \in [0,1]^{\kappa+1}$, *and where* $v(\mathbf{x},\mu)$ *is the* $\phi(\mathcal{X})^{\kappa+1}$ *random variable with distribution*

$$\bigotimes_{v=0}^{\kappa} \text{Categorical}(F_v^1(\mathbf{x},\mu),\dots,F_v^{|\mathcal{X}|}(\mathbf{x},\mu)) \tag{58}$$

and

$$\begin{cases} F_0^j(\mathbf{x},\mu) = P_0^{\lfloor |\mathcal{X}|\mathbf{x}_0 \rfloor, j} \\ F_v^j(\mathbf{x},\mu) = \mathbb{E}^{\mathbf{Y}\sim\mu}\left[P_v^{\lfloor |\mathcal{X}|\mathbf{x}_v \rfloor, j} \mid \mathbf{Y}_0 = \mathbf{x}_v, \mathbf{Y}_1 = \mathbf{x}_0\right] \quad \textit{if } v \neq 0. \end{cases} \tag{59}$$

We can think of $F_v^j(\mathbf{x},\mu)$ as the probability of transition of node v to state j given the value of $\tilde{X}[k]$ and its law. As an alternative to Definition 5.1, we can also characterize $\tilde{X}$ via its law at each time step. If $\mu_t := \text{Law}\,\tilde{X}(t)$, the sequence $\{\mu_t\}_t$ is defined recursively via the update

$$\mu_{t+1} = \Phi(\mu_t) := \mu_t \mathcal{P}^{\mu_t}, \tag{60}$$

where $\mu\mathcal{P}^{\mu}$ is the pushforward of μ by the transition kernel $\mathcal{P}^{\mu}$ (also dependent on μ). More explicitly, this can be written as

$$\Phi(\mu)(A) = \int_A \mathcal{P}^{\mu}(\mathbf{x},A)d\mu(x). \tag{61}$$

We refer to Φ is the *nonlinear Markov operator*.

Example 5.1. Continuing our example from 3.2, we present the nonlinear Markov reformulation of the Contact Process on $\mathbb{Z}$. Regarding $\mathbb{Z}$ as a 2-regular infinite tree, we consider the tree $\mathcal{T}$ as the subset of three consecutive vertices, where the root is indexed by 0 and the neighbors indexed by $1,2$. Then for $\mathbf{x} = (\mathbf{x}_0,\mathbf{x}_1,\mathbf{x}_2) \in [0,1]^3$ we can write the transition kernel, verbatim from (57), of the nonlinear Markov process as

$$\mathbb{P}^{\mu}(\mathbf{x},\cdot) = \text{Law}(\frac{1}{2}\mathbf{x} + \frac{1}{2}\mathbf{v}(\mathbf{x},\mu)), \tag{62}$$

where

$$\mathbf{v}(\mathbf{x},\mu) \sim \text{Bernoulli}(F_0(\mu,\mathbf{x})) \otimes \text{Bernoulli}(F_1(\mu,\mathbf{x})) \otimes \text{Bernoulli}(F_2(\mu,\mathbf{x})) \tag{63}$$

is a product measure. In this case we can use a Bernoulli distribution instead of a full Categorical one, since the state space of the contact process is $\mathcal{X} = \{0,1\}$. The functionals $F_v(\mu,\mathbf{x}), v \in V$ are defined as follows:

$$F_0^1(\mu,\mathbf{x}) = \frac{p}{2}(\lfloor 2\mathbf{x}_2 \rfloor + \lfloor 2\mathbf{x}_1 \rfloor)\mathbf{1}_{\{\lfloor 2\mathbf{x}_0 \rfloor = 0\}} + (1-q)\mathbf{1}_{\{\lfloor 2\mathbf{x}_0 \rfloor = 1\}} \tag{64}$$

$$F_1^1(\mu,\mathbf{x}) = \frac{p}{2}(\lfloor 2\mathbf{x}_0 \rfloor + \mathbb{E}^{\mathbf{Y}\sim\mu}[\ \lfloor 2\mathbf{Y}_1 \rfloor \mid \mathbf{Y}_0 = \mathbf{x}_1, \mathbf{Y}_2 = \mathbf{x}_0])\mathbf{1}_{\{\lfloor 2\mathbf{x}_1 \rfloor = 0\}} + (1-q)\mathbf{1}_{\{\lfloor 2\mathbf{x}_1 \rfloor = 1\}} \tag{65}$$

$$F_2^1(\mu,\mathbf{x}) = \frac{p}{2}(\lfloor 2\mathbf{x}_0 \rfloor + \mathbb{E}^{\mathbf{Y}\sim\mu}[\ \lfloor 2\mathbf{Y}_2 \rfloor \mid \mathbf{Y}_0 = \mathbf{x}_2, \mathbf{Y}_1 = \mathbf{x}_0])\mathbf{1}_{\{\lfloor 2\mathbf{x}_2 \rfloor = 0\}} + (1-q)\mathbf{1}_{\{\lfloor 2\mathbf{x}_2 \rfloor = 1\}} \tag{66}$$

The quantity F_0^1 is the exactly probability of transition of the root to the state 1 at the next time step. Since this probability depends on the state of the root, we have indicator functions corresponding to each case. To map the state space of the nonlinear Markov chain $[0,1]^3$ into the state space $\mathcal{X}$ of the dynamics, we apply the function $\lfloor 2\mathbf{x}_v \rfloor$ for every coordinate of $\mathbf{x}$. The other quantities $F_1^1(\mu, x), F_2^1(\mu, x)$ are the conditional expectations of the transition probabilities to state 1 for nodes $1, 2$, with appropriate terms taken out of the conditioning.

For the recasting of the local recursions of several interacting particle systems into nonlinear Markov processes, refer to the appendix.

Implementation

Algorithm 3 gives pseudocode for an implementation of updating a measure via the nonlinear Markov operator. Given a measure $\mu \in \mathcal{M}^{\kappa+1}_{|\mathcal{X}|,k}$ it outputs $\Phi(\mu) \in \mathcal{M}^{\kappa+1}_{|\mathcal{X}|,k+1}$. Simulating the local recursion up to time T then amounts to a T-fold application of algorithm 3, which then returns the full joint measure of $\bar{X}$ for times up to T. One can then extract the desired information from the full measure via marginalizing.

To represent a measure $\mu \in \mathcal{M}^{\kappa+1}_{|\mathcal{X}|,k}$, we use a Numpy array of side length $|\mathcal{X}|^k$, with dimension $\kappa + 1$. An index in $\{0, \ldots, |\mathcal{X}|^k - 1\}$ then represents the history of a node via its $|\mathcal{X}|$-ary encoding. Then $\Phi(\mu)$ can be implemented as an array of side length $|\mathcal{X}|^{k+1}$ of the same dimension.

Algorithm 3 Apply Nonlinear Markov Operator Φ

procedure NonlinearUpdate(μ) ▷ Here $\mu \in \mathcal{M}^{\kappa+1}_{|\mathcal{X}|,k}$.
  Precompute $\forall y_0, y_1$:

$$\mathbb{E}^{Y\sim\mu}\left[P_v^j(Y) \mid y_0, y_1\right]$$

  for Dyadic cubes $y \in \mathcal{D}^{\kappa+1}_{|\mathcal{X},k|}$ **do**
    for $v = 0, \ldots, d$ **do**
      Compute $F_v^j(x, \mu)$.
    Compute

$$v(x,\mu) \leftarrow \otimes_{v=0}^{d} \text{Categorical}(F_v^1(x,\mu), \ldots, F_v^q(x,\mu))$$

    Compute

$$\mathcal{P}^\mu(x, \cdot) \leftarrow \text{Law}(\frac{1}{|\mathcal{X}|}(x + v(x,\mu)))$$

    $\Phi(\mu)(x) \leftarrow \int_A \mathcal{P}^\mu(y,x) d\mu(y)$ ▷ Compute weighted sum
  return $\Phi(\mu)$.

We analyze the complexity of the main steps of the algorithm.

- The precomputation in step 2 of the algorithm calculates $|\mathcal{X}|^{2k}$ conditional expectations, each of which requires summing over an array of size $|\mathcal{X}|^{k(\kappa-1)}$.
- Computing $v(x,\mu)$ and $P^{\mu}(x,\cdot)$ in steps 6, 7 requires a cheap lookup of the conditional expectations, but we need to store $P^{\mu}(x,\cdot)$ as a random variable with state space of size $|\mathcal{X}|^{\kappa+1}$. We do this for every $x \in \mathcal{X}^{(\kappa+1)k}$ (the factor of k accounts for histories), and so the space complexity is $|\mathcal{X}|^{(k+1)(\kappa+1)}$.
- To calculate $\Phi(\mu)$ the integration is just a weighted sum over $|\mathcal{X}|^{k(\kappa+1)}$ values. This computation is done for each dyadic cube in $\mathcal{D}^{\kappa+1}_{|\mathcal{X}|,k}$, so that the complexity is $|\mathcal{X}|^{2k(\kappa+1)}$.

Hence the total time complexity of the updating a measure $\mu \in \mathcal{M}^{\kappa+1}_{|\mathcal{X}|,k}$ by the nonlinear Markov operator Φ is given by $O(|\mathcal{X}|^{2k(\kappa+1)})$. When running this for T time steps, with initial measure $\mu_0 \in \mathcal{M}^{\kappa+1}_{|\mathcal{X}|,k}$, the T-fold application of Algorithm 3 which computes $\mu_T = \Phi^{(T)}(\mu_0)$ has a time complexity of $O(|\mathcal{X}|^{2(T+k)(\kappa+1)})$, with a similar space complexity. It is apparent, given these time and space considerations, that the local recursion can only be run for small times: it is exponential in T and κ, and polynomial in the size of the state space of the particle dynamics, of high degree. While there are a few ways of parallelizing the algorithm (say at step 3), the time complexity remains the primary barrier to applying the local recursion for large times. As compared to more general methods such as MCMC, the local recursion is far slower, but has the advantage of being exact for dynamics on large regular trees. Indeed, it has been shown that the local recursion is exact for infinite regular trees, but for small times T, the local recursion on a regular tree of depth greater than T is also exact, since the nodes past depth T are independent of the root and its neighbors up to time T. See [32] for further details on this note. As an example, Figure 3 shows the local recursion with 10^6 Monte Carlo samples for the oscillating cellular automata model defined in Section 2. We plot the probabilities of observing the root and a given neighbor in the state $(0,0)$ versus time. In this simulation, we observe that the nonlinear Markov formulation of the local recursion is practically exact. Figure 4 plots the joint probabilities for the state $(0,1)$, showing fairly good agreement between the Monte Carlo sampled dynamics and the nonlinear Markov chain. We explore these tradeoffs further with simulations in Section 6 and compare the local recursion with the mean field approximation and Monte Carlo sampling via MCMC.

Theoretical Comments

As introduced in Section 3.2, a major theoretical question in this vein of research is when ergodic properties of an interacting particle system on an infinite regular tree are mirrored by those of the local recursion. While it has been proven that the local recursion is exact for any finite time interval, the question at stationarity is still

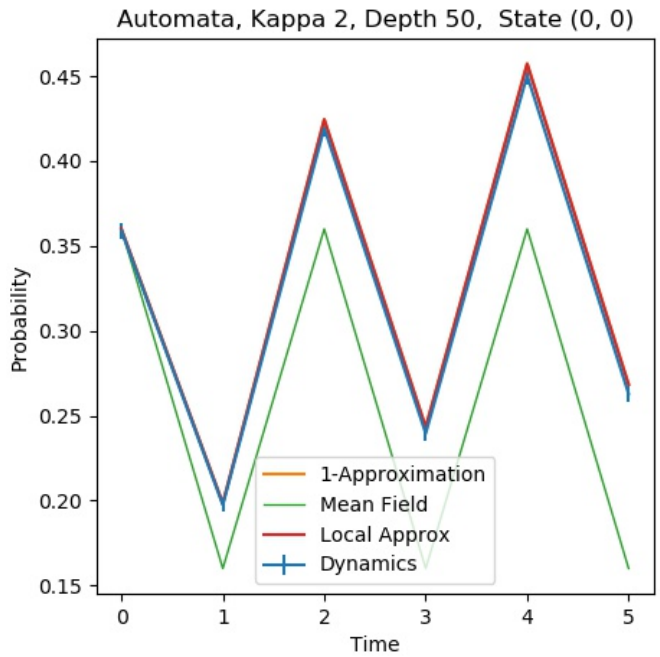


Figure 3: $p = 0.1, 10^6$ MC samples. $\mathbb{P}(X_v(0) = 1) = 0.4$.

unclear. In this setting, a stationary distribution is defined as in equation (49). In particular, we would like for the following conjecture to be true:

Conjecture 5.2.

- *Every stationary distribution $\bar{\mu}$ of $\{\bar{X}(t)\}$ is the marginal of a stationary distribution of $\{X(t)\}$, restricted to $\mathcal{T}$.*
- *If μ is a stationary distribution of the interacting particle system $\{X(t)\}$ then $\mu|_{\mathcal{T}}$ is a stationary distribution of $\{\bar{X}(t)\}$.*

Using the notation of Conjecture 5.2, another way to visualize the statement of the conjecture is that it makes the following diagram commute, which we can think of as an interchange property.

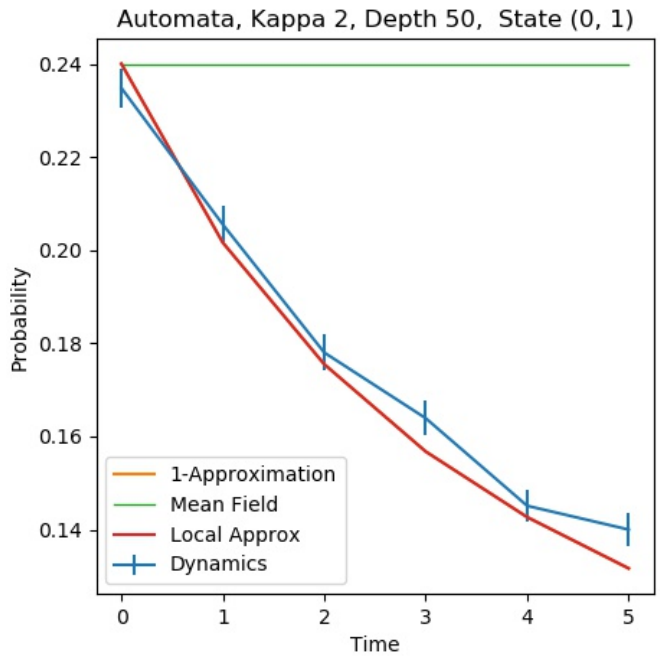


Figure 4: $p = 0.1, 10^6$ MC samples. $\mathbb{P}(X_v(0) = 1) = 0.4$.

$$\begin{tikzcd}
\{X(t)\} \arrow[r, "\lim_{t\to\infty} \bullet"] \arrow[d, "\text{Local Approx}"'] & \mu \arrow[d, "\bullet|_{\mathcal{T}}"] \\
\{\bar{X}(t)\} \arrow[r, "\lim_{t\to\infty} \bullet"] & \bar{\mu}
\end{tikzcd}$$

Figure 5: The conjectured interchange property.

To begin answering this question, we try to emulate general approaches from [7], [6]: we try to establish properties of the operator Φ and then appeal to functional analytic results, in particular fixed point theorems. Recall that in the finite history case, we can model the state space of the nonlinear Markov chain via n-ary measures on the measurable space $[0,1]^{\kappa+1}, \sigma(\bigcup_k \mathcal{D}^{\kappa+1}_{|\mathcal{X}|,k})$. We can then establish continuity of the nonlinear Markov operator in the following proposition:

Proposition 5.2. *For any fixed ϵ, let $U \subset \mathcal{M}^{\kappa+1}_{|\mathcal{X}|,k}$ be a subset of measures such that for $\mu \in U$, we have $\mu(x_{0,1}) \geq \epsilon$ for all $x_{0,1} \in \mathcal{X}^{[0,1]\times\{-k,\ldots,-1\}}$. Then nonlinear Markov*

operator of Eq. (61), viewed as

$$\Phi : U \to \mathcal{M}_{k+1}^{\kappa+1}$$

is continuous.

The proof of the proposition relies on the finiteness of the probability space $(\mathcal{D}_k^{\kappa+1}, \sigma(\mathcal{D}_k^{\kappa+1}))$, which allows us to express the conditional expectation in the local recursion as a sum. See the Proofs section for further details. We also tried exploring the continuity of the operator in the weak topology, and this problem boiled down to showing the weak convergence of the conditional expectations in Eq. (59): this should be explored further.

To develop a numerical understanding of the long time behavior of the local recursion, it is infeasible to naively evolve the nonlinear Markov chain for large times, as discussed in the complexity analysis of the algorithm. This leads to us discuss a natural, related process - the k-approximation - that can be simulated for large times. Theoretically, studying the k-approximation is simpler and offers insight into the long time behavior of the local recursion.

5.3 K-Approximation

As noted previously, the local recursion and its nonlinear Markov chain take exponential time and space to simulate. We now consider a related approximation to the local recursion, which is much easier to simulate. These dynamics, which we call the k-approximation, take inspiration from the literature on Markov approximation of infinite order chains [5, 10] as well as the related τ-approximation introduced in [32]. The idea of the k-approximation is to consider the dynamics local recursion, truncating past history of the process past the kth previous time step; intuitively we expect the k-approximation when we have a sort of continuity rate property, where the considering an extra time step in the history does not affect the process too much.

Definition 5.2 (k-Approximation)**.** *Let T_κ be a regular tree, and $\{X(t)\}_t$ an interacting particle system with state space $\mathcal{X}^{T_\kappa}$ and transition probabilities $P_v^{i,j}$. The k-approximation is the stochastic process $\{\bar{X}^{[k]}(t)\}_t$ defined on $\mathcal{T} = \{0, 1, \ldots, \kappa\}$ with state space $\mathcal{X}^{\mathcal{T}}$, whose transitions $\bar{P}_v^{[k],i,j}$ are given by*

- $\bar{P}_0^{[k]}(\bar{X}_{\mathcal{T}}^{[k]}[t]) = P_0(\bar{X}_{\mathcal{T}}^{[k]}(t))$.
- *For each $v \in \mathcal{T} \setminus \{0\}$,*

$$\bar{P}_v^{[k],\bar{X}_v^{[k]}[t],j} = \mathbb{E}^{Y[t]=^{(d)}\bar{X}^{[k]}[t]}\Big[P_0^j(Y_{\mathcal{T}}(k)) \,|Y_0[t,(t-k)\vee 0] = \bar{X}_v^{[k]}[t,(t-k)\vee 0], \\ Y_1[t,(t-k)\vee 0] = \bar{X}_0^{[k]}[t,(t-k)\vee 0]\Big]$$

Phrased in terms of the language of the previous section, we can also define the K-Approximation as a composition of Φ, the nonlinear Markov chain operator, and a "smoothing" operator S_k.

Definition 5.3. *Define the k-smoothing operator $S_k : \mathcal{M}^{\kappa+1}_{|\mathcal{X}|,k+1} \to \mathcal{M}^{\kappa+1}_{|\mathcal{X}|,k}$ to be the conditional expectation operator mapping*

$$\mu \mapsto \mathbb{E}\Big[\mu|\sigma(\mathcal{D}^{\kappa+1}_{|\mathcal{X}|,k})\Big].$$

At a more practical level, given $\mu \in \mathcal{M}^{\kappa+1}_k$ its image under the k-smoothing operator, $S_k(\mu)$, is a measure whose density is constant on cubes of the form

$$\prod_i^{\kappa+1}[a_i 2^{-k}, (a_i+1)2^{-k}).$$

To express $S_k(\mu)$ in terms of μ, first denote by $\mu((a_i)_i)$ the value of μ on the cube

$$\prod_i^{\kappa+1}[a_i 2^{-k-1}, (a_i+1)2^{-k-1}).$$

Then the constant value of $S_k(\mu)$ on a cube $\prod_i^{\kappa+1}[b_i 2^{-k}, (b_i+1)2^{-k})$ is given by

$$\frac{1}{2^{\kappa+1}} \sum_{x_i \in \{0,1\}^{\kappa+1}} \mu((2a_i + x_i)_i).$$

We can then define the k-Approximation equivalently as a nonlinear Markov chain:

Definition 5.4 (k-Approximation, nonlinear Markov chain)**.** *The K-Approximation is a nonlinear Markov chain whose transition operator is given by*

$$\Phi_k := S_k \circ \Phi.$$

Here we are using the fact that $\mathcal{D}^{\kappa+1}_{|\mathcal{X}|,n}$ naturally embeds into $\mathcal{D}^{\kappa+1}_{|\mathcal{X}|,m}$ for $n \le m$. Then to simulate the K-approximation up to time T, it suffices to consider a T-fold application of Φ_k. The advantage of the k-Approximation is that we associate the local recursion to a process evolving on a *finite* probability space, which is exactly of size $|\mathcal{X}|^{(\kappa+1)k}$, the histories of the root node and its neighbors, up to length k. Algorithm 4 provides a simple look at evolving the K-Approximation. To analyze the complexity of the algorithm, we first note that the operator S_k creates a measure in $\mathcal{M}^{\kappa+1}_{|\mathcal{X}|,k}$ via averaging the entries in the input measure $\mu \in \mathcal{M}^{\kappa+1}_{|\mathcal{X}|,l}$. Since each entry must be averaged, the time complexity of this procedure is given by $O(|\mathcal{X}|^{l(\kappa+1)})$. To analyze running the k-approximation for large times T, suppose for simplicity that the input measure μ_0 belongs to $\mathcal{M}^{\kappa+1}_{|\mathcal{X}|,k}$. Then since each application of Φ_k returns

another measure in the same space $\mathcal{M}^{\kappa+1}_{|\mathcal{X}|,k}$, it follows that the time complexity is $O(T|\mathcal{X}|^{(k+1)(\kappa+1)} + T|\mathcal{X}|^{2k(\kappa+1)}) = O(T|\mathcal{X}|^{2k(\kappa+1)})$ which is markedly *linear* in T as opposed to exponential with NonlinearUpdate. The first term is contributed by the application of the S_k, and the second terms comes from applying Φ. This allows us to numerically simulate long time behavior of the k-approximation; the hope is now that the ergodic properties of the k-approximation give information about those of the local recursion. We explore this numerically in later sections.

Algorithm 4 Apply K-Approximation Operator Φ_k

```
procedure KApproxUpdate(μ, k)                          ▷ Assume μ ∈ M^{κ+1}_{|X|,l}
    ν ← NonlinearUpdate(μ)
    if l ≥ k then
        Φ_k(μ) = S_k(ν)                                ▷ Φ_k(μ) ∈ M^{κ+1}_{|X|,k}
    else
        Φ_k(μ) = ν                                     ▷ Φ_k(μ) ∈ M^{κ+1}_{|X|,l+1}
    return Φ_k(μ).
```

Theoretical Comments

Theoretically, the fact that our probability space is finite is a boon to analysis; on a finite probability space, the topology induced by the total variation distance is actually equivalent to the weak topology and Wasserstein distance, simplifying analysis considerably. This allows us to prove that the set of probability measures on this measure space is compact under total variation distance. We record a few properties that derive from the finiteness of our space. See the proof section for details.

Lemma 5.1. *For any k, κ, the space $\mathcal{M}^{\kappa+1}_{|\mathcal{X}|,k}$ is convex and compact.*

Lemma 5.2. *The operator*

$$S_k : \mathcal{M}^{\kappa+1}_{|\mathcal{X}|,m} \to \mathcal{M}^{\kappa+1}_{|\mathcal{X}|,k} \tag{67}$$

is continuous, for all $m \geq k$.

Remark 5.1. *We require the restriction of the domain in this lemma, since the proof requires the conditional expectation in the local approximation to be well defined.*

Using these two lemmas and the Schauder fixed point theorem (see the appendix), we can establish existence of stationary distributions for $\bar{X}^{[k]}$ for each k.

Proposition 5.3 (Existence of invariant measures)**.** *Assume there exists a closed, convex subset $D \subset U$ with U as in Lemma 5.2 such that*

$$\Phi_k : D \to D.$$

Then there exists an invariant measure μ_k of the k-Approximation, i.e.

$$\Phi_k(\mu_k) = \mu_k.$$

Remark 5.2. *Further work is required to show existence of such a set D, although we conjecture that there does exist one.*

Ergodicity of k-approximations

In addition to existence of invariant measures, we are also interested in *uniqueness* of the k-Approximation, convergence of $\bar{X}^{[k]}$ to stationarity, and the rate of convergence. Numerically, we have observed that there is nonuniqueness of invariant distributions for some interacting particle systems. For example, for high values of the infection rate, the continuous-time contact process on the infinite regular tree has stationary measures other than the trivial point mass at 0. We can verify this numerically. For the contact process on T_3 with infection rate 0.9, recovery rate 0.1, we observe that the 1-approximation has multiple stationary measures. Two such stationary distributions are given in Table 2.

(a) Trivial invariant measure

State	Probability
(0,0,0,0)	1
(0,1,0,0)	0
(0,1,1,0)	0
(0,1,1,1)	0
(1,0,0,0)	0
(1,1,0,0)	0
(1,1,1,0)	0
(1,1,1,1)	0

(b) Non-trivial invariant measure

State	Probability
(0,0,0,0)	0.0003
(0,1,0,0)	0.0018
(0,1,1,0)	0.0110
(0,1,1,1)	0.0747
(1,0,0,0)	0.0012
(1,1,0,0)	0.0098
(1,1,1,0)	0.0778
(1,1,1,1)	0.6228

Table 2: Two invariant measures of the 1-approximation to the contact process, $p = 0.9, q = 0.1$, on the 3-regular tree T_3. The left hand side of each column represents the state of the nodes in $\mathcal{T} = (0, 1, 2, 3)$ where 0 is the root node of T_3 and $1, 2, 3$ are its children. By symmetry considerations we list only some of the states. Convergence was decided via a threshold of 10^{-8}.

This agrees with the theory established in [27], where for large enough values of the infection rate compared to the recovery rate, we have existence of nontrivial invariant distributions. See [27] for further theory on the ergodic behavior of the process under different regimes of the infection rate, although note that the model in [27] is

continuous-time and sets the recovery rate to 1, so roughly speaking we can use the ratio of the infection and recovery parameters p/q as a proxy in our discrete-time model.

As another example, it is known that the Ising model on a regular tree T_κ, for $\kappa \geq 3$. satisfies a phase transition as β, the inverse temperature varies. In the case of a zero external magnetic field, there exists a β_c^κ such that for $\beta > \beta_c^\kappa$ the Ising model has multiple stationary distributions, whereas for $\beta < \beta_c^\kappa$, the Ising model has a unique equilibrium measure. The quantity β_c^κ can be calculated explicitly as

$$\beta_c = \frac{1}{2} \ln \frac{\kappa}{\kappa - 2}.$$

For more details on the derivation of this critical value, see the notes [28] , which use the idea of *correlation decay*. The critical value is also verified in [23]. Numerically, we calculate the stationary distribution of the 1−approximations of the parallel Ising model with different values of the initial measure, and we recover nonuniqueness of the stationary distributions. See Table 3 for an example.

(a) Invariant measure, $\theta = 0.1$.

State	Probability
(0,0,0,0)	0.9839
(0,1,0,0)	0.0039
(0,1,1,0)	0.0001
(0,1,1,1)	0.0000
(1,0,0,0)	0.0040
(1,1,0,0)	0.0000
(1,1,1,0)	0.0000
(1,1,1,1)	0.0000

(b) Invariant measure, $\theta = 0.5$

State	Probability
(0,0,0,0)	0.1712
(0,1,0,0)	0.0262
(0,1,1,0)	0.0262
(0,1,1,1)	0.1712
(1,0,0,0)	0.0262
(1,1,0,0)	0.0262
(1,1,1,0)	0.0262
(1,1,1,1)	0.1712

Table 3: Two invariant measures of the 1-approximation to parallel Glauber dynamics for $\beta = 1, p = 1$ on the 3-regular tree T_3. The left hand side of each column represents the state of the nodes in $\mathcal{T} = (0, 1, 2, 3)$ where 0 is the root node of T_3 and $1, 2, 3$ are its children. By symmetry considerations we list only some of the states. Convergence was decided via a threshold of 10^{-8}. The initial distribution of each node was set to Bernoulli(θ) for the values of θ specified.

We explore these questions further in the Numerical Results section, but the evidence seems to suggest that the ergodic properties of the dynamics is mirrored by those of the k-approximation for each k.

Convergence of k-approximations

Then next natural question is whether the k-approximations $\bar{X}^{[k]}$ converge to $\bar{X}$ in some sense, and the rate of convergence. For a finite time $t \in \mathbb{Z}, t < \infty$, observe that

$$\bar{X}^{[k]}(t) \xrightarrow{k\to\infty} \bar{X}(t),$$

since for fixed k, $\bar{X}^{[k]}(t)$ agrees with $\bar{X}(t)$ for $t \in \{0, \ldots, k\}$. We can ask an analogous question at stationarity. If the k-approximations $\bar{X}^{[k]}$ have stationary distributions $\{\mu_k\}_k$, and if $\{\mu_k\}_k$, has a limit in an appropriate sense, is this a fixed point of the full nonlinear Markov chain, $\bar{X}$? Even further, is it a stationary distribution of the dynamics X on the infinite tree? Assuming uniqueness of these stationary distributions μ_k, an immediate corollary of our setup is that $\{\mu_k\}$ is precompact in the weak topology, so we have some candidates for the purported limit. This follows simply from the fact that the set of probability measures on $[0,1]^{\kappa+1}$ is tight, because the underlying space is compact. Standard weak convergence results then show that tightness is equivalent to weak precompactness; see for example Chapter 1, Section 2 of [4].

The analysis of this question touches on the perturbation analysis of nonlinear Markov chains. Indeed, we can abstract this problem as analyzing how the fixed points of an operator are perturbed upon changing the operator slightly. It should also be explored whether the Ornstein $\bar{d}$-distance for Markov approximations defined in [10] can be applied to this situation.

Our numerical results show that in many situations, convergence of the μ_k to the stationary distribution of the dynamics holds, although the rate of convergence as a function of k seems to vary. For example, using the k-approximation to analyze equilibrium behavior of the contact process on the line, we see that the k-approximations $\bar{X}^{[k]}$ converge quickly to the dynamics as a function of k. Figure 6 shows the probability of the root node and one of its neighbors at state $(1,1)$. The experiment was done for infection rate parameter $p = \frac{2}{3}$, and recovery rate parameter $q = \frac{1}{3}$, on a line of depth 200, with time up to $T = 200$. 10^5 Monte Carlo samples were taken to approximate the probabilities of the dynamics, and errorbars are shown. In this case, by choice of the parameters p, q, the mean field approximation gives a constant value to the joint probability. The probabilities of the joint states $(1,0), (0,1), (0,0)$ show similar convergence of the k-approximation to the dynamics. Figure 7 shows another example of the contact process on the same graph, where now the infection rate is 0.75 and recovery rate is 0.25. Convergence of the k-Approximations seems to be slightly slower than in the previous example.

In other instances, convergence can be much faster. Figure 8 gives an example where the k-approximations are nearly identical for $k \geq 1$, and in fairly good agreement with the dynamics. This experiment was done for the contrarian voter dynamics

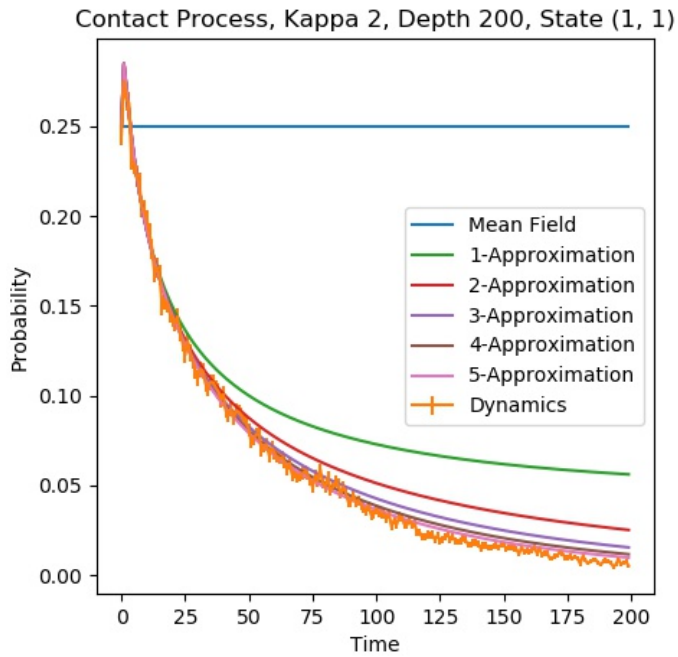


Figure 6: $p = 2/3, q = 1/3$, 10^5 MC samples. $\mathbb{P}(X_v(0) = 1) = 0.4$.

on the line graph of depth 100. The parameter p there was taken to be $p = 0.01$. To explain why the mean field gives constant results, it turns out that $\mathbb{P}(X_v^{MF}(t) = 1)$ oscillates between two values p_0, p_1 as a function of time, for any node v. Since the mean field approximation makes the dynamics of each particle independent, the joint probability of the root and its neighbor being at state $(1, 0)$ oscillates between $p_0 p_1$ and $p_1 p_0$. Hence the approximation is just constant. For other states $(0, 0), (0, 1), (1, 1)$ the agreement between the k-approximations as a function of k is fairly tight. See the numerical results section for further details and results on these experiments.

Given this evidence on uniqueness and convergence of the k-approximations, we make the following conjecture:

Conjecture 5.3. *Let $\{X_v(t)\}$ be some interacting particle system on a regular tree T_κ with initial distribution $X_v(0) \sim \nu$. Suppose that $\bar{X}_v(t)$ the local recursion with the same initial distribution converges to a stationary distribution μ. Then the stationary distributions μ_k of $\bar{X}^{[k]}$, each with initial distribution ν, satisfy*

$$\mu_k \Rightarrow \mu.$$

A converse question we can pose is:

Question 5.1. *What properties do the stationary distributions μ_k have when the*

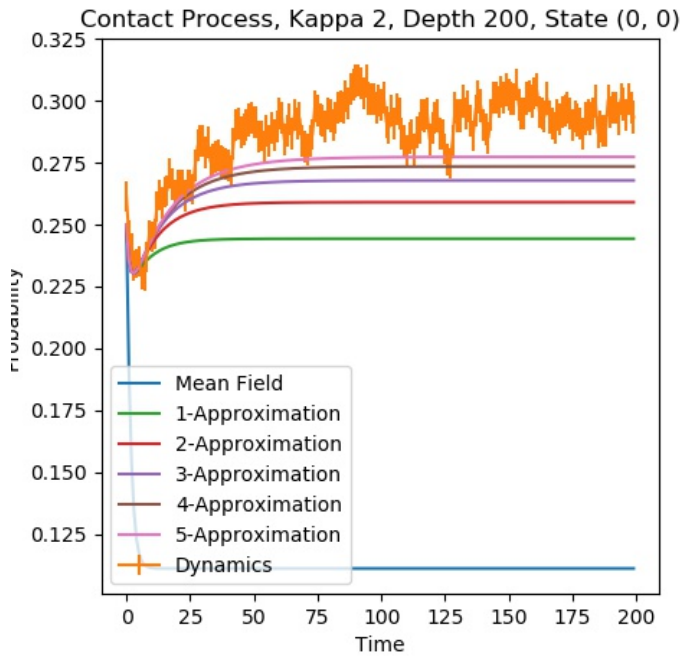


Figure 7: $p = 0.75, q = 0.25$, 10^4 MC samples. $\mathbb{P}(X_v(0) = 1) = 0.5$.

local recursion $\bar{X}_v(t)$ does not converge to an invariant distribution, or even when the interacting particle system does not converge to an invariant distribution?

6 Numerical Results

In this section we numerically explore the behavior of the local recursion and the k-approximations. One of the main questions driving this section is the comparison between the local recursion and k-approximation:

Question 6.1. *For which graphs and dynamics do we expect the local recursion to outperform the mean field approximation, and vice versa?*

To compare the efficacy of the local recursion versus the mean field simulation, we fix a graph G and an IPS $\{X(t)\}$ on G. We then assess the accuracy of both approximations in comparison to Monte Carlo-sampled estimates of the dynamics, with the following functionals:

- The probabilities of the root node
- The joint probabilities of the root node and a given neighbor
- The expected time spent at a given state for the root node.

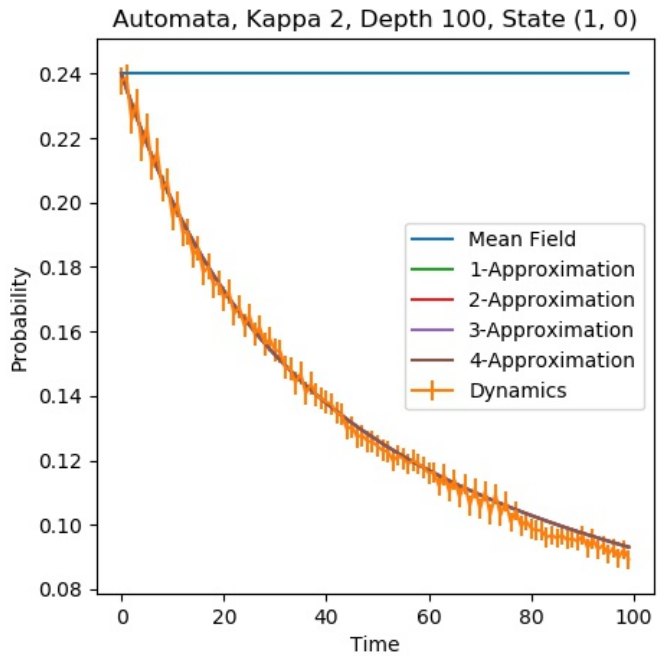


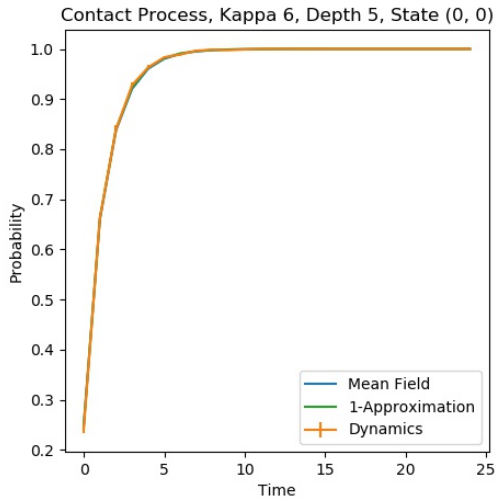
Figure 8: $p = 0.01, \mathbb{P}(X_v(0) = 1) = 0.4$. Dynamics averaged over 5000 MC samples.

We compare the above quantities mainly at large times, when the process in question is close to stationarity. We also begin to explore the ergodic properties of the k-approximations in relation to ergodic properties of the dynamics that they approximate. We end with a discussion on the evidence collected in this section. Throughout the section, we colloquially refer to the interacting particle system under question as 'dynamics'.

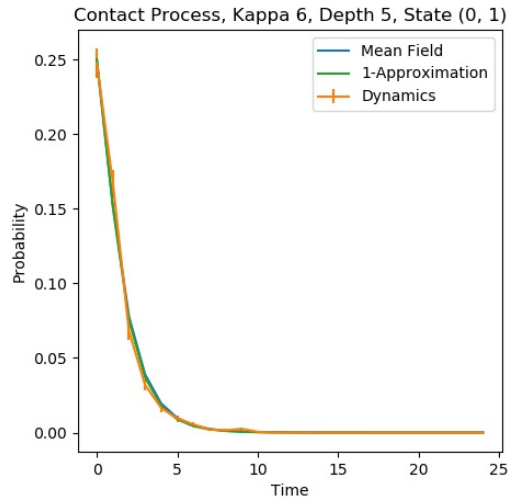
6.1 Contact Process

Low Infection Rate

The contact process with low infection rate seems to have the simplest relationship between the mean field and k-approximations. In this simulation, we took $p = 0.25, q = 0.75$ with $\mathbb{P}(X_v(0) = 1) = 0.5$ and considered the tree of degree 6 and depth 5. Figure 9 shows the joint probabilities of states $(0,0), (0,1)$ for the root and its neighbor over time. The results for the dynamics reflect the fast convergence to the death of the infection, by which we mean the process converges to the point mass at 0. It seems the 1-approximation and the mean field approximation agree very closely with the dynamics. The agreement between each approximation and the dynamics, as measured in expected fraction of time spent at a state and the probability of a single over time, were also very good.



(a) Probability of State $(0, 0)$ vs. Time

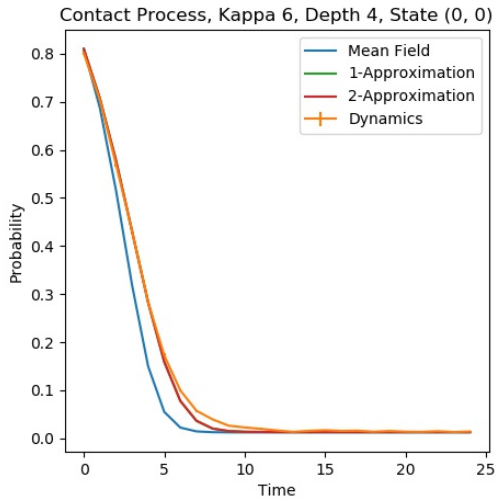


(b) Probability of State $(0, 0)$ vs. Time

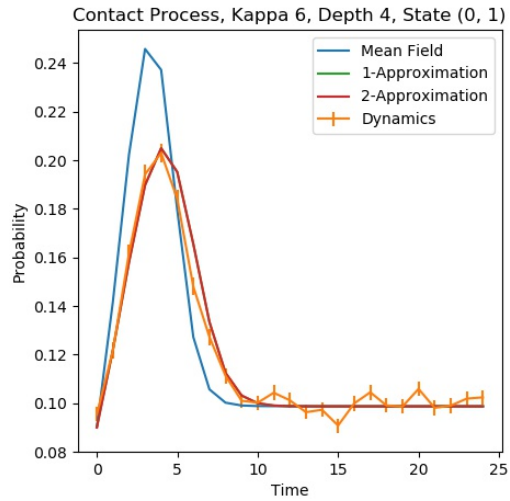
Figure 9: Contact Model model for $p = 0.25, q = 0.75$ and $\mathbb{P}(X_v(0) = 1) = 0.5$. Pictured are the probabilities of the states $(0, 0), (0, 1)$ vs time. 10^5 Monte Carlo samples were taken to generate the dynamics.

High Infection Rate

For high infection rates, we observe a similar story to the low infection rate case. We ran the contact process with $p = 0.9$ and $q = 0.1$, for graphs with $\kappa \in \{2, 4, 6\}, d \in \{2, 4\}$. The results for $\kappa = 6, d = 4$ are pictured in Figure 10.



(a) Probability of State $(0,0)$ vs. Time



(b) Probability of State $(0,0)$ vs. Time

Figure 10: Contact Model model for $p = 0.9, q = 0.1$ and $\mathbb{P}(X_v(0) = 1) = 0.1$. Pictured are the probabilities of the states $(0,0), (0,1)$ vs time. 5000 Monte Carlo samples were taken to generate the dynamics.

Both the mean field approximation and k-approximations agree with the dynamics nearly exactly at stationarity (in this case, $T > 10$) and capture the same general behavior of the dynamics: for state $(0,1)$ of Figure 10b, a trend upwards followed by a trend downwards before stabilizing at roughly 0.1, and for state $(0,0)$ of Figure 10a, a steady decrease to the limiting value.

However, at small times before stationarity, the mean field is not nearly as accurate as the k-approximation, whose relative error with the dynamics is small. This observation is also reinforced by the expected fraction of time spent at the state $(0,1)$ for the root and its neighbor. Figure 11 shows that for the dynamics of the contact process on trees of depth 4, the k-approximations agree with these dynamics much better than the mean field approximation, and the approximation gets increasingly accurate as κ grows. As expected, as the size of the graphs grow, i.e. as both κ, d grow, the mean field approximation becomes more accurate.

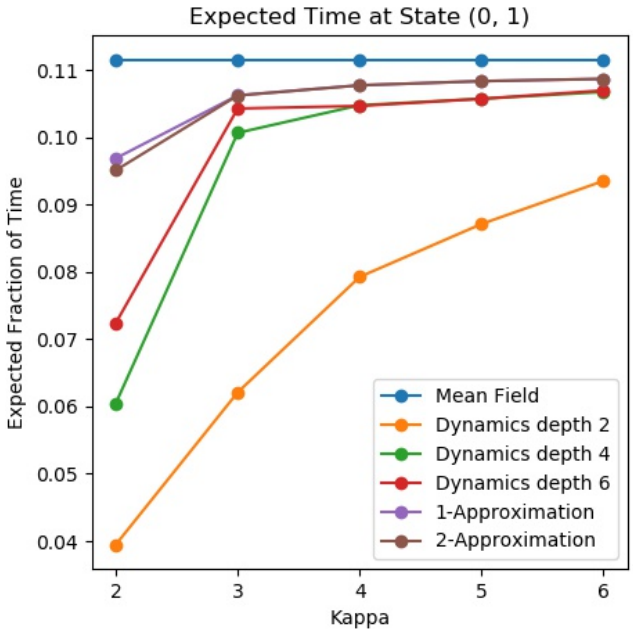


Figure 11: Expected Fraction of Time spent at State $(0, 1)$ vs. size of the tree. $p = 0.9, q = 0.1$, and $\mathbb{P}(X_v(0) = 1) = 0.1$.

For completeness, we also include some behavior of the local recursions and mean field approximation on smaller graphs. Figure 12 shows the k-approximations and mean field approximation on the 2-regular graph of depth 2, that is the chain graph with 5 nodes. We see that neither the k-approximations nor the mean field approximations model the asymptotic value for the joint probability, and the agreement for small times is also poor. It is interesting to node that the the limiting values for each of the mean field approximation and the k-approximation seem to agree well. In general, for dynamics on small graphs, we observe that for purposes accuracy the mean field and the k-approximations should be eschewed in favor of Monte Carlo simulation. In both the study of long term and short term behavior of the dynamics, both approximations do not perform well.

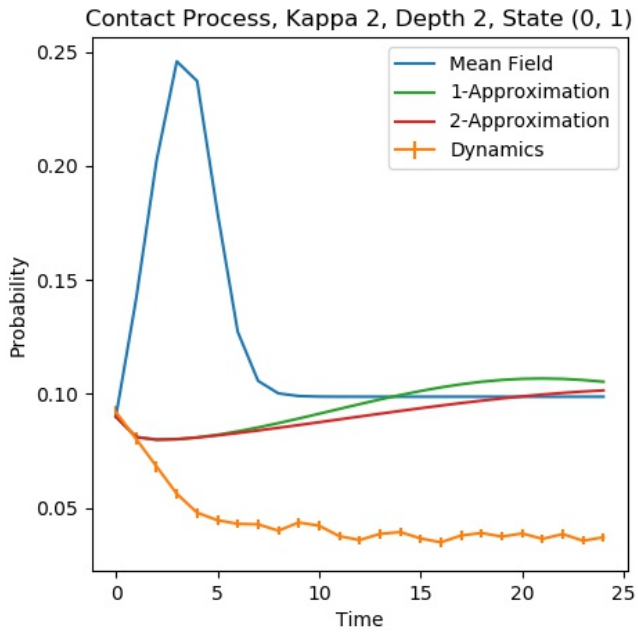


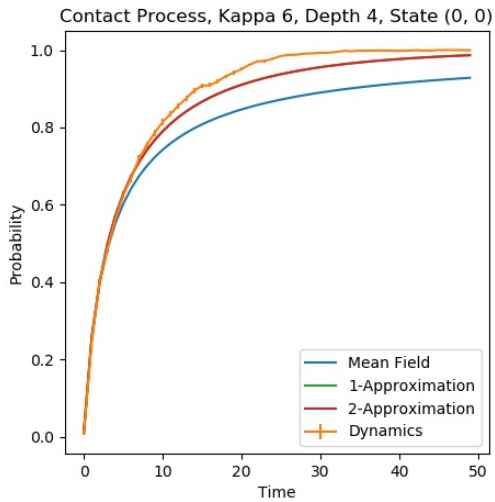
Figure 12: Contact Model model for $p = 0.9, q = 0.1$ and $\mathbb{P}(X_v(0) = 1) = 0.1$. Pictured are the probabilities of the states $(0, 1)$ vs time. 5000 Monte Carlo samples were taken to generate the dynamics.

Medium Infection Rate

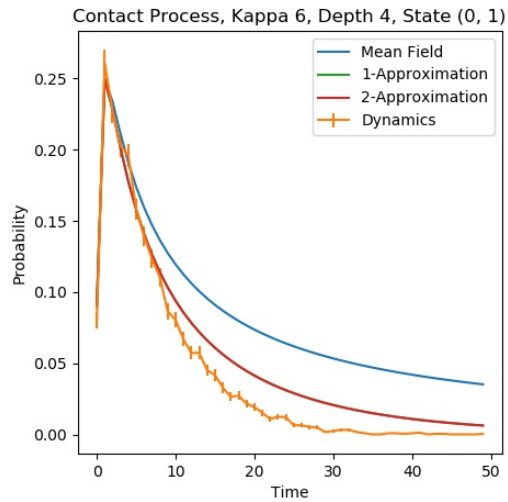
We explore the setting when the contact process has similar infection and recovery probabilities $p = 0.5, q = 0.5$. Throughout these simulations we set $\mathbb{P}(X_v(0) = 1) = 0.9$, and explore the k-approximations and mean field approximations for regular trees of degree κ and depth d. For simplicity we cover the case $\kappa \in \{2, 4, 6\}$ and depth= $\{2, 4\}$.

Figure 13 displays the joint probabilities for a node and its neighbor for $\kappa = 6$, depth 4. For each of the states, we observe generally that each of the k-approximations and the mean field approximations agree closely with the dynamics for small times, but they begin to diverge for times past $T = 10$. The local recursions seem to be more accurate than the mean field approximations, although not perfect, for Figures 13a, 13b, 13c. However, the local recursion seems to asymptote to the correct limit values for each of these states (0 for each of $(0, 1), (1, 0), (1, 1)$ and 1 for $(1, 1)$). In this case, the 1-approximation and 2-approximation are overlapping, indicating fast convergence of the k-approximations in k. The probabilities of state $(1, 1)$ show good agreement between each of the approximations, possibly due to the fast convergence to 0. The story for the smaller graphs is similar, although we omit the figures here.

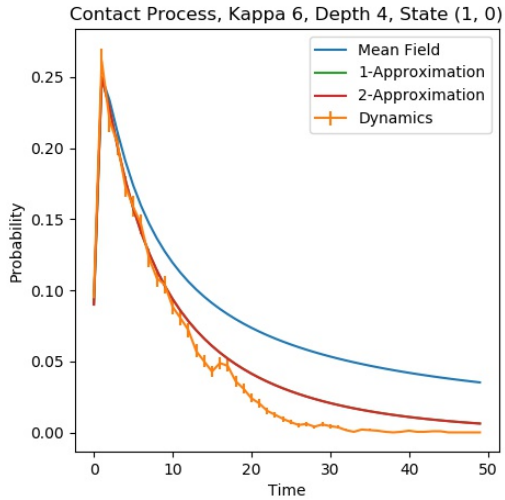
If we analyze the approximations via expected time spent at joint state, we see



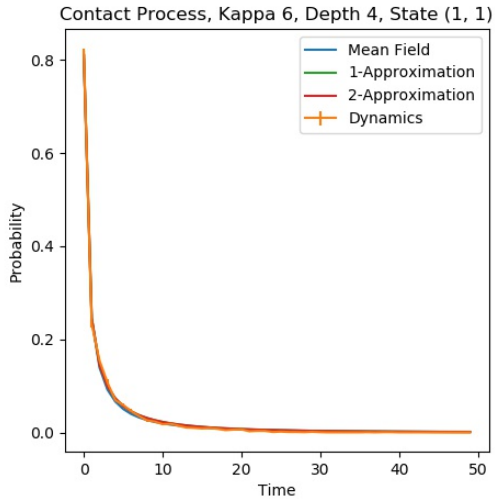
(a) Probability vs. Time of State $(0, 0)$



(b) Probability vs. Time of State $(0, 1)$

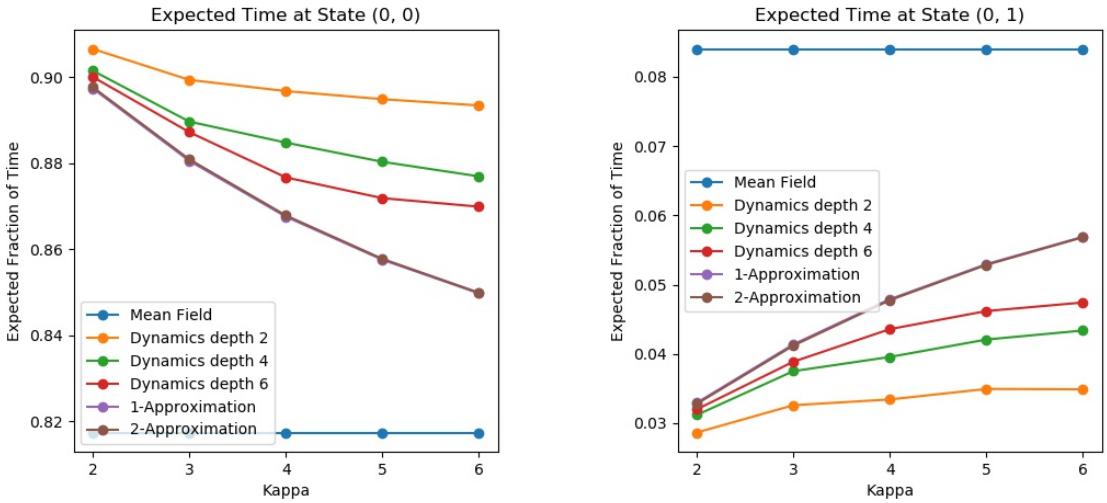


(c) Probability vs. Time of State $(1, 0)$



(d) Probability vs. Time of State $(1, 1)$

Figure 13: Contact Process for $p = 0.5, q = 0.5, \mathbb{P}(X_v(0) = 1) = 0.9$, and $\kappa = 6, d = 4$. Pictured are the joint probabilities of the states $(0, 0), (0, 1), (1, 0), (1, 1)$ for the root and a given neighbor. 10^5 Monte Carlo samples were taken to generate the dynamics.



(a) Degree of tree vs. Expected Fraction of Time at State $(0, 0)$

(b) Degree of tree vs. Expected Fraction of Time at State $(0, 1)$

Figure 14: Contact Model model for $p = 0.5, q = 0.5$ and $\mathbb{P}(X_v(0) = 1) = 0.9$. Pictured are the expected fraction of time spent at states $(0, 0)$ and $(0, 1)$ for the root and a given neighbor. We plot the expected fraction of time spent at these states up to time $T = 50$ versus the degree of the tree. 10^5 Monte Carlo samples were taken to generate the dynamics.

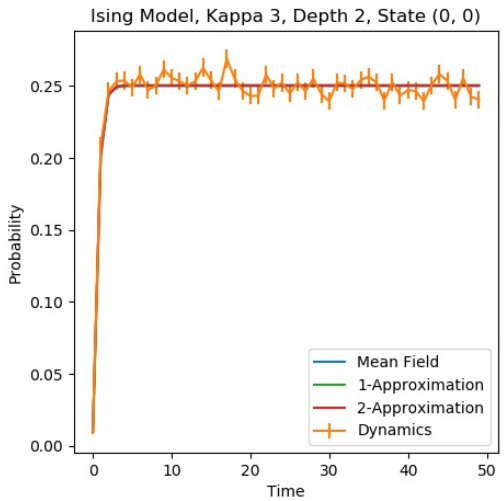
that the local recursions outperform the mean field approximations, although the relative errors for both approximations are significant. See Figure 14. As the depth of the trees grows, while fixing the kappa, it seems that the expected fraction of time at these states grows closer to those predicted by the local recursion. However, we also notice that for larger degree graphs, the local approximation seems to be more accurate for higher depth graphs. Note that the mean field approximation for the contact process has no dependence on the degree of the tree, as the recursion in Example 3.1 shows. It would be interesting to analyze this relationship as the degree and depth of the graph are much larger.

6.2 Ising Model

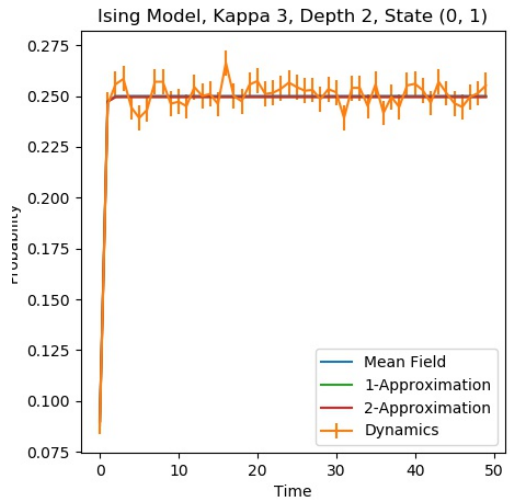
Low Beta

The low beta regime for the Ising model is also interesting for our experiments: we ran the Ising model for $\beta = 0.01$, and the dynamics indicate that asymptotically, neighboring nodes become independent. Indeed, Figure 15 shows that each of the joint states has a limiting probability of 0.25, indicating that any given node and its neighbor are independent, since this is simply a product measure on $\{0, 1\}^2$. In this case we see that the mean field and k-approximations are in near-perfect agreement on the joint probabilities with the dynamics. Agreement in the other metrics are

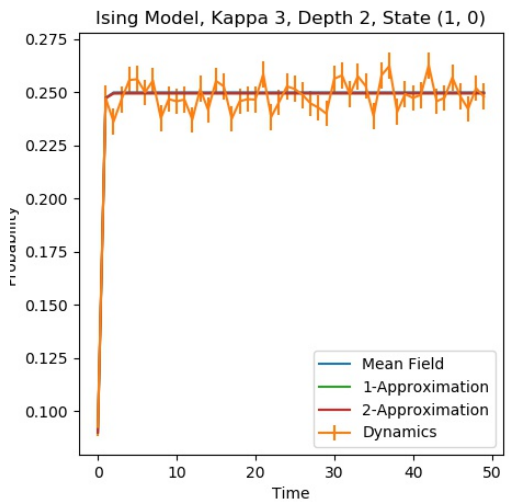
equally as good. Figure 15 analyzes these dynamics on a tree of $\kappa = 3$ and depth 2; the observations made in this case extend as well to larger graphs, of higher degree and depth.



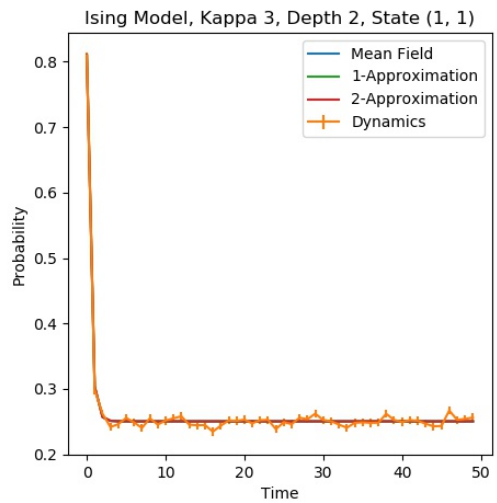
(a) Probability of State $(0, 0)$ vs Time



(b) Probability of State $(0, 1)$ vs Time



(c) Probability of State $(1, 0)$ vs Time



(d) Probability of State $(1, 1)$ vs Time

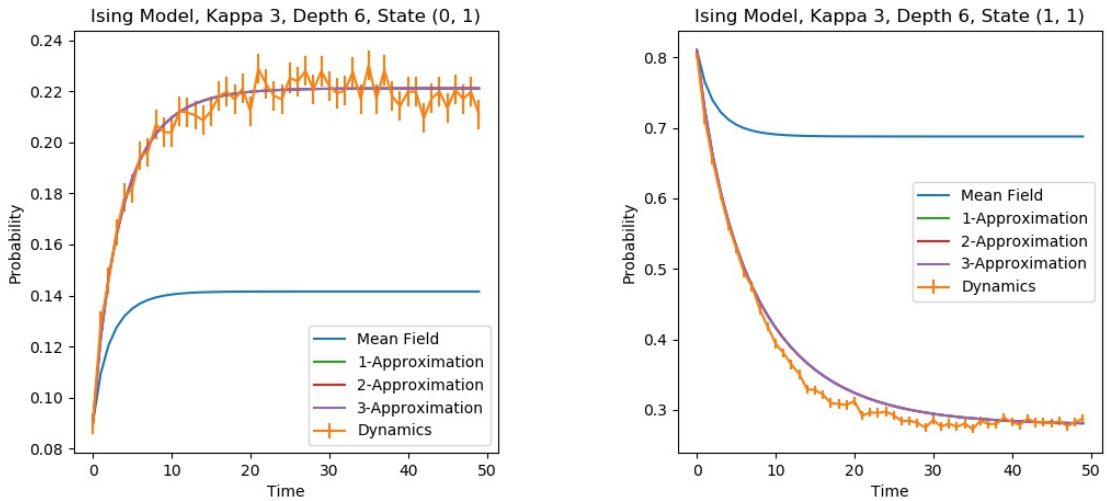
Figure 15: Parallel Glauber Dynamics for $p = 0.9, \beta = 0.01$ and $\mathbb{P}(X_v(0) = 1) = 0.9$. 10^5 Monte Carlo samples were taken to generate the dynamics.

While it is unusual to expect extremely good agreement between the mean field approximation and the k-approximations, especially for small graphs, we conjecture that this behavior is due to the asymptotic independence property of the Ising Model for very low inverse temperatures. It would be interesting to see if any such interacting

particle system whose nodes are asymptotically independent (deemed the propagation of chaos property), or at least approximately, has good agreement between each of the mean field, the k-approximations, and the dynamics.

Medium Beta

We also conducted experiments on the Ising model for intermediate values of β. On T_3, we took $p = 0.9, \beta = 0.4$, which is below the critical value for these dynamics. As opposed to the previous case of low β, the nodes are not asymptotically independent, and we see that the mean field predicts markedly different limiting values for the joint states than the dynamics. The k-approximations succeed in capturing the correct asymptotic value, and in this case we see fast convergence in k of these approximations. Figure 16 pictures these results. In fact, for times before stationarity, the k-approximations track the dynamics very closely, although in Figure 16b the dynamics seems to decrease more quickly than the k-approximations.



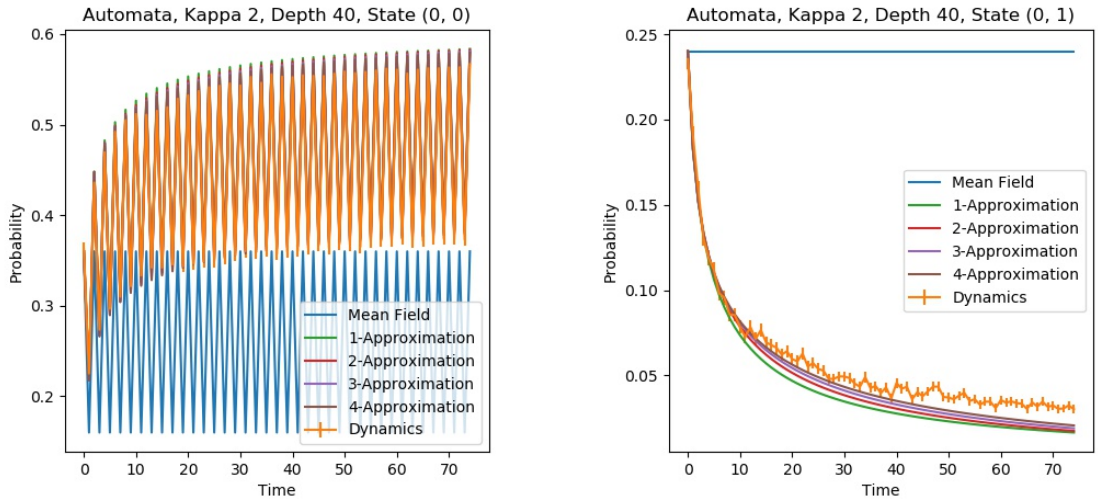
(a) Probability vs. Time of State $(0, 1)$

(b) Probability vs. Time of State $(1, 1)$

Figure 16: Parallel Glauber dynamics for $p = 0.9, \beta = 0.4, \kappa = 3$ and $\mathbb{P}(X_v(0) = 1) = 0.9$. Pictured are the joint probabilities of the state $(0, 1)$ and $(1, 1)$ for the root and a given neighbor. 10^5 Monte Carlo samples were taken to generate the dynamics.

6.3 Oscillator Dynamics

Finally, we consider the same simulation scheme for the oscillator dynamics defined in Equation (16), first on the ring graph, or T_2, and the 4-regular tree T_4. We plot the joint probabilities of the the states $(0, 1), (0, 0)$ for the root and a neighbor. The results for $\kappa = 2$ the line are pictured in Figure 17, and the results for $\kappa = 4$ in Figure 18.



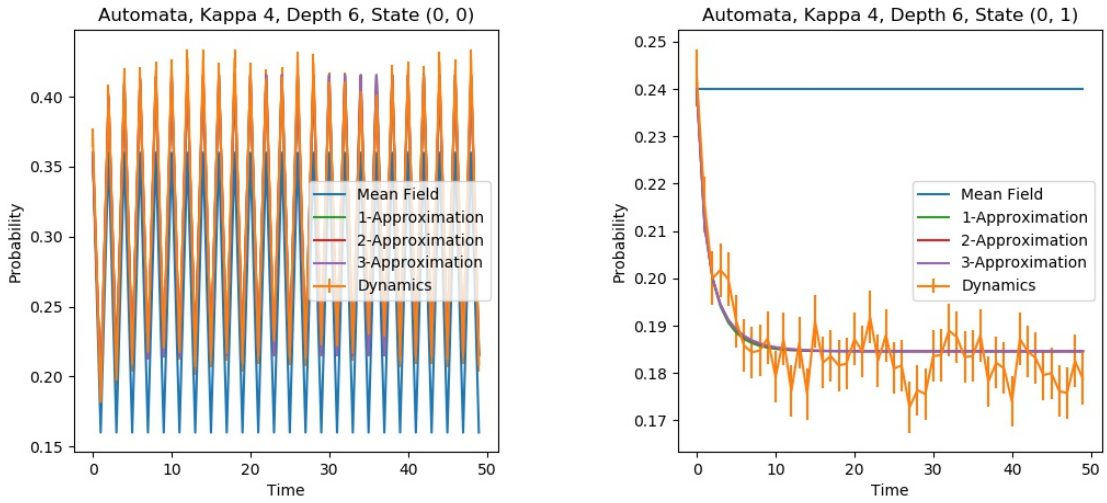
(a) Probability vs. Time of State $(0,0)$

(b) Probability vs. Time of State $(0,1)$

Figure 17: Binary Oscillator C.A. model for $p = 0.6, \kappa = 2$ and $\mathbb{P}(X_v(0) = 1) = 0.4$. Pictured are the joint probabilities of the state $(0,0)$ and $(0,1)$ for the root and a given neighbor. We do not picture the other joint states, by symmetry. 10^5 Monte Carlo samples were taken to generate the dynamics.

From Figure 17a, we see that the dynamics of the process nearly overlap with the k-approximations, and they seem to be agreeing very closely. The k-approximations capture both the oscillating behavior of the dynamics - at the same phase - along with the general trend of the dynamics upwards. The mean field approximation also succeeds in capturing the phase of the dynamics and the oscillating behavior, but the upwards drift is not modeled: the mean field approximation oscillates between two static values.

Figure 17b highlights the comparative advantage of the k-approximations over the mean field approximation to model the correlation structure between two neighboring nodes. Note that the we expect the probability of the joint state $(0,1)$ in the dynamics to asymptote to zero, since we know that the process converges to a state where all the nodes agree, either the all 0 or all 1 state. The local recursions trend toward 0, and agree relatively well with the dynamics. Note in this case the k-approximations become significantly more accurate for medium times as k grows. As noted previously in the discussion of Figure 8, we expect the mean field approximation to yield a constant approximation for the joint probability of $(0,1)$. As we saw in Figure 3, the mean field approximation assigns the $\mathbb{P}(X_v(2t)) = p_0$ for all v, t for some p_0, and $\mathbb{P}(X_v(2t+1)) = p_1$ for all p_1 (note we can extrapolate this information from the figure since the mean field recursion only depends on the previous time step, and the figure establishes periodicity). Then the joint probability is exactly $p_0 p_1$ and hence is constant.



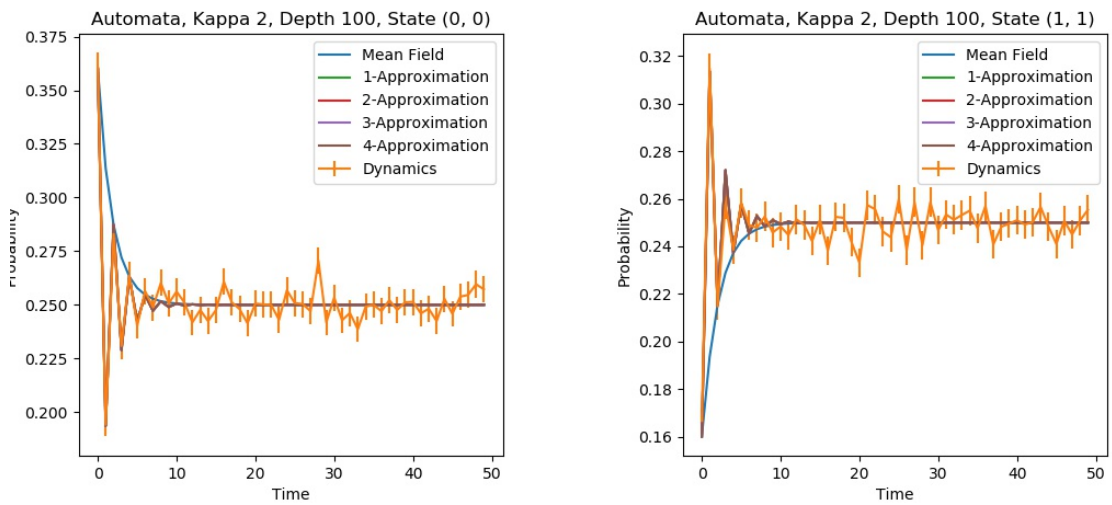
(a) Probability vs. Time of State $(0, 0)$

(b) Probability vs. Time of State $(0, 1)$

Figure 18: Binary Oscillator C.A. model for $p = 0.6, \kappa = 4$ and $\mathbb{P}(X_v(0) = 1) = 0.4$. The depth of the tree was taken to be 6 generations. Pictured are the joint probabilities of the state $(0, 0)$ and $(0, 1)$ for the root and a given neighbor. We do not picture the other joint states, by symmetry. 10^5 Monte Carlo samples were taken to generate the dynamics.

The story for the case of $\kappa = 4, d = 6$ as pictured in 18 is similar. Figure 18a shows that both the mean field and k-approximations capture the phase and oscillation of the dynamics, although the mean field approximation seems to capture different values for the oscillating probabilities. As in the previous case, Figure 18b shows that the joint probability for state $(0, 1)$ cannot be captured by the mean field approximation, while the k-approximations succeed in doing so. In this case, it seems like we have the k-approximations converge rapidly in k to the dynamics, as opposed to the $\kappa = 2$ case. This is more evidence that there may be different properties required for convergence of the k-approximations when $\kappa > 2$ versus $\kappa = 2$.

The stochastic frequency model introduced in Equation (18) exhibits similar oscillating behavior for small times. The results for the ring are pictured in Figure 19. We took $p = 0.6$, and $\mathbb{P}(X_v(0) = 0.4)$. We can see that for small times $T < 10$, the dynamics of the joint probabilities exhibit an oscillating behavior before converging to 0.25. The k approximations all capture this oscillation behavior very well, while the mean field does not capture this at all. At stationarity, both approximations perform very well, and agree with the stationary value of 0.25. Our results for the other joint states actually indicate that the process has the propagation of chaos property, where the nodes are asymptotically independent as in the case of the Ising model with low beta. Just as in Figure 15, we observe that each of the mean field and k



(a) Probability vs. Time of State $(0,0)$ (b) Probability vs. Time of State $(0,1)$

Figure 19: Stochastic Frequency C.A. model for $p = 0.6, \kappa = 2$ and $\mathbb{P}(X_v(0) = 1) = 0.4$. Pictured are the joint probabilities of the state $(0,0)$ and $(0,1)$ for the root and a given neighbor. 10^4 Monte Carlo samples were taken to generate the dynamics.

approximations do very well in capturing the stationary properties of the dynamics.

6.4 Comparison of Ergodic Properties

In this section, we test whether the ergodic properties of the k-approximation are inherited from dynamics on the full tree. We test this in a few separate examples. This provides evidence that not only the interchange property in Conjecture 5.2 should hold, but that the k-approximation is often good enough to analyze ergodicity of the local recursion.

Ising Model

Recall that in the Ising Model on T_κ exhibits a phase transition as β increases: there is a critical β_k^c such that for inverse temperatures below this critical value, there is a unique stationary distribution for the Glauber dynamics, and for larger β there exist unique stationary distributions. The quantity β_c^κ can be calculated explicitly as

$$\beta_c = \frac{1}{2} \ln \frac{\kappa}{\kappa - 2}.$$

In this section we test whether the k-approximation satisfies a similar phase transition property. We test this on the trees T_κ for $\kappa = 3, 4$ and for completeness, we test this on $\kappa = 2$.

For each of these simulations, we run the 1-approximation until stationarity for various values of β and θ, where $\mathbb{P}(X_v(0) = 1) = \theta$. Stationarity was decided by running the process for 500 time steps, and we summarize the invariant measure via the probability of the all zeros state or the $\mathbb{L}_2$ norms of the measures. Each of the points in the figure are the probabilities of the zero state for each combination of θ, β, and we set $p = 0.9$ for the parallel Glauber dynamics. The results for $\kappa = 2$ - Figure 20, $\kappa = 3$ in Figure 21, and $\kappa = 4$ in Figure 22.

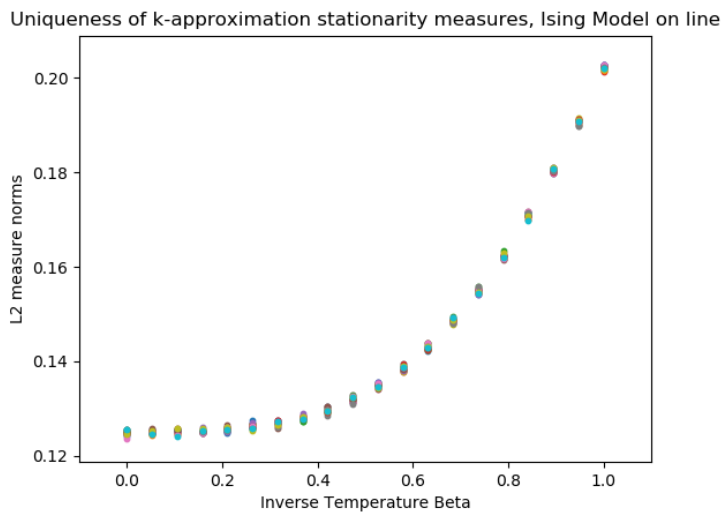


Figure 20: Ising Model $p = 0.9$ on the line. We plot the $\mathbb{L}_2$ norms of the 1-approximation stationary measures for various levels of β and θ the initial parameter. We took $\theta \in \{0, 0.05, 0.1, \ldots, 1\}$ and jittered the points slightly. We took $k = 1$.

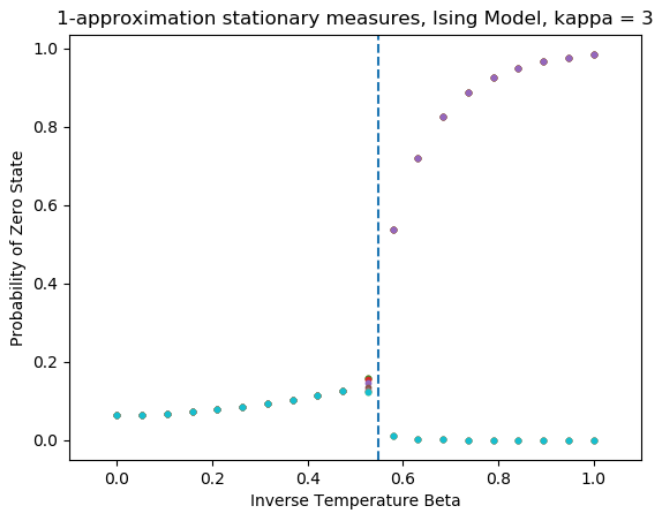


Figure 21: Ising Model $p = 0.9$ on T_3. We plot the probabilities of the zero state of the 1-approximation stationary measures for various levels of β and θ the initial parameter. We took $\theta \in \{0, 0.1, 0.2, \ldots, 1\}$. A vertical line corresponding to the exact, theoretical phase transition value $\beta = \frac{1}{2}\ln 3 \approx 0.549306$ is also plotted.

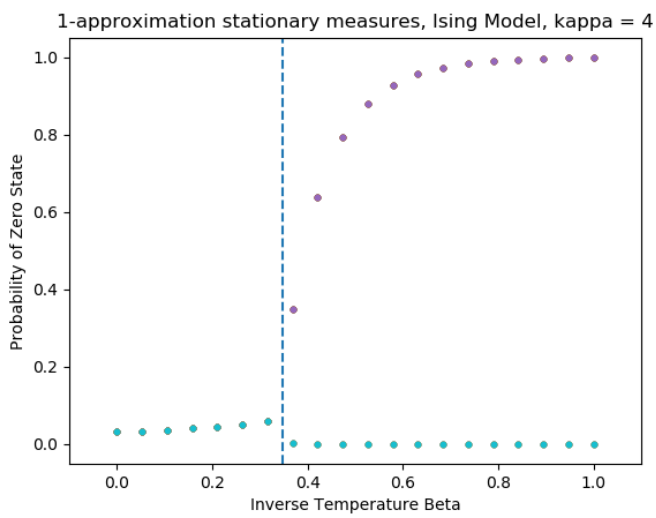


Figure 22: Ising Model $p = 0.9$ on T_4. We plot the probabilities of the zero state of the 1-approximation stationary measures for various levels of β and θ the initial parameter. We took $\theta \in \{0, 0.1, 0.2, \ldots, 1\}$. The line $\beta = \frac{\ln 2}{2} \approx 0.346573$ is plotted corresponding to the theoretical value of the critical value.

Interestingly, we do not see a phase transition for the Ising model on the line for the values of $\beta \in [0, 1]$. However, for $\kappa = 3, 4$ our results indicate that there is a phase

transition for the stationary distributions of the Ising model, with $\beta_3^c \approx 0.55, \beta_4^c \approx 0.34$: at these values of beta, the processes seem to converge to different stationary distributions given different initial probabilities - in particular there seem to be two invariant measures. These critical values line up fairly well with those predicted by the theory. In particular, when $\kappa = 3$, the $\beta_3^c = \frac{1}{2}\ln 3 = 0.5493, \beta_4^c = \frac{1}{2}\ln\frac{4}{2} = 0.3466$. This indicates that the ergodic properties of the k-approximations accurately reflect those of Glauber dynamics.

Contact Process

We conduct a similar investigation as with the last section. For the continuous-time contact process on regular trees T_κ, with $\kappa \geq 3$, the reference [27] offers a characterization of the phase transition properties of the contact process. In this framework, the infection parameter λ is varied and the recovery rate is fixed at 1. For $\lambda < \lambda_1 \approx \frac{1}{\kappa-1}$, the process converges to the zero state - i.e. the infection dies out. For $\lambda > \lambda_1$, the infection lives forever with positive probability. For larger λ the process converges to a non-trivial invariant distribution. In the experiments we conducted, we considered the contact process with recovery rate $q = 0.1$ and ran the 1-approximation until stationarity for different values of p the infection rate and the initial probability of 1, θ.

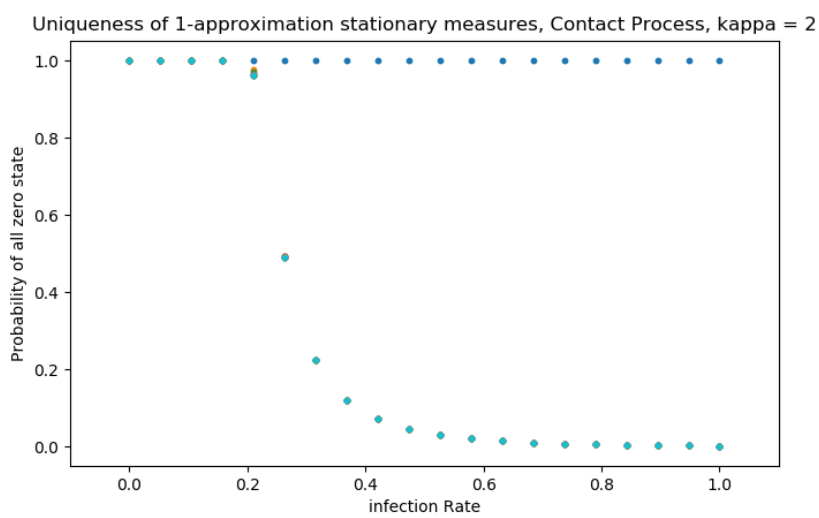


Figure 23: Contact Process $q = 0.1$ on $\mathbb{Z}$. We plot the probabilities of the zero state of the stationary measures for dynamics at various levels of p the infection rate and θ the initial parameter. We took $p \in \{0, 0.1, 0.2, \ldots, 1\}$.

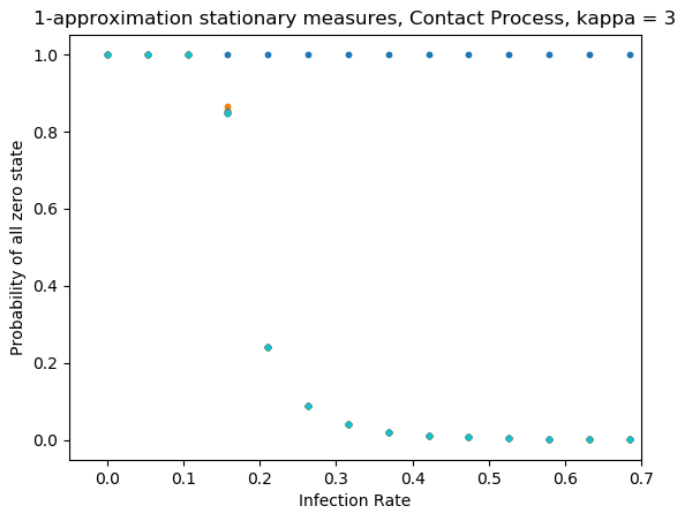


Figure 24: Contact Process $q = 0.1$ on T_3. We plot the probabilities of the zero state of stationary measures for various levels of β and θ the initial parameter. We took $\theta \in \{0, 0.05, 0.1, \ldots, 1\}$.

While the critical values of the infection parameter does not seem to line up with the theory - which is reasonable given that we are working with a discrete time model - it seem as if there is similar behavior: below a certain threshold for p, approximately 0.15 for the $\kappa = 3$ case, the only stationary distribution is the zero state, where the infection dies out. Additionally, past $p \approx 0.2$, there seems to be a unique nonzero stationary distribution. The other stationary distributions are exactly obtained when $\theta = 0$, since then the initial measure is the point mass at zero. It is unclear what it is happening in the regime in between these two parameters, although it is possible that this corresponds to the intermediate regime discussed in [27], where there are nonzero probabilities of the infection persisting for all time.

6.5 Mean Field vs. Local Approximation Comparison

In this section, we present some numerical results on the accuracy of the mean field approximation in relation to the local approximation and k-approximations. We focus on studying the total variation distance between marginals of the mean field process and marginals of the local recursion at low time. We also study the same comparison between the k-approximations for large times. We consider this the very beginning of an analysis of this comparison, and hope to expand the results shown here to obtain a more complete characterization of performance tradeoffs of the mean field approximation.

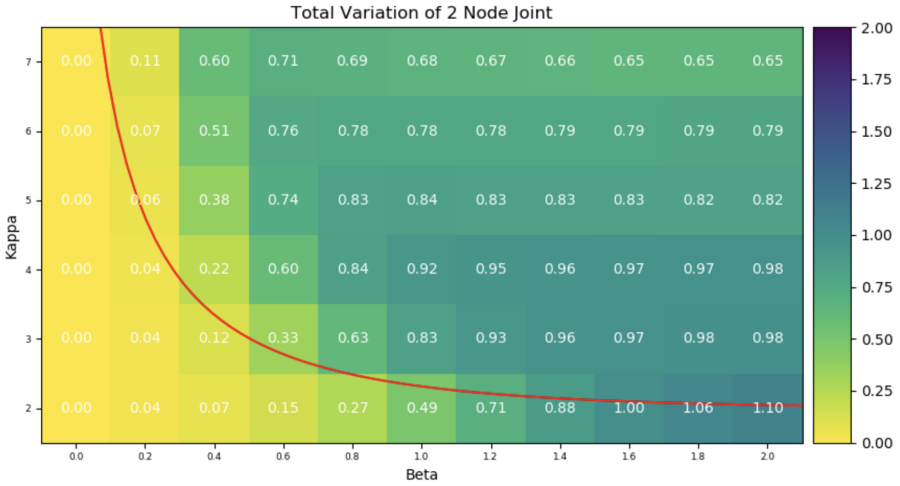


Figure 25: Total variation distance between marginal of root and neighbor for the mean field process and the local approximation at time $T = 2$. The initial value for each of these processes was taken to be $\mathbb{P}(X_v(0) = 1) = 0.4$. We plot this difference for various values of κ the degree of the infinite tree and β the inverse temperature parameter for the parallel Glauber dynamics. The critical values for phase transition in the Ising model are plotted in red.

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Figure 25 compares the mean field process to the local approximation for small time, $T = 2$. We considered the joint distribution of the root and a given neighbor, and plotted the total variation distances between the joint marginal of the mean field process and that of the local approximation. Recalling that the local recursion is exact as an approximation to the marginal of the dynamics on the infinite regular tree, we can view this as a comparison of the mean field process to the dynamics on the infinite tree. We make this comparison for several values of kappa and β, the inverse temperature parameter in the Ising model. We immediately note that for low β, the mean field approximation is fairly exact, and by earlier results, e.g. Figure 15, we know the dynamics are asymptotically independent. This matches the intuition that the mean field approximation performs well when the process has the propagation of chaos property.

An interesting feature is that the critical values of β as a function of κ seem to demarcate two regimes for the performance of the mean field approximation. Indeed, for fixed kappa, the mean field approximation for beta less than the critical value seems to perform very well. For values of beta past this critical value, the total variation distance seems to steeply increase. We conjecture that this could be due to long-range correlation present when β is large, that is not captured by the mean field approximation. When β is large and κ increases, we observe in general that the total variation distance decreases, which matches the intuition behind the mean field

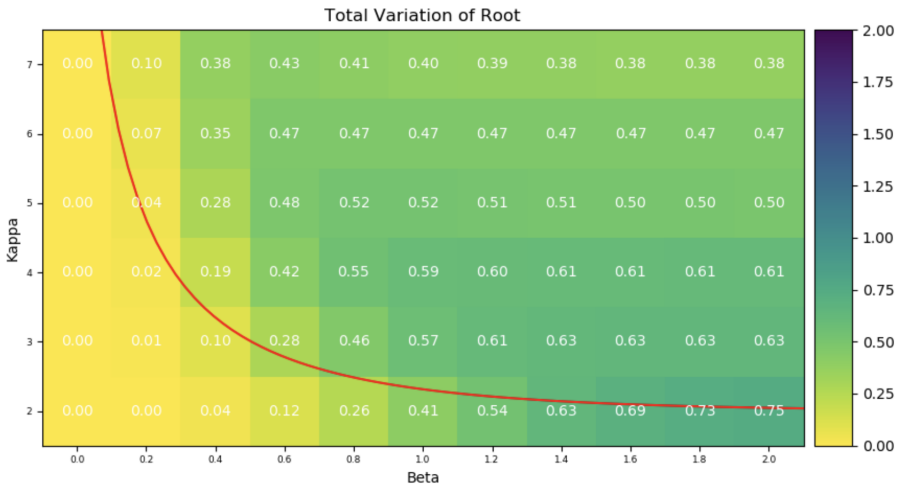

Figure 26: Total variation distance between joint of root for the mean field process and the local approximation at time $T = 2$. $\mathbb{P}(X_v(0) = 1) = 0.4$. The critical values for phase transition in the Ising model are plotted in red.

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approximation - the denser the graph, the more accurate the mean field approximation is. For smaller values of β the total variation distance does not monotonically decrease with increasing κ. These observations carry over when looking at the distribution of the root, as well as the joint distribution of the root and 2 neighbors. The results for these cases are pictured in Figures 26, 27. We do observe that the more neighbors we consider, the larger the total variation distance between the mean field approximation and the local recursions.

In fact, if we consider the marginal of $\mathcal{T}$, the root with all its neighbors, the total variation increases as a function of κ regardless of the value of beta. We conjecture that this could be due to more correlation between the nodes which is not captured in the mean field approximation. This is pictured in Figure 28. Overall, the total variation distance is still higher than in the previous cases where fixed a constant number of neighbors. Figure 29 details the total variation distance between the distributions of the trajectories of the root and its neighbors for the mean field and local approximation. Interestingly, for low beta, the total variation distance is significantly larger than 0, in contrast to the fixed time case. Furthermore, as κ increases, we see an monotone increase in the total variation distance. This indicates that the distribution of histories of the mean field approximation is inaccurate for small times, and that there is a fundamental difference between the evolution of the mean field approximation and the local approximation.

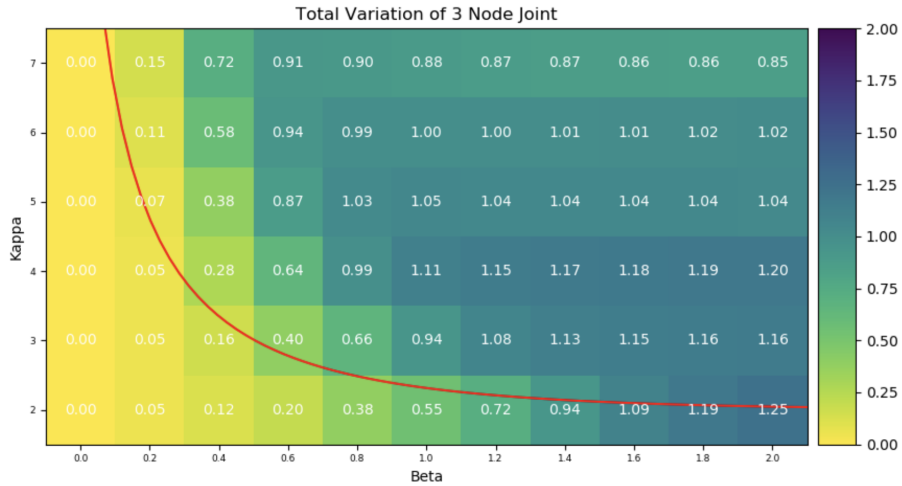

Figure 27: Total variation distance between joint distribution of the root and two of its neighbors for the mean field process and the local approximation at time $T = 2$. $\mathbb{P}(X_v(0) = 1) = 0.4$. The critical values for phase transition in the Ising model are plotted in red.

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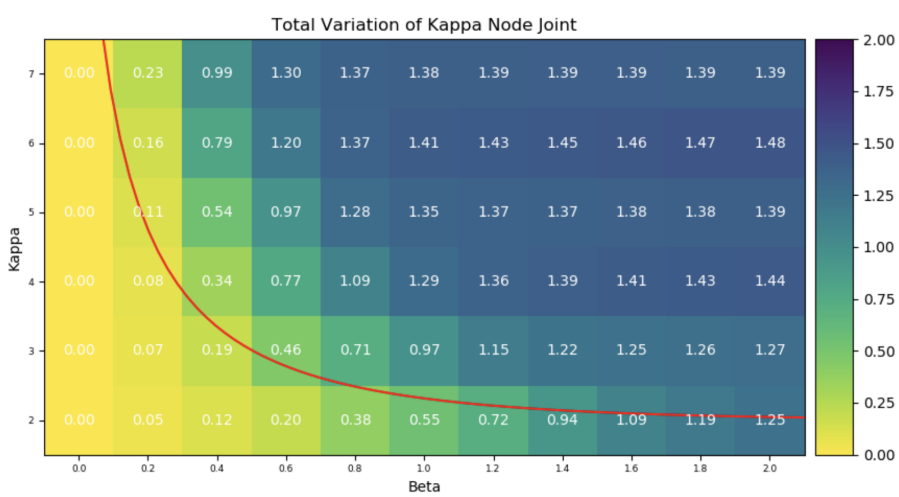

Figure 28: Total variation distance between joint distribution of the root and its neighbors for the mean field process and the local approximation at time $T = 2$. $\mathbb{P}(X_v(0) = 1) = 0.4$. The critical values for phase transition in the Ising model are plotted in red.

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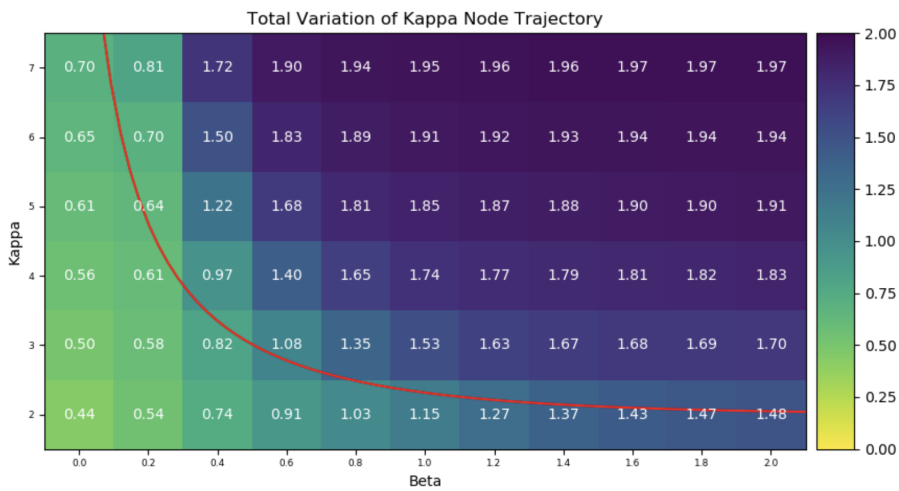


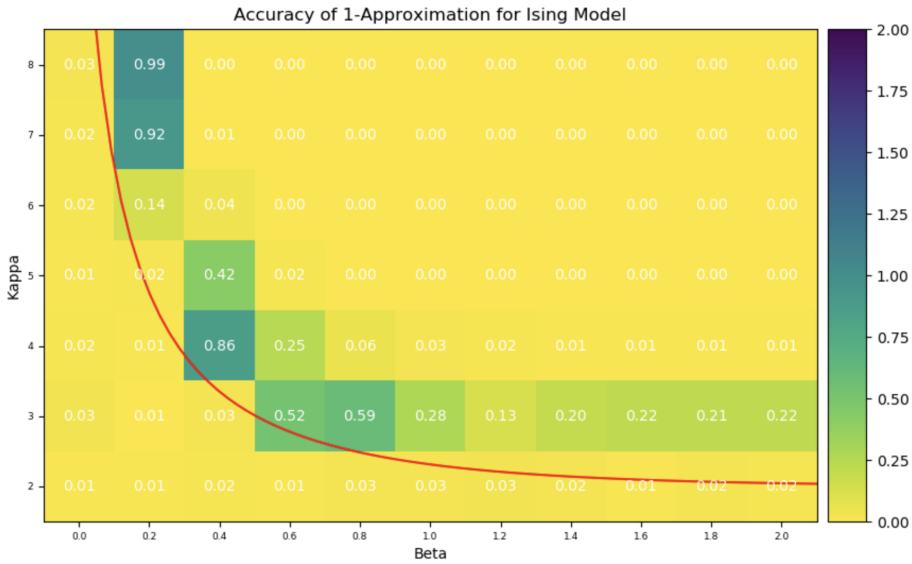
Figure 29: Total variation distance between joint distribution of the root and its neighbors for the mean field process and the local approximation for histories up to time $T = 2$. $\mathbb{P}(X_v(0) = 1) = 0.4$. The critical values for phase transition in the Ising model are plotted in red.

.

Long Time Results

We also compare the k-approximations and mean field approximation for large times. We compare both of these comparisons to the Ising model on a trees of degree κ for varying κ and fixed depth $d = 8$. We also vary the values of the inverse temperature β, and we set $T = 50, p = 0.9, \mathbb{P}(X_v(0) = 1) = 0.4$. Figure 30 pictures the fidelity of both approximations for the joint distribution of the root and a given neighbor. The behaviors of both the 1-approximation and the mean field seem to be similar. It seems like both approximations work very well: the only cases in which these algorithms seem to underperform are for values of β and κ near the critical regime. We conjecture that this could be due to a longer convergence time for the dynamics on the finite tree, or perhaps a longer convergence time for the 1-approximation and mean field approximation. We note that the region of underperformance by the 1-approximation and the boundary region for the mean field approximation do not match up precisely with the critical values for the Ising model - a deeper investigation is required to understand the exact behavior in the critical regime. These results suggest that the 1-approximation would be a strong algorithmic candidate to investigate long term behavior of processes like the Ising model for various values of the parameters, especially for values of β not near the critical regime. In fact, it seems like the 1-approximation outperforms the mean field approximation in terms of accuracy in almost every case. One significant advantage that the 1-approximation holds over the mean field approximation seems to be better performance in the low κ regime; this

matches the intuition that the 1-approximation was designed for dynamics on *sparse* graphs.



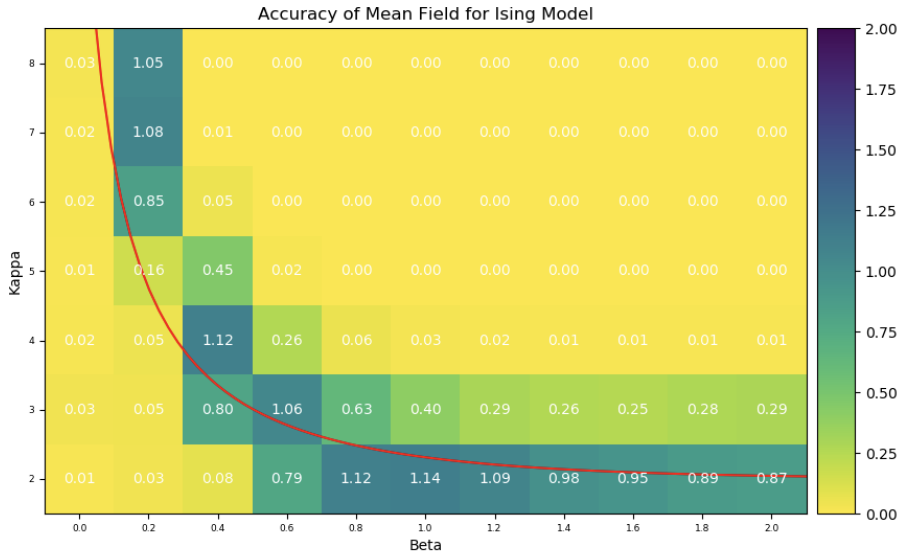


Figure 30: Total variation distance between the joint distribution of the root and one of its neighbors for the mean field process and Monte-Carlo sampled dynamics up to $T = 50$, on the tree of depth 8 and degree κ, for various values of κ, β. The same comparison is done for the 1-approximation instead of the mean field. $\mathbb{P}(X_v(0) = 1) = 0.4, p = 0.9$. The critical values for phase transition in the Ising model are plotted in red. 10^4 Monte Carlo values were taken for the dynamics.

6.6 Discussion

To understand when one should use MCMC, the mean field approximation, or the local recursion, we must understand the tradeoffs in accuracy and simulation time in a variety of examples. When considering accuracy of the results, we know that MCMC is asymptotically exact for every case, given that the number of samples is taken large enough. The mean field, on the other hand, is observed to work for large, dense graphs, for example in the complete graph or the regular tree with high degree. In many other cases, however, the mean field approximation fails to capture the correct stationary distributions: its largest shortcoming is the independence assumption, which makes the mean field approximation quite poor at capturing joint distributions between a node and its neighbor. Figure 18 exemplifies a case where capturing the correlation between a node and its neighbor is important, and in which the mean field approximation fails in this respect. Finally, the local recursion is known to be asymptotically exact in the limit of larger and larger graphs, and exact on the infinite graph, so simulating this - if one cared about accuracy only - would be optimal. However, for computational reasons, we consider the k-approximation, which seems to be fairly accurate in many cases. Based on the examples in this section, it seems that convergence of the k-approximations to the stationary distribution of the dynamics is hopeful and seems relatively quick. This, however, seems to work best given κ fixed for trees of large depth. To add to this point, notice that for some dynamics such as the contact process, the mean field approximation has no dependency on the degree and depth of the graph, while the local recursion depends on the degree of the graph. In this way we can expect the local recursion to be more accurate for graphs of fixed degree and large depth, say for the ring or the line graph. One such example of this is in Figure 7: the mean field approximation asymptotes to a severely incorrect limiting value as compared to the Monte Carlo estimate, while the k-approximations successively approach the Monte Carlo dynamics. In the case of smaller graphs, of small depth, the k-approximation and local recursions do not seem as accurate as MCMC. One issue with the use of the k-approximation, while being a very practical tool in many cases where convergence in k seems quick, there is no (yet known) theoretical convergence result showing that the k-approximations give the correct stationary distribution as $k \to \infty$, and neither is there a strong understanding of the convergence rate in k. The theories for the mean field approximation and MCMC are better developed.

When considering computational expense, undoubtedly the mean field approximation is the fastest and simplest to simulate, amounting to a recursion via a matrix equation. In particular, the independence assumption of the mean field approximation allows the user to simulate only a single node at a time and compute the joint of several nodes via the product measure. In the cases where the dynamics simulated occur on large graphs and only the distribution of a single node is required, it seems most efficient to use the mean field. As analyzed in Section 4.1.1 the nonlinear Markov chain associated with the local recursion is extremely intensive to simulate, being ex-

ponential in time and space in the number of time steps run, preventing the algorithm from being run for investigations at stationarity. To investigate dynamics over long times, we can use the k-approximations, which is linear in the number of time steps run, but exponential in k and κ the degree of the graph. Hence the k-approximation is relatively fast for graphs of smaller degree, and slows down considerably as κ increases, and similarly with k. MCMC algorithms are easily parallelizable and fast to simulate, but the issue is in general we lack a rate a convergence - it is difficult to answer how many MCMC samples are required to get an accurate result.

Overall, it seems that for dynamics on large dense graphs (as the degree gets large), one should use the mean field approximation: the error is relatively low in each of the metrics we considered, and the speed of computation is extremely fast. Compared with the local recursions and k approximations, the mean field accuracy is similar, but the speed is much faster. For large graphs, simulating Monte Carlo especially for large times is also very computationally expensive, suggesting that the mean field should be used in this case.

The issue with the mean field appears on smaller, less dense graphs. In this case, it becomes apparent that the independence assumption of the mean field approximation leads to a deterioration in accuracy, especially in computing the joint. In this case, the local recursion and k-approximations are capable of capturing the correlation quite well; one good example of this is in the oscillator dynamics presented above: the mean field approximation predicts a constant value for the joint probability but the k-approximations correctly capture the trend towards zero. For large times, the nonlinear Markov chain is infeasible to run, so we must rely on the k-approximation or Monte Carlo. In all cases for the k-approximation, choosing the largest such k provides the most accurate information on the long time behavior of the dynamics, although the time complexity must be noted, especially for larger graphs. In some cases, which we conjecture is when the dynamics converge quickly to stationarity, the k-approximations agree well with the dynamics for all k. In these situations running the 1-approximation provides adequate accuracy without too much computational burden. As mentioned before however, there is no theoretical guarantee that the k-approximations give the asymptotically correct information on the stationary distributions of the dynamics, and how quickly this convergence occurs. On the other hand, the Monte Carlo estimate takes a moderate amount of time to simulate, and we do have a convergence guarantee, but knowing exactly how many samples to take to get an estimate of the desired quantities is difficult. In some cases, taking a large enough number of samples to get below a desired level of variance might take longer than running the k-approximation, with no tradeoff in accuracy. In the small (finite) time regime, running the local recursion proper is provably exact for large enough graphs, and so this provides a solid alternative to Monte Carlo sampling. However beyond $T = 5, 6$ the nonlinear Markov chain implementation often becomes too slow

to practically run, and so past these times, the k-approximation or Monte Carlo offer accurate results without prohibitively large time costs.

In the intermediate regime with small degree and large depth, or large depth but small degree, it seems like Monte Carlo is the most accurate procedure, without a large time cost. Several examples in the previous sections show that neither the k-approximations or mean field approximations are very accurate in the stationary case.

7 Discussion and Further Work

In this thesis we study the *local recursion* and its ergodic properties, and analyzed the performance of the local recursion. To theoretically and numerically study the long term behavior of the local recursion, we introduced novel reformulations of the local recursion as a chain of infinite order and nonlinear Markov chain and introduced the k-approximations and frozen local recursions. We proved rudimentary properties concerning the ergodicity of the k-approximations and the frozen local recursions, and presented numerical investigations on the long term behavior of the k-approximations and local recursions.

We realize that this thesis poses more questions than answers, and much more work is needed to fully understand the ergodic properties of the local recursions. Among the most important properties to establish are:

- Investigations of ergodic properties of the k-approximations and local recursions in the *weak* topology, rather than the total variation topology.
- The convergence of the stationary distributions of the k-approximations to that of the local recursions, and the rate of such convergence.
- The correspondence between invariant measures of the local recursions and those of the full dynamics.
- A systematic comparison between the local recursion and other approximations for interacting particle systems.

The numerical evidence we collected in Section 6 indicate that the k-approximation is robust, in that it captures the ergodic properties of the local recursions quite well, and that convergence of the k-approximations at stationarity is hopeful. For a number of dynamics on regular trees, we see that it often suffices to run the 1-approximation to simulate the stationary distribution of the dynamics at a neighborhood of a node, and for most other cases it seems sufficient to run the k-approximation for small values of k. As another check of the richness of the k-approximations, we saw that

the stationary distributions exhibited similar behaviors to those of the full dynamics, indicating that the interchange property of Fig. 5.2 might hold: we in particular recovered phase transitions of the stationary distributions upon varying parameters of the models, and saw nonuniqueness of invariant distributions in the expected parameter regimes. The end goal is to analyze when the k-approximation is a good practical algorithm to understand the invariant distributions of the local recursion as well as those of the full dynamics.

In comparing the mean field approximation to the local recursion, we saw in several dynamics a very good agreement between the Monte-Carlo estimate of node probabilities, the mean field approximation, and the local recursion, especially in the cases where dynamics converged quickly. In other examples, we saw that the mean field approximation does not capture the correct stationary distributions; its main shortcoming, which is well-addressed by the local recursion, is the lack of dependence between neighboring nodes. In dynamics such as the oscillating models we defined, the mismatch between the mean field approximation and the Monte-Carlo estimate was quite large. On the other hand, the local recursion was able to capture this dependence structure quite accurately. For completeness, further work would also compare the local recursions to other methods such as the dynamics cavity method [17] and moment methods. See the results section of [32] for some analysis on this question.

Other questions we would have liked to explore include testing the local recursion on non-regular trees or even general graphs, to see when standard approximations like the mean field approximation outperform the local recursion. There are also numerous extensions one could make to this model: considering continuous time, infinite state space, or general graphs. However, many questions remain in the setting studied in this thesis. Based on the results provided in Section 6.5, further exploration is needed into the behavior of the 1-approximation near the critical values of the parameters for the Ising model. This suggests looking at the *boundary conditions* of the dynamics, since these are known to affect the convergence time of MCMC algorithms. At the same time, it is also known that fixing the boundary condition while simulating dynamics leads to convergence to different stationary measures, particularly in the case of the Ising model. An interesting question to then ask is which stationary distribution is picked out by the local approximation, and how does the initial measure affect this stationary distribution? We look forward to a complete understanding of the ergodic properties of the local recursion, as well as an understanding of the practical significance of the local recursion.

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A Appendix

A.1 Proofs

Progress on Conj. 5.1. Let β_s^μ denote the continuity rates for μ. Letting x, y be histories such that $(x = y)_m$, $m \geq 1$, we can observe the following (since, at least, the previous time steps agree):

$$\begin{cases} \mathbb{P}(\bar{X}_0(n) = 1|x) = \mathbb{P}(\bar{X}_0(n) = 1|y) = 1 - q & \text{if } \bar{X}_0(n-1) = 1 \\ \mathbb{P}(\bar{X}_0(n) = 1|x) = \mathbb{P}(\bar{X}_0(n) = 1|y) = \frac{p}{2}(\bar{X}_{-1}(n-1) + \bar{X}_1(n-1)) & \text{if } \bar{X}_0(n-1) = 0 \end{cases} \tag{68}$$

Hence $|\mathbb{P}(\bar{X}_0(n) = i\,|x) - \mathbb{P}(\bar{X}_0(n) = i\,|y)| = 0$. Doing a similar analysis with $v = \{-1, 1\}$, we see that (for $v = \{-1, 1\}$)

$$|\mathbb{P}(\bar{X}_v(n) = i|x) - \mathbb{P}(\bar{X}_v(n) = i|y)| \leq \frac{p}{2}|\mathbb{E}^\mu[X_v(n-1)|x] - \mathbb{E}^\mu[X_v(n-1)|y]|, \tag{69}$$

for all histories x, y with $(x = y)_m, i \in \{0, 1\}$.

Since $X_v(n)$ is $0, 1$-valued, we have

$$|\mathbb{P}(\bar{X}_v(n) = i|x) - \mathbb{P}(\bar{X}_v(n) = i|y)| \leq \frac{p}{2}|\mathbb{P}^\mu(\bar{X}_v(n) = i|x) - \mathbb{P}^\mu(\bar{X}_v(n) = i|y)|. \tag{70}$$

We would then like the following estimate, or something similar, to be true:

$$|\mathbb{P}^\mu(\bar{X}_v(n) = i|x) - \mathbb{P}^\mu(\bar{X}_v(n) = i|y)| \leq \frac{p}{2}\beta_s^\mu, \tag{71}$$

for each $v \in \{-1, 0, 1\}$. Using this we can bound:

$$|\mathbb{P}(\bar{X}(n)|x) - \mathbb{P}(\bar{X}(n)|y)| \leq p\beta_s^\mu \tag{72}$$

using the fact that $\mathbb{P}(\bar{X}(n)|x) = \prod_{v \in \{V\}} \mathbb{P}(\bar{X}_v(n)|x)$ by independence of the noise, and the inequality

$$|a'b'c' - abc| \leq |a - a'| + |b - b'| + |c - c'|, \tag{73}$$

which holds when each of $a, b, c, a', b', c' \leq 1$. Given this, we can take the continuity rates $\beta^{\bar{X}^\mu}(s) = p\beta_s^\mu$, which would gives summable continuity rates for the frozen local recursion.

□

Proof of Prop. 5.2. The goal is to prove for measures in the set $U \subset \mathcal{M}_{|\mathcal{X},k|^{\kappa+1}}$ that the nonlinear Markov chain operator $\Phi : \mu \mapsto \mu\mathbb{P}^\mu$ is continuous with respect to total variation distance. The approach will be to analyze the conditional expectation in Eq. (59) showing that the bounds on these conditional expectations will imply convergence of the measures $\mathbb{P}^\mu(x, \cdot), \mathbb{P}^\nu(x, \cdot)$ for any x. We then show that

$d_{TV}(\mu\mathbb{P}^{\mu}, \nu\mathbb{P}^{\nu}) \to 0$ as $d_{TV}(\mu,\nu) \to 0$. This will show continuity of the map Φ.

We now show that

$$d_{TV}(\mathbb{P}^{\mu}(x,\cdot), \mathbb{P}^{\nu_n}(x,\cdot)) \to 0 \tag{74}$$

as $d_{TV}(\mu,\nu_n) \to 0$, for measures ν_n, μ in $\mathcal{D}^{\kappa+1}_{|\mathcal{X}|,k}$. Note that $\mathbb{P}^{\mu}(x,\cdot) : \sigma(\mathcal{D}^{\kappa+1}_{|\mathcal{X}|,k}) \to [0,1]$. To bound the total variation distance between the kernels, we used the general facts that

$$d_{TV}(\mu_1 \otimes \cdots \otimes \mu_n, \nu_1 \otimes \cdots \otimes \nu_n) \le \sum_{i=1}^{n} d_{TV}(\mu_i, \nu_i) \tag{75}$$

and

$$d_{TV}(\text{Categorical}(p_1,\ldots,p_k), \text{Categorical}(q_1,\ldots,q_k)) = \frac{1}{2}\sum_{i=1}^{k} |p_i - q_i| \tag{76}$$

Eq. (76) follows from the isometry between d_{TV} and $\mathbb{L}^1$ on countable state probability spaces. Given Equations (76), (75), we can bound

$$d_{TV}(\mathbb{P}^{\mu}(x,\cdot), \mathbb{P}^{\nu_n}(x,\cdot)) \le \frac{1}{2}\sum_{v=0}^{\kappa}\sum_{i=1}^{|\mathcal{X}|} |F_v^i(\mathbf{x},\mu) - F_v^i(\mathbf{x},\nu_n)|, \tag{77}$$

so to prove the convergence (74), we need to show that $|F_v^i(\mathbf{x},\mu) - F_v^i(\mathbf{x},\nu_n)| \to 0$ as $\nu_n \to \mu$.

Note that for $v = 0$ this property is satisfied since $F_0^i(\mathbf{x},\nu_n)$ does not depend on μ. For $v = 1,\ldots,\kappa$, we can write the conditional expectation in Eq. (59) as

$$\sum_{\mathbf{z}} P_v^{\lfloor |\mathcal{X}_{\mathbf{x}_v}|, i\rfloor}(\mathbf{x}_v, \mathbf{x}_0, \mathbf{z}) \frac{\mu(\mathbf{x}_v, \mathbf{x}_0, \mathbf{z})}{\sum_{\mathbf{z}} \mu(\mathbf{x}_v, \mathbf{x}_0, \mathbf{z})}, \tag{78}$$

for some vector $\mathbf{z}$.

It suffices then to bound the difference of this term and itself, with a perturbed measure ν_n :

$$\left| \sum_{\mathbf{z}} P_v^{\lfloor |\mathcal{X}_{\mathbf{x}_v}|, i\rfloor}(\mathbf{x}_v, \mathbf{x}_0, \mathbf{z}) \frac{\mu(\mathbf{x}_v, \mathbf{x}_0, \mathbf{z})}{\sum_{\mathbf{z}} \mu(\mathbf{x}_v, \mathbf{x}_0, \mathbf{z})} - \sum_{\mathbf{z}} P_v^{\lfloor |\mathcal{X}_{\mathbf{x}_v}|, i\rfloor}(\mathbf{x}_v, \mathbf{x}_0, \mathbf{z}) \frac{\nu_n(\mathbf{x}_v, \mathbf{x}_0, \mathbf{z})}{\sum_{\mathbf{z}} \nu_n(\mathbf{x}_v, \mathbf{x}_0, \mathbf{z})} \right|. \tag{79}$$

Calling the normalization term on the left Z_μ and the other Z_{ν_n}, we can bound this term above by

$$\begin{aligned}
&\frac{1}{Z_\mu Z_{\nu_n}} \sum_{\mathbf{z}} P_v^{\lfloor |\mathcal{X}_{\mathbf{x}_v}|, i\rfloor}(\mathbf{x}_v, \mathbf{x}_0, \mathbf{z}) \left| \mu(\mathbf{x}_v, \mathbf{x}_0, \mathbf{z}) Z_{\nu_n} - \nu_n(\mathbf{x}_v, \mathbf{x}_0, \mathbf{z}) Z_\mu \right| \\
\le &\frac{1}{Z_\mu Z_{\nu_n}} \sum_{\mathbf{z}} P_v^{\lfloor |\mathcal{X}_{\mathbf{x}_v}|, i\rfloor}(\mathbf{x}_v, \mathbf{x}_0, \mathbf{z}) \left(|\mu(\mathbf{x}_v, \mathbf{x}_0, \mathbf{z}) - \nu_n(\mathbf{x}_v, \mathbf{x}_0, \mathbf{z})| + |Z_\mu - Z_{\nu_n}| \right) \\
\le &\frac{1}{Z_\mu Z_{\nu_n}} (2 d_{TV}(\mu,\nu_n) + 2 d_{TV}(\mu,\nu_n)).
\end{aligned}$$

The last line follows by definition of total variation distance and the fact that the functional $P_v^{[|\mathcal{X}_{\mathbf{x}_v}|,i]} \in [0,1]$. Here, we assume that Z_μ, Z_{ν_n} are nonzero, so that the conditional distribution is well defined. Hence, for μ fixed and $\nu_n \to \mu$ in total variation, the above quantity goes to zero.

The next step is to analyze the quantity $d_{TV}(\mu\mathbb{P}^\mu, \nu_n\mathbb{P}^{\nu_n})$, which by triangle inequality we can bound above by the sum of $d_{TV}(\mu\mathbb{P}^\mu, \mu\mathbb{P}^{\nu_n})$ and $d_{TV}(\mu\mathbb{P}^{\nu_n}, \nu_n\mathbb{P}^{\nu_n})$. Consider first the former quantity. It suffices to get bounds on

$$\left|\int_\Omega \mathbb{P}^\mu(\mathbf{x}, A) - \mathbb{P}^{\nu_n}(\mathbf{x}, A) d\mu(\mathbf{x})\right|, \tag{80}$$

but by the last result we had, we know that for every fixed x,

$$|\mathbb{P}^\mu(\mathbf{x}, A) - \mathbb{P}^{\nu_n}(\mathbf{x}, A)| \to 0 \tag{81}$$

as $d_{TV}(\mu, \nu_n) \to 0$. Combined with the finiteness of the space, this is enough to conclude that the integral converges to 0 (essentially, the integral above is actually a sum). Consider now the latter quantity, where it suffices to bound

$$\left|\int \mathbb{P}^{\nu_n}(x, A) d\mu(x) - \int \mathbb{P}^{\nu_n}(x, A) d\nu_n(x)\right|. \tag{82}$$

But again, these integrals are just sums, and so we can write the above as bounded above by

$$\sum_{\mathbf{x}} \mathbb{P}^{\nu_n}(\mathbf{x}, A) \left|\mu(\{\mathbf{x}\}) - \nu_n(\{\mathbf{x}\})\right|, \tag{83}$$

which goes to zero as $d_{TV}(\mu, \nu_n) \to 0$, since the terms $\mathbb{P}^{\nu_n}(\mathbf{x}, A) \in [0,1]$. Hence $d_{TV}(\mu\mathbb{P}^{\nu_n}, \nu_n\mathbb{P}^{\nu_n})$ as $d_{TV}(\mu, \nu_n) \to 0$. Putting these two results together, we obtain that as $d_{TV}(\mu, \nu_n) \to 0$,

$$d_{TV}(\mu\mathbb{P}^\mu, \nu_n\mathbb{P}^{\nu_n}) \le d_{TV}(\mu\mathbb{P}^\mu, \mu\mathbb{P}^{\nu_n}) + d_{TV}(\mu\mathbb{P}^{\nu_n}, \nu_n\mathbb{P}^{\nu_n}) \to 0 \tag{84}$$

□

Proof of Lemma 5.1. Since $\mathcal{M}_k^{\kappa+1}$ is a set of measures on a finite probability space, the total variation distance between $\mu, \nu \in \mathcal{M}_k^{\kappa+1}$ is given by

$$\frac{1}{2}\|p - q\|_{\mathbb{L}^1}, \tag{85}$$

where p, q. Hence $\mathcal{M}_k^{\kappa+1}$ under the total variation distance is isometric to a simplex under the $\mathbb{L}^1$ distance, which is compact since it is compact under the equivalent Euclidean metric.

Convexity of the space is simple to establish: if $\mu, \nu \in \mathcal{M}_k^{\kappa+1}$, then

$$(1-\alpha)\mu + \alpha\nu \tag{86}$$

is a probability measure constant on cubes of side length $|\mathcal{X}|^{-k}$, since each of μ, ν are. □

Proof of Lemma 5.2. We in fact have more then continuity; $S_k : \mathcal{M}^{\kappa+1}_{|\mathcal{X}|,m} \to \mathcal{M}^{\kappa+1}_{|\mathcal{X}|,k}$ is in fact distance non-increasing. To establish this, note that for all $A \in \sigma(\mathcal{D}^{\kappa+1}_{\mathcal{X},k})$, for any measure $\mu \in \mathcal{M}^{\kappa+1}_{|\mathcal{X}|,m}$

$$\mu(A) = S_k \circ \mu(A). \tag{87}$$

Hence,

$$\sup_{A \in \sigma(\mathcal{D}^{\kappa+1}_{\mathcal{X},k})} |S_k \circ \mu(A) - S_k \circ \nu(A)| = \sup_{A \in \sigma(\mathcal{D}^{\kappa+1}_{\mathcal{X},k})} |\mu(A) - \nu(A)|. \tag{88}$$

But this last expression is less than

$$\sup_{A \in \sigma(\mathcal{D}^{\kappa+1}_{\mathcal{X},l})} |\mu(A) - \nu(A)|, \tag{89}$$

since we are taking the infimum over a smaller set. Then it follows that

$$d_{TV}(S_k\mu, S_k\nu) \leq d_{TV}(\mu, \nu), \tag{90}$$

as desired. Of course this implies continuity since this shows that S_k is 1-Lipschitz. □

Proof of Prop. 5.3. We apply the Schauder Fixed Point Theorem, recorded in Thm. A.1, on the operator

$$S_k \circ \Phi : D \to D \tag{91}$$

which will then produce a fixed point of the k-approximation.

To check the assumptions of the theorem, note that $X = \mathcal{M}^{\kappa+1}_m$ is a Banach space under total variation distance: indeed the space is complete since it is isometric to a simplex with $\mathbb{L}^1$ distance - see the proof of Lemma 5.1. Now $S_k \circ \Phi$ is a continuous operator via Lemma 5.2, Prop 5.2, and that the space X is convex and compact. Since we assumed the subset D is closed it is also compact since X is isometric to Euclidean space, and we also assumed convexity. Further, D is bounded since the total variation of a probability measure is simply 1. Then the assumptions of Thm. A.1 are satisfied, producing existence of a fixed point. □

A.2 Fixed point theorems

In the proof of existence of a fixed point, we utilized the Schauder fixed point theorem, given below:

Theorem A.1 (Schauder Fixed Point Theorem)**.** *Given X a Banach space (complete, normed vector space), and $M \subset X$ a bounded, nonempty, convex, compact subset of X, and if $T : M \to M$ is continuous, there exists a fixed point of T.*

Its proof relies on the Banach fixed point theorem; see Zeidler [33] for further details.

A.3 Review of Distances and Convergence of Probability Measures

While there are numerous notions of probability distances and convergence, we review only weak convergence and total variation distance here.

Definition A.1 (Total Variation Distance)**.** *Let μ, ν be two measures on a common probability space $(\Omega, \mathcal{F})$. The total variation distance $d_{TV}(\mu, \nu)$ is defined by*

$$\sup_{A \in \mathcal{F}} |\mu(A) - \nu(A)|. \tag{92}$$

It is well known that for Ω a countable space then the total variation distance can be expressed as

$$\frac{1}{2} \sum_{\omega \in \Omega} |\mu(\{\omega\}) - \nu(\{\omega\})|. \tag{93}$$

Definition A.2 (Weak Convergence of Random Variables)**.** *Let $\{X_n\}_{n \in \mathbb{Z}}, X$ be random variables on probability spaces $(\Omega_n, \mathcal{F}_n), (\Omega, \mathcal{F}, \mathbb{P}_n)$ respectively, all with image (S, ρ) a metric space. We say that X_n converges weakly to X or $X_n \Rightarrow X$ if*

$$\mathbb{E}_n[f(X_n)] \to \mathbb{E}[f(X)] \tag{94}$$

for every $f : (S, \rho) \to \mathbb{R}$ bounded and continuous.

If $\{\mu_n\}_n$ is a sequence of measures on a probability space $(\Omega, \mathcal{F})$, then μ_n converges weakly to μ a measure on $(\Omega, \mathcal{F})$ if

$$\int f d\mu_n \to \int f d\mu \tag{95}$$

for every f bounded continuous function $f : \Omega \to S$, a metric space.

There are equivalent definitions when the probability distributions are defined on Euclidean space such as convergence in distribution; see the content of the well-known Portmanteau Lemma for related results. We note briefly that the topology induced by weak convergence (known as the weak-* topology in analysis) can be metrized, for example by the Levy-Prohorov metric. For further details see the general reference [4].

A.4 Nonlinear Local Equations and Mean Field Update

For completeness, we detail the nonlinear Local Equations for the processes listed in the thesis here, and their associated conversions to nonlinear Markov chains. These are formulated on a d-regular tree. We omit the contact process since the examples provided throughout the thesis handle these dynamics.

Parallel Ising Model

Recall that the dynamics for the node $v = 0$ is given by the same dynamics is defined in a previous section. Recall, for $v = 1, \ldots, d$

$$\mathbb{P}(\bar{X}_v^{t+1} = 1) = 1 - p\frac{\exp\left(-\beta \sum_{w=1}^d \bar{X}_0^t \bar{X}_w^t\right)}{\exp\left(-\beta \sum_{w=1}^d \bar{X}_0^t \bar{X}_w^t\right) + \exp\left(\beta \sum_{w=1}^d \bar{X}_0^t \bar{X}_w^t)\right)} \quad \text{if } \bar{X}_v^t = 1$$
$$\mathbb{P}(\bar{X}_v^{t+1} = 1) = p\frac{\exp\left(\beta \sum_{w=1}^d \bar{X}_0^t \bar{X}_w^t\right)}{\exp\left(-\beta \sum_{w=1}^d \bar{X}_0^t \bar{X}_w^t\right) + \exp\left(\beta \sum_{w=1}^d \bar{X}_0^t \bar{X}_w^t\right)} \quad \text{if } \bar{X}_v^t = -1.$$

Of course, since $|\mathcal{X}| = 2$, we have $\mathbb{P}(\bar{X}_v^{t+1} = 1) = 1 - \mathbb{P}(\bar{X}_v^{t+1} = 1)$.

Let X^{MF} denote the mean field approximation to the parallel Ising model. Its evolution is given by

$$\mathbb{P}(X^{MF}(t+1) = 1) =$$
$$1 - p\frac{\exp\left(-d\beta X^{MF}(t)(2X^{MF}(t) - 1)\right)}{\exp\left(-d\beta X^{MF}(t)(2X^{MF}(t) - 1)\right) + \exp\left(d\beta X^{MF}(t)(2X^{MF}(t) - 1)\right)} \quad \text{if } X^{MF}(t) = 1,$$
$$\mathbb{P}(X^{MF}(t+1) = 1) =$$
$$p\frac{\exp\left(d\beta X^{MF}(t)(2X^{MF}(t) - 1)\right)}{\exp\left(-d\beta X^{MF}(t)(2X^{MF}(t) - 1)\right) + \exp\left(d\beta X^{MF}(t)(2X^{MF}(t) - 1)\right)} \quad \text{if } X^{MF}(t) = -1.$$

We can rewrite the Parallel Ising Model as a nonlinear Markov chain with the following functionals for the Bernoulli terms. Here $\phi : \mathbb{R} \to \mathbb{R}$ is a function converting $\mathcal{X} = \{\pm 1\} \to 0, 1$ via $x \mapsto 2x - 1$. For notational simplicity, denote $\psi(x) = \phi \circ \lfloor 2x \rfloor$. For $v = 1, \ldots, d$, we then have:

$$\mathbb{E}^{Y \sim \mu}\left[1 - p\frac{\exp\left(-\beta \sum_{w=1}^d \psi(Y_0)\psi(Y_v)\right)}{\exp\left(-\beta \sum_{w=1}^d \psi(Y_0)\psi(Y_v)\right) + \exp\left(\beta \sum_{w=1}^d \psi(Y_0)\psi(Y_v)\right)} \mid Y_0 = x_v, Y_1 = x_0\right] \quad \text{if } \lfloor 2x_v \rfloor = 1$$
$$\mathbb{E}^{Y \sim \mu}\left[p\frac{\exp\left(\beta \sum_{w=1}^d \psi(Y_0)\psi(Y_v)\right)}{\exp\left(-\beta \sum_{w=1}^d \psi(Y_0)\psi(Y_v)\right) + \exp\left(\beta \sum_{w=1}^d \psi(Y_0)\psi(Y_v)\right)} \mid Y_0 = x_v, Y_1 = x_0\right] \quad \text{if } \lfloor 2x_v \rfloor = 0.$$

The functional for $v = 0$ is simply the expression inside the conditional expectation above.

voter Process

The dynamics of the voter process is given by:

$$\mathbb{P}(\bar{X}_v^{t+1} = 1) = \mathbf{1}\{\sum_{w\in\mathcal{N}_v} \bar{X}_w^t > 0\} + \frac{1}{2}\mathbf{1}\{\sum_{w\in\mathcal{N}_v} \bar{X}_w^t = 0\}. \tag{96}$$

$$\mathbb{P}(\bar{X}_v^{t+1} = -1) = \mathbf{1}\{\sum_{w\in\mathcal{N}_v} \bar{X}_w^t < 0\} + \frac{1}{2}\mathbf{1}\{\sum_{w\in\mathcal{N}_v} \bar{X}_w^t = 0\}. \tag{97}$$

The mean field approximation X^{MF} is given simply by:

$$\mathbb{P}(X^{MF}(t+1) = 1) = \mathbf{1}\{\mathbb{P}(X^{MF}(t) = 1) > \frac{1}{2}\} + \frac{1}{2}\mathbf{1}\{\mathbb{P}(X^{MF}(t) = 1) = \frac{1}{2}\}. \tag{98}$$

$$\mathbb{P}(X^{MF}(t) = -1) = \mathbf{1}\{\mathbb{P}(X^{MF}(t) = 1) < \frac{1}{2}\} + \frac{1}{2}\mathbf{1}\{\mathbb{P}(X^{MF}(t) = 1) = \frac{1}{2}\}. \tag{99}$$

Its nonlinear Markov chain has functionals

$$F_v(\mu, x) := \mathbb{P}^{Y\sim\mu}\left[\sum_{w=1}^{d} \psi(Y_w) > 0 \mid Y_0 = x_v, Y_1 = x_0\right] + \mathbb{P}^{Y\sim\mu}\left[\frac{1}{2}\sum_{w=1}^{d} \psi(Y_w) \mid Y_0 = x_v, Y_1 = x_0\right]. \tag{100}$$

Here, $\psi(x) := \phi \circ \lfloor 2x \rfloor = 2\lfloor 2x \rfloor - 1$.

Potts Model

Recall we consider the synchronous update Potts model, where each node on the graph is updated simultaneously. We assume that the state space of the Potts model is given by $\mathcal{X} = \{1, \ldots, q\}$, and its Hamiltonian be

$$H(\sigma) = \sum_{(i,j)\in E} \delta_{\sigma_i,\sigma_j}$$

for simplicity. Then the probability of a node v transitioning to state $l \in \mathcal{X}$ is given by

$$\frac{1}{q}\frac{\exp\left(-\frac{\beta}{2}\sum_{w\sim v}\delta_{l,\sigma_w}\right)}{\exp\left(-\frac{\beta}{2}\sum_{w\sim v}\delta_{l,\sigma_w}\right) + \exp\left(-\frac{\beta}{2}\sum_{w\sim v}\delta_{\sigma_v,\sigma_w}\right)}.$$

For the mean field update, we replace $\sum_{w\sim v}\delta_{l,X_w(t)}$ with $|\mathcal{N}_v|\sum_{w\sim v}\mathbb{P}(M(t) = l)$. Using some casework we can then write

$$\mathbb{P}(X^{MF}(t+1) = l) = \tag{101}$$

$$\sum_{i=1}^{q}\frac{1}{q}\frac{\exp\left(-\frac{\beta}{2}|\mathcal{N}_v|\mathbb{P}(X^{MF}(t) = l)\right)}{\exp\left(-\frac{\beta}{2}|\mathcal{N}_v|\mathbb{P}(X^{MF}(t) = l))\right) + \exp\left(-\frac{\beta}{2}|\mathcal{N}_v|\mathbb{P}(X^{MF}(t) = i)\right)}\mathbb{P}(X^{MF}(t) = i) \tag{102}$$

$$+\sum_{i=1}^{q}\frac{1}{q}\frac{\exp\left(-\frac{\beta}{2}|\mathcal{N}_v|\mathbb{P}(X^{MF}(t) = l)\right)}{\exp\left(-\frac{\beta}{2}|\mathcal{N}_v|\mathbb{P}(X^{MF}(t) = l))\right) + \exp\left(-\frac{\beta}{2}|\mathcal{N}_v|\mathbb{P}(X^{MF}(t) = i)\right)}\mathbb{P}(X^{MF}(t) = l) \tag{103}$$

The first term accounts for the probability of transition to l in a different state, and the second term accounts for the event that $M(t) = l$ and the transition to a different state fails.

The nonlinear Markov chain formulation of the Markov chain formulation has transition kernel

$$\mathbb{P}^{\mu}(x, \cdot) = \mathrm{Law}\left(\frac{1}{q}x + \frac{1}{q}v(x, \mu)\right)$$

where

$$v(x, \mu) \sim \bigotimes_{i=0}^{d} \mathrm{Categorical}(F_i^1(x, \mu), \ldots, F_i^q(x, \mu)),$$

and the functionals $F_v^j(x, \mu)$ are the transition probabilities of node $v \in \{0, \ldots, d\}$ to state $j \in \mathcal{X}$. For $v \neq 0$, these are given by

$$\mathbb{E}^{Y\sim\mu}\left[\frac{1}{q}\frac{\exp\left(-\frac{\beta}{2}\sum_{w=1}^{d}\delta_{j,Y_w}\right)}{\exp\left(-\frac{\beta}{2}\sum_{w=1}^{d}\delta_{j,Y_w}\right) + \exp\left(-\frac{\beta}{2}\sum_{w=1}^{d}\delta_{l,Y_w}\right)} \mid Y_0 = x_v, Y_1 = x_0\right] \text{ if } \lfloor qx_v \rfloor = l \neq j \tag{104}$$

$$\mathbb{E}^{Y\sim\mu}\left[\frac{1}{2q} + \frac{1}{q}\sum_{i=1}^{q}\frac{\exp\left(-\frac{\beta}{2}\sum_{w=1}^{d}\delta_{j,Y_w}\right)}{\exp\left(-\frac{\beta}{2}\sum_{w=1}^{d}\delta_{i,Y_w}\right) + \exp\left(-\frac{\beta}{2}\sum_{w=1}^{d}\delta_{j,Y_w}\right)} \mid Y_0 = x_v, Y_1 = x_0\right] \text{ if } \lfloor qx_v \rfloor = j \tag{105}$$

$$\tag{106}$$

Oscillating Stochastic CA Model

Recall the dynamics for the binary oscillating c.a. model:

$$\mathbb{P}(X_v(t+1) = 0 \mid X_v(t) = 1) = 1 - \frac{p\sum_{w:w\in\partial v}(1 - X_w(t))}{|\partial v|} \tag{107}$$

$$\mathbb{P}(X_v(t+1) = 1 \mid X_v(t) = 0) = 1 - \frac{p\sum_{w:w\in\partial v} X_w(t)}{|\partial v|}. \tag{108}$$

The mean field approximation X^{MF} is given by the evolution

$$\mathbb{P}(X^{MF}(t+1) = 0 \mid X^{MF}(t) = 1) = 1 - p\mathbb{P}(X^{MF}(t) = 0) \tag{109}$$

$$\mathbb{P}(X^{MF}(t+1) = 1 \mid X^{MF}(t) = 0) = 1 - p\mathbb{P}(X^{MF}(t) = 1), \tag{110}$$

and the nonlinear Markov chain has functionals, for $v \neq 0$, given by $F_v(\mathbf{x}, \mu) =$

$$\mathbb{E}^{Y\sim\mu}\left[\frac{p\sum_{w=1}^{d}(1 - \lfloor 2Y_w \rfloor)}{|\partial v|} \mid Y_0 = x_v, Y_1 = x_0\right] \text{ if } \lfloor 2x_v \rfloor = 1 \tag{111}$$

$$\mathbb{E}^{Y\sim\mu}\left[1 - \frac{p\sum_{w=1}^{d}\lfloor 2Y_w \rfloor}{|\partial v|} \mid Y_0 = x_v, Y_1 = x_0\right] \text{ if } \lfloor 2x_v \rfloor = 0 \tag{112}$$

$$\tag{113}$$

Code Organization

The code for the local recursion and k-approximation was implemented in Python with heavy use of Numpy, and black-box optimized using Numba, a JIT compiler. For install instructions, see http://numba.pydata.org. To preserve fidelity with Numba, implementations of the dynamics and local recursions for each of the models discussed in Section 2 are done separately, although many parts of the algorithm can be abstracted. The code can be found on https://github.com/tsudijon.